DALHOUSIE UNIVERSITY

FACULTY OF GRADUATE STUDIES

A thesis submitted to the Faculty of Graduate Studies of
Dalhousie University in partial fulfillment of the requirements
for the degree of Doctor of Philosophy

Department of Electrical and Computer Engineering

Cognitive Autonomic Power System Management: A Rigorous Experimental Study of Neural, Quantum-Inspired and Metacognitive Control against Classical Planning for FACTS-Coordinated EV Charging

by

«Candidate Name»

Dissertation defended by proxy of the evidence-integrated manuscript — see the Completion Matrix

Submitted: «Month Year»

Halifax, Nova Scotia — «Month Year»

ABSTRACT

This dissertation reports a fully reproducible, statistically pre-registered experimental study of a three-layer cognitive control architecture for electric power system operation — the Cognitive Autonomic Power System Manager (CAPSM) — and answers, with paired episode-level inference on a frozen final benchmark, whether any of its components improves on strong classical control. The testbed is a 24-hour sequential FACTS/EV operation problem on the IEEE 39-bus network: a validated Q-limited AC power-flow engine (reproducing a 384-row PSS®E 33.4 replay audit and cross-checked against PYPOWER on IEEE 9/14/39/118 to under 5×10^{-6} pu) dispatches a shared charging command to three aggregated EV fleets (state-of-charge, deadline and headroom constraints), an SVC at bus 14 (±50 MVAr, one level per hour) and a TCSC on the 2-3 corridor (0.70-1.00× compensation, one level per hour), under five stress classes including N-1 line trips. System 1 is a CNN-LSTM imitation policy with a cost-to-go value head trained on a deployable MPC teacher; System 2 is a quantum-inspired imaginary-time candidate search over the 75 joint actions; a metacognitive Arbiter gates System 2 by predicted value of computation. Against deadline-aware greedy, tabular cost-Q reinforcement learning, gradient-boosted trees, budget-matched population search and genuine time-varying receding-horizon DP (H=6/12/24), the pre-declared primary endpoint (paired median episode cost, clustered bootstrap 95% CI, sign-flip test, Holm-corrected over 24 comparisons) yields a scientific negative for every CAPSM component on the 75-episode profile-family-separated final corpus: the CNN-LSTM is not significantly better (median difference +181 MW-eq, Holm p=0.20), the quantum-inspired search is indistinguishable from matched-budget random search at N=8/16/32/64 (all median differences ≈ 0), and the Arbiter adds no value over its constituents because its uncertainty and risk signals fire on 93.6% of steps. The classical MPC planner, by contrast, improves ≈700 MW-eq per episode (≈9.4%, Holm p=0.0011, d=−0.66) and recovers 99.3% of the perfect-foresight oracle gap at 465 ms CPU per episode. The dissertation documents every repair that produced these results — a cost/reward sign-invariant reformulation, honest budget accounting of draws versus distinct AC evaluations, a nine-way failure taxonomy separating solver non-convergence from proven infeasibility, and a sealed one-run evaluation protocol — and preserves the negative findings with the same prominence as the positive control.

TABLE OF CONTENTS

ABSTRACT.....

ABSTRACT.....2

TABLE OF CONTENTS.....3

LIST OF FIGURES.....3

LIST OF TABLES.....4

NOMENCLATURE AND ABBREVIATIONS.....7

Chapter 1 — Introduction.....9

1.1 Operating context and motivation.....9

1.2 The CAPSM hypothesis.....9

1.3 Research questions and hypotheses.....10

1.4 Scope and contributions.....10

1.5 Evidence tiers and honesty contract.....11

1.6 Dissertation organisation.....12

Chapter 2 — Literature Review and Gap Analysis.....12

2.1 Scope and method of the review.....12

2.2 Optimal power flow, MPC and planning-based control.....13

2.3 Reinforcement learning and multi-agent control for power systems.....13

2.4 Deep sequence learning: forecasting, imitation and the closed-loop gap.....14

2.5 Quantum and quantum-inspired optimisation.....14

2.6 Metacognition, metareasoning and agentic architectures.....14

2.7 Physics-anchored validation, uncertainty and real-time constraints.....15

2.8 Synthesis: the gap this dissertation fills.....15

2.9 The grid-forming and inverter-control context.....16

2.10 The EV-charging coordination literature in depth.....16

2.11 The FACTS literature in depth....16
2.12 The deep-RL scheduling literature in depth....17
2.13 The forecasting and sequence-model literature in depth....17
2.14 The quantum and quantum-inspired literature in depth....17
2.15 The metacognitive and agentic literature in depth....17
2.16 Synthesis revisited....18
Chapter 3 — Physical System, Constraints and Objective....18
3.1 Overview and design principles....18
3.2 Network, base case and AC power flow....18
3.3 AGC re-dispatch....19
3.4 EV fleets and the energy-balance ledger....19
3.5 FACTS devices....20
3.6 Action space and legality....20
3.7 Objective function and failure taxonomy....20
3.8 Documented simplifications....21
3.9 Newton-Raphson engine and the batch solver....22
3.10 Network matrices and branch model....23
3.11 The Q-limit enforcement semantics in equations....24
Chapter 4 — CAPSM: Formulation of the Cognitive Control Stack....24
4.1 The sequential decision problem....24
4.2 System 1: CNN-LSTM reflex policy with cost-to-go head....25
4.3 System 2: quantum-inspired sequential search....25
4.4 Metacognitive Arbiter: the value-of-computation gate....26
4.5 The sign invariant....27
4.6 Complexity, latency and deployment posture....27
4.7 Algorithm summaries....28

Chapter 5 — The Planning Core: Sign Invariant, Exact DP and Classical Baselines.....29
5.1 The analytical two-step example (E01).....29
5.2 Exact dynamic programming on the joint lattice.....30
5.3 The offline oracle and honest labelling.....30
5.4 MPC: receding-horizon deployment of the planner.....31
5.5 Classical baselines.....31
5.6 What the planning core buys.....32
5.7 Exactness verification against exhaustive enumeration.....32
Chapter 6 — Testbed, Data Provenance and Reproducibility.....33
6.1 Network cases and provenance.....33
6.2 Independent AC cross-checks (E03).....33
6.3 Corpus design.....34
6.4 The freeze protocol and the sealed run.....34
6.5 Run provenance.....35
6.6 Software architecture and test suite.....36
6.7 Reproduction entry points.....36
6.8 Test-suite inventory.....36
6.9 A worked episode from the sealed ledger.....37
6.10 The frozen controller suite.....37
6.11 The freeze amendment log.....38
Chapter 7 — System 1: CNN-LSTM Reflex Control.....38
7.1 Experimental design.....38
7.2 Dataset and training.....39
7.3 Architecture ablation (E06).....39
7.4 Closed-loop behaviour and the drift signature.....40
7.5 Latency and deployment posture.....40

7.6 Chapter summary.....40
7.7 Training records.....41
7.8 System 1 on the final corpus, by class.....41
7.9 Distributional analysis of the System-1 deficit.....42
7.10 Summary of the System-1 evidence.....42
Chapter 8 — System 2: Quantum-Inspired Sequential Search.....42
8.1 Experimental design.....42
8.2 Honest budget accounting (E04).....43
8.3 Quantum-inspired vs uniform search at matched budgets.....44
8.4 The value-augmentation ablation.....44
8.5 Chapter summary.....44
8.6 Step-status inventory of the sealed run.....45
Chapter 9 — The Metacognitive Arbiter.....45
9.1 Experimental design.....45
9.2 Tuning outcome.....46
9.3 Signal calibration.....46
9.4 Arbiter vs controls.....47
9.5 What would a working gate require.....47
9.6 Chapter summary.....47
9.7 The complete tuning grid.....47
9.8 Decision-log statistics.....48
Chapter 10 — Final Evaluation: The Sealed Run and Statistical Verdict.....48
10.1 The sealed run.....48
10.2 Primary endpoint (PE1).....49
10.3 Reading of the primary endpoint.....50
10.4 Marginal medians and per-class structure.....50

10.5 Compute and latency.....51
10.6 The verdict.....51
10.7 Per-profile-family results.....51
Chapter 11 — Robustness, Network Qualification and the Dynamic-Security Boundary.....52
11.1 Robustness across stress classes (E09).....52
11.2 Five-network qualification (E11).....53
11.3 The dynamic-security boundary.....53
11.4 Threats to validity.....54
11.5 Summary.....54
11.6 Complete development-test comparison.....55
Chapter 12 — Discussion, Conclusions and Future Work.....55
12.1 Reconciling the findings with the hypotheses.....55
12.2 Engineering versus algorithmic interpretation.....56
12.3 Practical deployability.....56
12.4 Limitations.....56
12.5 Future work.....57
12.6 Conclusions.....57
12.7 Contribution-evidence map.....58
12.8 Closing statement.....58
12.9 Hypothesis verdicts.....58
REFERENCES.....59

Note: page numbers in the table of contents are refreshed by Word/WPS: right-click the table and choose Update Field.

LIST OF FIGURES

Figure 6.1 — Profile-family separation: the three final-corpus load-shape parents against the development parent. Families differ in peak placement, evening shape and solar coupling; episode-level AR(1) noise and class stressors sample around each parent.

Figure 7.1 — System-1 architecture ablation (dev val medians). The CNN-LSTM composite is the strongest architecture; all neural policies remain in the same band as the greedy baseline.

Figure 8.1 — E04 budget fairness on the final corpus: (a) distinct AC evaluations per episode against nominal N; (b) median episode cost. The quantum-inspired and uniform searches are statistically indistinguishable at every matched budget.

Figure 8.2 — Value-head calibration on the validation split (log-log axes): rank discrimination is strong, absolute calibration is weak. The colour is proportional density of 2,700 state visits.

Figure 9.1 — Arbiter behaviour on the final corpus (all three variants): (a) System-2 invocation fraction by hour; (b) the decision signals, with the invoke/accept split. The gate is effectively always-closed.

Figure 10.1 — PE1 forest plot on the final corpus: paired median differences with 95% clustered-bootstrap confidence intervals. Blue: significant improvement (Holm-adjusted); red: significant worsening; grey: not significant.

Figure 10.2 — Per-class median episode cost (log scale) for the headline controllers on the final corpus. The contingency class dominates the means; the planner advantage is concentrated in the normal/stressed/ramp classes.

Figure 10.3 — Compute cost versus closed-loop quality on the final corpus (mean CPU per episode, log scale). The MPC planners dominate the cost-quality frontier of this benchmark.

Figure 11.1 — Five-network base-case voltage profiles (sorted by magnitude) under MATPOWER runpf semantics, with the 0.94-1.06 source band shaded. All five converge with published-level losses; cross-network closed-loop control is NOT EXECUTED.

LIST OF TABLES

Table 2.1 — Literature coverage matrix (2020-2026). ✓ = addressed by multiple peer-reviewed studies; (✓) = addressed partially or only in simulation; ✗ = not addressed in the reviewed literature.

Table 5.1 — Development validation medians (dev val, 75 episodes) for the planning core and baselines, MW-eq per 24-h episode. Lower is better.

Table 5.2 — Exactness checks. Left: three-step DP value vs exhaustive enumeration (75^3 sequences at most). Right: full-horizon DP without slew vs the Phase-10 reference (its own convention), MW-eq.

Table 6.1 — Evidence artefacts and their roles. All paths are relative to the released repository root.

Table 6.2 — Test-suite inventory (63 tests, all green on the release hardware in under five seconds).

Table 6.3 — Worked episode from the sealed ledger (mpc_H12, V27_FINAL__E_NORMAL_000). Cost in MW-eq; latency in ms. Status codes as in Section 3.7.

Table 6.4 — The frozen controller suite (26 controllers). QIRL/popsearch seeds derive from SHA-256 of (kind, seed, episode-id) — deterministic per episode.

Table 6.5 — Freeze amendments A1-A4. In every case the scientific impact assessment is none: no evaluated code path, model, corpus, table or statistic changed.

Table 7.1 — System-1 architecture ablation, dev val split (75 episodes). Median closed-loop episode cost in MW-eq; latency is batch-size-1 CPU inference per step, legality screening included.

Table 7.2 — Training records. Validation cross-entropy (CE) over 75 action classes; chance = log 75 ≈ 4.32. The value head’s rank discrimination is Spearman ρ (within-episode).

Table 7.3 — System-1 variants on the final corpus: per-class medians (MW-eq). Lower is better within a column.

Table 7.4 — Distributional statistics on the final corpus (MW-eq): medians, means, upper quantiles, maxima and shortfall inventories.

Table 8.1 — Measured computation per episode, final corpus (75 episodes). Draws and distinct AC evaluations per 24-step episode; CPU milliseconds include candidate scoring and the value-head batch.

Table 8.2 — Step-status inventory of the sealed run (counts per controller over 1,800 steps; the four statuses that occurred are shown).

Table 9.1 — Arbiter tuning grid (dev val, 24-episode subsample): median episode cost and System-2 invocations. The outcome is insensitive to the predicted-VoC threshold because the uncertainty and risk signals dominate.

Table 9.2 — Complete arbiter tuning grid (dev val, 24-episode subsample; 576 steps).

Table 9.3 — Arbiter decision-log statistics, final run (5,402 rows, all three variants).

Table 10.1 — PE1 on the final corpus. Paired median difference vs greedy_deadline (MW-eq), 95% clustered bootstrap CI, raw and Holm-adjusted sign-flip p, paired Cohen’s d. Decision rule: Holm $p < 0.05$ and CI entirely below zero.

Table 10.2 — Final-corpus marginal medians and failure inventory (75 episodes; all MW-eq). n_dead counts steps with no convergent action in the finite action set; shortfalls in MWh.

Table 10.3 — Final-corpus medians by profile family (MW-eq). Each cell is the median over the 25 episodes of one family.

Table 11.1 — Development test split (E09): paired median differences vs greedy_deadline with Holm adjustment over the 22-comparison family (MW-eq).

Table 11.2 — E09 complete: paired median differences vs greedy_deadline on the dev test split (MW-eq), Holm-adjusted p over the 22-comparison family.

Table 12.1 — Contribution-evidence map. Gates: G02 (sign consistency), G03 (compute fairness), G04 (statistics), G05 (physical safety), G10 (final lockbox), G11 (statistical verdict), G16 (complete package).

Table 12.2 — Hypothesis verdicts under the pre-registered decision rule.

NOMENCLATURE AND ABBREVIATIONS

The following notation is used throughout the thesis. Units follow SI conventions; per-unit (pu) quantities use a 100 MVA base.

V_o, θ_o — bus voltage magnitude (pu) and angle (rad); **Y** — bus admittance matrix; **S_i = P_i + jQ_i** — net complex injection at bus *i* (MVA).

P^d, Q^d — branch active/reactive flow from-end (MW/MVAr); **P^d^max** — branch rate A (MVA).

r ∈ {0, 0.25, 0.5, 0.75, 1} — commanded EV charging rate; **B ∈ {±50, ±25, 0, +25, +50} MVAr** — SVC injection; **k ∈ {0.70, 0.85, 1.00}** — TCSC reactance multiplier.

SoC^f — fleet *f* state of charge (pu); **E^f** — fleet energy (MWh); **η^f, P^f^max, C^f** — fleet efficiency, max charging power (MW), capacity (MWh).

c_t(a) — step cost of action *a* at hour *t* (MW-eq); **J** — 24-hour episode objective (MW-eq); **π_t** — normalised energy price (MW-eq per MWh).

Q_C(s,a), V_C(s) — cost-to-go action/state values under minimisation; **γ** — discount factor (=1 episodic here).

ψ ∈ $\mathbb{C}^{75}$ — System-2 complex amplitude vector; **β** — imaginary-time suppression rate; **N** — candidate budget per step.

$\hat{H}$(s) — learned cost-to-go value head; **VoC** — predicted value of computation (MW-eq).

Chapter 1 — Introduction

1.1 Operating context and motivation

Electric transmission systems are operated closer to their physical limits than at any point in the last three decades. Three structural changes drive this. First, inverter-based renewable generation has displaced synchronous inertia and made net load both more volatile and less predictable, with wind and solar forecast errors propagating directly into dispatch and voltage-control decisions [2,5,10]. Second, transportation electrification concentrates large, schedulable but deadline-bound charging loads at distribution interfaces: fleets of electric vehicles introduce flexibility that is valuable only if it is orchestrated under state-of-charge (SoC), departure-deadline and feeder-capacity constraints [1,23]. Third, flexible AC transmission system (FACTS) devices — static VAR compensators (SVC) and thyristor-controlled series compensators (TCSC) — give operators fast, discrete compensation levers whose effectiveness depends on the coordination of slow setpoint changes with the hourly evolution of the network state [38,45].

The operational problem that emerges is a sequential decision problem under uncertainty: at every hour the operator (or an automated controller) selects joint commands — an EV charging rate, an SVC injection level, a TCSC compensation level — whose immediate AC power-flow consequences and whose future feasibility both matter. A command that is cheap now may strand a fleet below its departure target; a compensation setting that relieves one corridor may push another past its thermal rating after a line trip. Classical practice decomposes this problem: energy management systems re-dispatch on 5-15 minute cycles, automatic generation control absorbs fast deviations, and special protection schemes react to contingencies. The question this dissertation pursues is whether a cognitive, layered control architecture can do better within the same information set — and, equally importantly, to quantify precisely when it cannot.

1.2 The CAPSM hypothesis

The Cognitive Autonomic Power System Manager (CAPSM) is a three-layer control architecture proposed in the earlier phases of this research programme. Its design borrows from dual-process theories of cognition. System 1 is a fast reflexive policy — a

convolutional-recurrent neural network (CNN-LSTM) mapping the observable operating state directly to a joint action in a single forward pass. System 2 is a slower deliberative search — in the version studied here, a quantum-inspired candidate generator that maintains complex amplitudes over the discrete action set and suppresses high-cost actions through an imaginary-time update, scored by a learned cost-to-go value model. A metacognitive Arbiter sits above both and decides, step by step, whether the fast policy suffices or the deliberative system should be invoked, using an explicit model of the value of computation.

The hypothesis motivating CAPSM is that layered cognition buys operational efficiency: reflex where the state is familiar, deliberation where it is stressed. The literature contains encouraging signals for each ingredient in isolation — deep reinforcement learning for scheduling [34,53], quantum-inspired metaheuristics for dispatch-class problems [73], metareasoning for compute allocation in sequential decision tasks [82] — but remarkably few studies the composed architecture to a pre-registered, statistically disciplined comparison against a strong classical planner on the same physical engine. This dissertation does exactly that, and reports the outcome whether positive or negative.

1.3 Research questions and hypotheses

Five research questions structure the study. Each is paired with a falsifiable hypothesis and a pre-declared decision rule evaluated in Chapters 7-10.

(1) **RQ1** (System 1). Can a CNN-LSTM policy trained by imitation of a deployable planner match or beat a deadline-aware greedy controller in closed loop, at millisecond inference latency? *H1: the CNN-LSTM improves the median episode cost by at least 2% with Holm-adjusted significance.*

(2) **RQ2** (System 2). Does quantum-inspired imaginary-time candidate search improve action selection over (a) myopic table greedy and (b) uniform population search at matched measured computation? *H2: at N=32 the search improves the median cost over both comparators with Holm-adjusted significance.*

(3) **RQ3** (Arbiter). Does value-of-computation gating produce a better cost-compute operating point than always invoking System 1, always invoking System 2, or random switching at the same invocation rate? *H3: the arbiter dominates all three controls on the cost-compute frontier.*

(4) **RQ4** (Baselines). How much of the gap between myopic greedy and the perfect-foresight offline optimum can a deployable receding-horizon planner recover on

this benchmark? *H4 (positive control): MPC with a genuinely time-varying multi-period DP and obligation-aware terminal value recovers at least 80% of the oracle gap.*

(5) **RQ5** (Methodology). Can the full pipeline — physics engine, controllers, statistics — be repaired to a state where every claim is reproducible, every failure is classified honestly, and the final evaluation is sealed before execution? *H5: a one-run sealed protocol with pre-registered endpoints is achievable in practice, and its verdict is stable across development and final corpora.*

1.4 Scope and contributions

The scope is deliberately bounded. The physical engine is a quasi-steady-state hourly AC power-flow benchmark on the IEEE 39-bus New England system with three aggregated unidirectional EV fleets, one SVC, one TCSC, AGC-style proportional re-dispatch and N-1 line contingencies from a frozen critical list. Dynamic security (RMS simulation), hardware-in-the-loop validation, bidirectional V2G discharge and distribution-level network detail are outside the executed scope; Chapter 11 documents these boundaries explicitly and labels the corresponding work NOT EXECUTED or EXTERNALLY BLOCKED rather than describing design intent as completed evidence.

The contributions are methodological as much as algorithmic:

(1) A **cost-sign-invariant** reformulation of the sequential control problem: every learned quantity is a cost-to-go under minimisation, enforced by an analytical two-state, two-action, two-step example in which the myopic rule provably selects the wrong first action (Chapter 5), with the v2.6 reward/cost mixture defect class shown to be impossible by construction.

(2) An **exact** backward-induction dynamic program over the joint cumulative-rate/device lattice of the EV-FACTS action space, verified against exhaustive enumeration, that serves both as the offline oracle and as the deployable MPC planner.

(3) An **honest computation-accounting** protocol that separates nominal candidate draws, distinct proposals, distinct AC evaluations, cache hits and measured CPU time, making budget-matched comparisons between stochastic searches meaningful.

(4) A **nine-way failure taxonomy** that distinguishes AC solver non-convergence from proven infeasibility from controller illegality, and a sealed one-run evaluation

protocol with a pre-registered primary endpoint, clustered bootstrap inference, a sign-flip randomization test and Holm correction.

(5) A **complete negative-and-positive verdict**: no CAPSM component beats the classical baseline under the pre-registered rule, while the classical MPC planner recovers 99.3% of the oracle gap — a result the author considers as valuable as a positive one, because it is the first result in this programme that survives its own statistical protocol.

1.5 Evidence tiers and honesty contract

Every quantitative claim in this dissertation belongs to one of five declared evidence tiers, and the tier is stated wherever a number appears. Tier E1 (executed, this work): the v2.7 sealed run and its supporting development experiments, all reproducible from the released software. Tier E2 (executed, historical): the Phase 1-11 evidence base, including the 384-row PSS®E 33.4 replay audit and the Phase 8 MATPOWER cross-check, preserved immutably. Tier E3 (independently cross-checked): results reproduced by an independent solver in this work — notably the PYPOWER agreement on four IEEE cases. Tier E4 (design-only): interfaces and models specified but not executed (PSS®E 35 native, RMS dynamics, SIL/HIL). Tier E5 (blocked): work that could not run in this environment, with the precise blocker recorded. Claims never silently cross tiers: a design-only capability is never narrated as a measurement, and a historical measurement is never relabelled as a current result.

1.6 Dissertation organisation

Chapter 2 reviews the 2020-2026 literature across optimisation, learning, quantum-inspired methods and metacognition, and derives the gap this work fills. Chapter 3 specifies the physical model, its constraints, its objective and every simplification. Chapter 4 formalises CAPSM as a constrained sequential decision problem and derives the mathematical form of each layer. Chapter 5 develops the planning core: the sign invariant, the exact DP, the MPC formulation and the classical baselines. Chapter 6 describes the testbed, data provenance, the five-network validation and the reproducibility protocol. Chapters 7, 8 and 9 report the System 1, System 2 and Arbiter experiments respectively. Chapter 10 reports the sealed final evaluation and the statistical verdict. Chapter 11 covers

robustness, network qualification and the dynamic-security boundary. Chapter 12 discusses the findings, states the limitations honestly and closes with the conclusions.

The remainder of the dissertation is intentionally written so that every table and figure can be regenerated from the released software with a single command, and every number can be traced to a row of the released evidence ledger. The thesis is, in this sense, the narrative layer of an executable artefact.

Chapter 2 — Literature Review and Gap Analysis

2.1 Scope and method of the review

This chapter reviews the 2020-2026 literature that bears on the CAPSM hypothesis along six axes: (i) classical optimisation and model-predictive control of power systems; (ii) reinforcement learning and multi-agent control; (iii) deep sequence learning for state forecasting; (iv) quantum and quantum-inspired optimisation; (v) metacognition, metareasoning and agentic architectures; and (vi) physics-anchored validation, uncertainty quantification and real-time constraints. The review draws on 200 individually Crossref-verified sources (all DOIs validated and audit-trailed in bibliography/reference_audit.csv); it privileges peer-reviewed journal and proceedings records, and it includes balanced coverage of negative and methodological results where they exist. Foundational pre-2020 sources — Bellman’s optimality principle, the IEEE test-case archive, the dual-process cognitive literature — are cited by name in the text and are deliberately excluded from the 2020-2026 bibliometric core, per the bibliography contract.

The review serves a specific function: to locate, for each CAPSM ingredient, the strongest published evidence that it helps on power-system-class problems, so that the hypotheses of Chapter 1 are stated against the best case the literature makes, not against straw baselines. This is why the experimental chapters compare against a genuine receding-horizon dynamic-programming planner rather than against fixed schedules or naive random agents: the review shows that planning-based baselines dominate the published learning results more often than not.

2.2 Optimal power flow, MPC and planning-based control

Optimal power flow (OPF) remains the canonical formulation for steady-state network optimisation, and the last six years of research have consolidated both its solver technology and its approximation theory [142,143,145,149]. Convex-relaxation approaches — second-order-cone and semidefinite reformulations — give certified bounds for distribution-class problems [144,147,148], while interior-point solvers handle security-constrained transmission instances at operational scale [146]. Two properties of this literature matter for this dissertation. First, the objective is always a cost to be minimised under explicit physical constraints — the sign convention is fixed by the physics community's OPF heritage, and the v2.6 reward/cost mixture defect documented in Chapter 5 is precisely a violation of this discipline. Second, feasibility is a certificate, not a solver exit code: an unconverged Newton iteration is not evidence of infeasibility, a distinction formalised in the nine-way failure taxonomy of Chapter 3.

Model predictive control translates the OPF idea to the sequential setting: re-solve a finite-horizon optimisation every step, apply the first action, shift the horizon. For power systems, MPC has matured from a promising idea into deployed practice for building-climate co-optimisation [48], microgrid energy management [47,50] and transmission-level voltage control [17,21]. The published evidence consistently shows that the terminal value function — what the truncated horizon assumes about the future beyond its edge — separates good MPC implementations from mediocre ones. Phase 10 of this research programme reproduced exactly this sensitivity: an MPC with zero terminal value ignored departure obligations at the horizon edge, and repairing the terminal value to charge remaining deadline obligations immediately (Chapter 5) recovered several hundred MW-equivalents per episode. The MPC baseline used throughout this dissertation therefore represents the repaired, obligation-aware variant — the strongest classical comparator the literature supports.

The planning literature also supplies the honest labelling discipline this dissertation adopts. An offline policy computed with realised future information is an oracle — an upper reference — and must never be compared as a deployable method. The Phase-10 legacy controller labelled dp_clairvoyant, which selected device setpoints myopically while planning only the rate trajectory, was renamed in this work to reflect exactly what it computes; Chapter 5 formalises the distinction between perfect-foresight offline references and forecast-only deployed policies.

2.3 Reinforcement learning and multi-agent control for power systems

Reinforcement learning is the most active of the reviewed axes. For economic dispatch and unit commitment, deep RL policies achieve competitive operating cost with constraint-handling mechanisms of varying rigour — action masking, penalty shaping, safety layers and Lagrangian reformulations [34,53,150,151]. For voltage control, safe RL with explicitly validated constraint satisfaction has produced policies that respect device limits in closed loop [155,156]. Multi-agent formulations decompose the control problem spatially, with consensus and communication architectures receiving the bulk of recent attention [153,154,169].

Three patterns recur across this literature and shape the design of the experiments in Chapters 7-9. First, baselines are weak more often than not: a substantial fraction of published improvements are measured against fixed schedules or unstructured random policies rather than against a planning baseline of equal information. Second, evaluation is overwhelmingly point-estimate: mean or median improvement over a modest episode sample, without paired episode-level inference, without multiplicity control across the many reported comparisons, and without a pre-declared primary endpoint. Third, the tabular and function-approximation variants are rarely compared on the same engine with the same information set, which is precisely the comparison Chapter 7 runs (tabular cost-Q against CNN-LSTM imitation against gradient-boosted trees, all on identical episodes).

The classical-RL baseline in this dissertation is deliberately modest — tabular Q-learning with a cost-to-go update and a factored state of a few thousand cells — because its role is diagnostic: it isolates how much of the neural policies' performance is available to any value-agnostic memoriser of the state-action landscape, and how much genuinely requires function approximation over raw features.

2.4 Deep sequence learning: forecasting, imitation and the closed-loop gap

The forecasting literature provides the strongest published evidence for deep sequence models in power systems. CNN-LSTM and related architectures dominate short-term load and renewable forecasting benchmarks [150,152,154,157], and probabilistic variants now quantify their own uncertainty [151,156]. The lesson the CAPSM System 1 inherits from this

line is architectural: convolutional feature extraction over the cross-sectional observation (bus loads, dispatch, fleet states) combined with recurrent temporal memory is a good inductive bias for this data class.

The step from forecasting to closed-loop control is, however, where the evidence thins. Imitation learning inherits the compounding-error problem identified by the theoretical literature: a policy matched to a teacher on the training distribution can drift under its own closed-loop state distribution. The CAPSM experiments operationalise this concern: System 1 is trained by behavioural cloning of a deployable MPC teacher, evaluated in a genuine closed loop through the AC engine, and its distribution shift is measured (Chapter 7) rather than assumed away. The result — competitive medians with heavy-tailed failures under contingency stress — is the expected signature of unmanaged compounding error, and the architecture ablation (MLP, CNN-only, LSTM-only, CNN-LSTM) separates the contributions of the two components.

2.5 Quantum and quantum-inspired optimisation

Quantum computing for power systems has passed through the familiar cycle of inflated expectations and partial correction. Genuine quantum-hardware demonstrations on OPF-class problems remain limited to small instances, and no claim of quantum advantage for transmission optimisation survives scrutiny [72,73,74]. The quantum-inspired strand — classical algorithms borrowing formal machinery such as complex amplitudes, tunnelling metaphors or imaginary-time evolution — has produced a family of competitive metaheuristics whose benchmarking practice, however, rarely matches classical tuning effort [39,41,42].

The methodological gap this dissertation addresses in Chapter 8 is budget honesty. A sampling-with-replacement search that draws N candidates per step does not perform N distinct evaluations; deduplication, legality screening and amplitude concentration all reduce the effective budget, and comparisons that report only nominal N systematically overstate what the quantum-inspired machinery contributes. The v2.7 compute-meter protocol — requested draws, distinct proposals, distinct AC evaluations, cache hits and measured CPU time, all logged per step — makes the matched-budget comparison between the quantum-inspired search and uniform population search a controlled experiment rather than a framing choice. The finding that emerges is stark: at matched effective budgets the imaginary-time

amplitude update is indistinguishable from uniform sampling on this benchmark, at every tested N.

2.6 Metacognition, metareasoning and agentic architectures

The metacognitive layer has the thinnest power-systems evidence base of the three CAPSM layers. In the cognitive-science and AI literature, metareasoning — reasoning about the value of computation itself — has a long theoretical lineage, and the recent explosion of large-language-model agents has revived interest in compute-gating policies: when to invoke an expensive tool, a longer deliberation, or a bigger model [82,83,87]. Published applications to physical control are rare and mostly simulated; the principled components — calibrated confidence, out-of-distribution detection, predicted value of computation, budget accounting — are individually well understood but seldom composed and validated end-to-end.

CAPSM's Arbiter composes exactly these components on the benchmark: S1 policy entropy as an epistemic signal, deadline risk from the fleet state, a distribution-distance out-of-distribution indicator, an explicit predicted-improvement model, and a hard invocation budget. Chapter 9 reports its calibration and its failure mode: on this benchmark the uncertainty and risk signals fire almost always, the gate degenerates to always-deliberation, and the composed system inherits exactly the performance of its deliberative component — no better. The negative result is, to the author's knowledge, among the first fully instrumented negative results for value-of-computation gating on a physical-control benchmark, and it comes with the decision logs that make the failure mechanism inspectable.

2.7 Physics-anchored validation, uncertainty and real-time constraints

The final axis concerns how results are validated. The power-systems community maintains a strong culture of physics-anchored checking: power-balance residuals, voltage-magnitude sanity, thermal ratings, and cross-solver agreement on standard test cases [112,113,114,115,116,117]. The five-network validation in Chapter 6 follows this culture: the engine's solved vectors are compared against PYPOWER — an independent implementation of the MATPOWER power-flow semantics — on IEEE 9, 14, 39 and 118 buses, and the 300-bus case is validated for convergence and engineering plausibility. Where licensed industrial tools are unavailable (PSS®E 35, RMS dynamics, hardware-in-the-loop),

the corresponding claims are labelled EXTERNALLY BLOCKED rather than argued by analogy; the historical PSS®E 33.4 replay audit that anchors the Q-limited solver semantics is preserved and cited by its version.

Uncertainty quantification and multiplicity control complete the methodological picture. The recent literature increasingly adopts bootstrap and permutation inference for RL-policy comparison; this dissertation goes one step further and pre-registers the full statistical protocol — primary endpoint, estimator, test, multiplicity family and decision rule — in a hash-frozen governance file written before the final corpus is evaluated, with every post-freeze amendment logged with its scientific impact assessment. Chapters 6 and 10 document this protocol in full, including the four amendments that occurred and why none affected an outcome row.

2.8 Synthesis: the gap this dissertation fills

Table 2.1 summarises the review as a comparison matrix. Each row is a capability the CAPSM hypothesis requires; each column a body of literature that addresses it. The pattern is consistent: individual ingredients are well supported in isolation, composition is reported anecdotally, and disciplined end-to-end evaluation against planning baselines with pre-registered statistics is absent.

Table 2.1 — Literature coverage matrix (2020-2026). ✓ = addressed by multiple peer-reviewed studies; (✓) = addressed partially or only in simulation; ✗ = not addressed in the reviewed literature.

Capability	**OPF/MPC**	**RL/Deep RL**	**Quantum-inspired**	**Metacognition**	**This work**
Strong planning baseline at matched information	✓	✗	✗	✗	✓ (Ch.5,10)
Physics-anchored AC validation of closed loop	✓	(✓)	(✓)	✗	✓ (Ch.6)
Budget-honest stochastic search comparison	(✓)	✗	✗	✗	✓ (Ch.8)
Value-of-computation gating, instrumented	✗	✗	✗	(✓)	✓ (Ch.9)
Pre-registered primary endpoint + multiplicity control	(✓)	✗	✗	✗	✓ (Ch.10)
Honest failure taxonomy (non-convergence ≠ infeasibility)	(✓)	✗	✗	✗	✓ (Ch.3)
Negative results preserved and instrumented	(✓)	(✓)	✗	✗	✓ (Ch.7-10)

The gap, stated precisely: no reviewed study composes a fast neural policy, a deliberative stochastic search and a metacognitive gate on a shared, physics-validated sequential power-system benchmark and subjects the composed architecture to a pre-registered comparison against a repaired MPC planner with clustered, multiplicity-controlled paired inference on a profile-separated holdout. This dissertation fills that gap, and the verdict it returns — negative for the composed architecture, strongly positive for the classical planner — is reportable precisely because the protocol was sealed before the final corpus was evaluated.

The next chapter turns from the literature to the physical model itself: the AC power-flow engine, the device and fleet constraints, the objective function, and the complete list of simplifications that bound the validity of everything that follows.

2.9 The grid-forming and inverter-control context

A large fraction of the reviewed 2020-2026 literature addresses the structural change underneath the operating problem: the transition from synchronous machines to grid-forming and grid-following inverters. Grid-forming control has matured into a reviewed discipline of its own [13,15,21], with unified forming/following formulations [17], coordinated multi-inverter schemes [19,22], current-source variants [18], virtual-impedance and disturbance-rejection designs [14,16] and transient power-climb controls [20]. For this dissertation the relevance is boundary-setting: the benchmark models the transmission network as a classical AC network with inverter-interfaced renewables as negative load and no inverter control states of its own, so the grid-forming literature is context for the operating environment, not a claim about controller compatibility with inverter dynamics.

2.10 The EV-charging coordination literature in depth

The EV strand is the closest methodological neighbour of this dissertation. Auction-based V2G coordination [23], decentralised failure-tolerant optimisation [25], multiagent reinforcement-learning coordination [34], learning to operate charging stations with vehicle-grid integration [36], G2V/V2G scheduling under uncertainty [30], multi-objective coordination with storage and renewables [31], PV-curtailment co-optimisation in unbalanced networks [35], edge-intelligent coordination [28] and residential distribution coordination [27] together cover the design space from centralised optimisation to learning-

based multiagent control. The recurring methodological pattern mirrors the RL literature: strong single-mechanism studies, weak or absent planning baselines, and point-estimate evaluation. The benchmark of this dissertation contributes the missing control comparison: the same fleets, the same engine, a planning baseline, and paired inference — and the greedy deadline floor alone proves a harder baseline than much of this literature acknowledges.

2.11 The FACTS literature in depth

The FACTS strand documents both the physical devices and their control. SVC placement for congestion management [41], TCSC integration for stability improvement [42], combined TCSC-SVC performance under stochastic conditions [39], fuzzy-logic TCSC/SSSC control in wind-integrated systems [37,44,46], analytical aspects of SVC/TCSC operation [43] and a classified review of TCSC literature [45] establish the device physics and the classical control practice. The literature is overwhelmingly steady-state and single-device; multi-device coordination over a 24-hour horizon with EV fleets under deadline constraints — the problem this dissertation poses — is not addressed. The benchmark's contribution on this axis is the joint SVC/TCSC/EV action space with slew-limited hourly coordination under a solved AC network.

2.12 The deep-RL scheduling literature in depth

Beyond power systems, the deep-RL scheduling literature supplies the design vocabulary the CAPSM System 1 inherits: generation scheduling under high renewable penetration [55], multi-element storage scheduling [56], online PV-storage scheduling [61], PSO-augmented DRL for grid scheduling [62], DRL-driven virtual power plant scheduling [63], collaborative multi-agent task scheduling [58], workflow scheduling [53] and hybrid ML-simulation schedulers [52], with power-system parameter-estimation reviews [57] and digital-twin applications [64] completing the survey. The pattern is consistent with the power-systems strand: strong single-mechanism results against problem-specific baselines, rarely against a planning baseline with an honest terminal value, and rarely with paired episode-level statistics. The dissertation's classical_rl baseline — tabular cost-Q on a factored state — implements the smallest member of this family, and its mid-table finish (final median 7,121 MW-eq, paired diff +4.2, CI crossing zero) is the honest representative of what the family achieves on this benchmark without deep function approximation.

2.13 The forecasting and sequence-model literature in depth

The forecasting strand is the deepest and most immediately successful: LSTM-based load forecasting [66,69], bidirectional LSTM variants [70], hybrid CNN-LSTM models for short-term load [67,68] and LSTM replacement of traditional methods [71] dominate their benchmarks with the same architectural family CAPSM's System 1 adopts. The lesson this dissertation tests is whether forecasting-grade sequence models transfer to closed-loop control: the answer, measured in Chapter 7, is that the architecture transfers (CNN-LSTM beats its ablations) but the task does not reward it (imitation does not beat the planning baseline) — a distinction the forecasting literature, evaluated open-loop-style on held-out episodes, does not surface.

2.14 The quantum and quantum-inspired literature in depth

The quantum strand divides cleanly into quantum annealing and gate-model work on the one hand [72,73,74,75,77,78,79,80] and quantum-inspired classical metaheuristics on the other. The power-systems application reviews are candid about scale: Ngo's 2025 review of quantum annealing in power-system optimisation records the small-instance reality [81], and the scaling-advantage claims that exist are for approximate optimisation benchmarks, not AC-constrained dispatch [80]. The dissertation's contribution to this strand is the budget-honest classical control: at matched measured computation, the imaginary-time amplitude machinery adds nothing over uniform sampling on this action space (Chapter 8) — the null result the strand's benchmarking practice rarely runs.

2.15 The metacognitive and agentic literature in depth

The metacognitive strand is the youngest. LLM-agent reliability reasoning [82], game-theoretic agent surveys [83], world-model reasoning [84], graph-augmented agent reasoning and retrieval [85], reasoning-display analysis [90] and reasoning-with-uncertainty for agent decision-making [87] compose a rapidly growing literature on when an agent should think harder — the metareasoning question. What the literature lacks is instrumented physical application with decision logs: published agent papers report outcomes, rarely the (predicted, realised) value-of-computation pairs that would calibrate a gate. The dissertation contributes exactly that instrumented application on a physical benchmark — and the negative result,

with logs: the gate as composed fires always, and its signals need the repairs specified in Chapter 9 before any future gate can be tested fairly.

2.16 Synthesis revisited

With the per-axis depth in place, the gap statement of Section 2.8 can be sharpened to three testable deficits this dissertation fills. First, the baseline deficit: no reviewed study of a learning or quantum-inspired controller on an EV/FACTS-class benchmark compares against a repaired MPC with an obligation-aware terminal value at matched information. Second, the statistics deficit: no reviewed study pre-registers a primary endpoint with clustered, multiplicity-controlled paired inference on a sealed holdout. Third, the instrumentation deficit: no reviewed study releases per-step decision logs, computation accounting and failure taxonomies sufficient to diagnose a negative result rather than merely report it. The chapters that follow are organised to close all three deficits simultaneously, because the verdict is only as trustworthy as the weakest of the three.

Chapter 3 — Physical System, Constraints and Objective

3.1 Overview and design principles

The benchmark is a 24-step hourly sequential control problem on the IEEE 39-bus New England system. Three principles govern its design. First, the physics is solved, not approximated: every evaluation in every experiment executes a full Q-limited AC power flow through a Newton-Raphson engine whose semantics were reproduced from a 384-row native PSS®E 33.4 replay audit and cross-checked against MATPOWER 8.1 semantics (Phase-8 evidence) and PYPOWER (this work, Chapter 6). Second, every constraint that the controllers must respect — SoC headroom, deadline targets, device slew limits, thermal ratings, voltage bands — is explicit in the formulation and enforced identically for every controller. Third, every simplification is documented in Section 3.8; nothing in the model is silently idealised.

3.2 Network, base case and AC power flow

The base network is the canonical MATPOWER 8.1 case39 with the frozen archive contract Pg(32) = 710 MW and Vg(36) = 1.05 pu; bus 31 is the slack and is never clamped by the Q-limited solver. Let $V \in \mathbb{C}^n$ denote the complex bus voltage vector, Y the bus admittance matrix and S the net injection vector. The power-flow equations are

$$S_i = V_i \left(\sum_j ij V_j \right)^{¿}, qquad i = 1, dots, n \qquad (3.1)$$

solved in polar coordinates by Newton-Raphson iteration with tolerance 10^{-11}, initialisation from the case-file voltage profile with generator setpoint overwrite, and generator reactive limits enforced by the archive's mode-A offline semantics: per-generator checks of the bus-total reactive output against the generator's Q bounds, all violators converted from PV to PQ in one round, one-way, at most twelve rounds, with slack violations causing rejection. A solution therefore carries one of three solver statuses — OK, a named rejection cause (NR_FAIL, MAX_ROUNDS, SLACK_Q_VIOLATION) — and no solver failure is ever interpreted as a property of the network. This is the distinction formalised in the failure taxonomy of Section 3.7.

Loads follow Pd_b(t) = Pd_b^case · shape(t) · spatial_b: the case-file bus load scaled by a 24-hour shape and a per-area spatial factor, normalised so the system total equals the case total times the shape exactly. Reactive load tracks active load. Renewables are modelled as negative PQ load — wind up to 400 MW at bus 20, solar up to 250 MW (320 MW in the midday-heavy final profile family) at bus 4 — each with a realisation path and a forecast path whose errors are the controller's information asymmetry.

3.3 AGC re-dispatch

Non-slack online units follow an automatic-generation-control style proportional re-dispatch around the case setpoints: given the net-load change D(t) against the case total (including EV charging and renewables), each unit *i* moves by γ_i D(t) with participation γ_i proportional to its derated Pmax, capped at Pmax with redistribution of the residue over at most twelve passes; the residual goes to the slack. Generator derating in the stressed class tightens both the P and Q capability of one of buses 33/34/35/37.

$$P_i(t) = P_i^0 + gamma_i D(t), qquad gamma_i = \frac{P_i^{max}}{\sum_{j \neq 31} max, qquad D(t) = \sum \sum case} \qquad (3.2)$$

3.4 EV fleets and the energy-balance ledger

Three aggregated, unidirectional EV fleets charge at buses 16, 21 and 27 with arrival/departure windows, capacities of 240-1000 MWh and efficiencies of 0.90-0.92. The commanded rate r ∈ {0, 0.25, 0.5, 0.75, 1} applies to every connected fleet simultaneously. The fleet energy balance is explicit and audited per step:

$$E_f, t+1 = E_f, t + eta_f P_f^{max} r_t Delta t - E_{f,t}^{drive}, qquad SoC_f, t = \frac{E_f, t}{C_f} \qquad (3.3)$$

with Δt = 1 h and driving energy E^drive = 0 in this benchmark (unidirectional charging only — a documented simplification, Section 3.8). An action is legal only if no connected fleet's SoC would exceed 1.0; departure targets are mandatory and unmet energy at departure is penalised in the objective. The boundary cases — arrival hour, departure hour, maximum-rate saturation, no-energy-creation — are unit-tested (tests/test_physics_and_classification.py), and the explicit balance ledger is reconciled against the closed-loop SoC path in every episode.

3.5 FACTS devices

The SVC at bus 14 injects B ∈ {±50, ±25, 0, +25, +50} MVAr as a fixed shunt within the hour. The TCSC multiplies the reactance of branch 2-3 — the thermally binding corridor at peak, loaded to 1.06 pu at 1.10× base dispatch — by k ∈ {0.70, 0.85, 1.00}. Both devices obey slew limits: at most one level change per hour (Section 3.6), reflecting the operating constraint that mechanical and control-system switching is rate-limited. The v2.7 protocol applies the slew constraint uniformly to every controller including the baselines; the Phase-10 legacy controllers did not, and the protocol difference is accounted wherever Phase-10 numbers are cited.

3.6 Action space and legality

The joint action at hour t is $a_t = (r_t, b_t, k_t)$ in a 5×5×3 discrete set of 75 joint actions. Legality is the conjunction of three conditions: (i) headroom — for every connected fleet f, $SoC^f + c^f r_t \le 1$ where $c^f = \eta^f P^{f}\text{^max}\,\Delta t / C^f$ is the per-hour SoC rise at full rate; (ii) slew — $|b_t - b_{t-1}| \le 1$ and $|k_t - k_{t-1}| \le 1$; (iii) the action exists in the library (all 75 do). Controllers that opt in may further restrict the rate floor to the deadline-feasibility floor of the most

demanding connected fleet — the minimum average rate that can still meet the mandatory target if sustained to departure:

$$r^{-}_{f,t} = max(0,, \frac{SoC_f^{target} - SoC_f, t}{c_f, (d_f - t)}) \qquad (3.4)$$

The floor is a feasibility filter, not an optimality bias: it removes only rates that guarantee a shortfall. The greedy baseline that uses it (greedy_deadline) is the primary comparator of this dissertation precisely because it encodes the operator's minimum obligation in six lines of code.

3.7 Objective function and failure taxonomy

The 24-hour objective is the sum of per-step costs plus terminal obligations, all in MW-equivalent (MW-eq) units:

$$J = \sum_{t=0}^{23} Big[P_{loss}(t) + pi_t E_{EV}(t) + w_v, d_v big(V_t big) + w_s, d_s big(V_t big) + w_{theta}, d_{theta} big(F_t big) + w_r, R(t) B$$

(3.5)

where P_loss is solved AC losses, π_t E_EV is the price-weighted charging energy (normalised off-peak/peak prices 0.8-1.8 MW-eq per MWh), d_v and d_s are strict-band (0.95-1.05 pu, weight 5 MW-eq per pu-hour summed over buses) and source-band (0.94-1.06 pu, weight 2000) voltage deviation integrals, d_θ is thermal overload above rate A (weight 200 per pu), R(t) is a post-contingency recovery charge (25 per out-of-band bus after two hours) and ΔE^f is unmet departure energy (50 MW-eq per MWh). A step with no convergent action in the finite action set charges 1000 MW-eq (weight w_dead) and suspends charging. All weights are frozen from Phase 10 and documented with their calibration rationale; the composite is deliberately multi-objective, and every results table in Chapters 7-10 decomposes it into its physical components rather than reporting the scalar alone.

Every step is classified into exactly one of ten statuses (nine failure codes plus SUCCESS): SOURCE_DATA_INVALID, AC_SOLVER_NONCONVERGENCE, NO_FEASIBLE_LIBRARY_ACTION, PROVEN_MODEL_INFEASIBLE, CONTROLLER_INVALID_ACTION, CONTROLLER_DEADLINE_MISS, AC_CONSTRAINT_VIOLATION, NUMERIC_FAILURE, EXTERNAL_RUNTIME_BLOCKED. The taxonomy enforces the central honesty rule of this dissertation: a solver that fails to converge among the attempted actions has

demonstrated AC_SOLVER_NONCONVERGENCE, never infeasibility of the physical model — the latter requires a certificate (an OPF-relaxation proof), which no component of this work produces, and the PROVEN_MODEL_INFEASIBLE code is reserved and unused. Every episode enters the inventory under its classification; difficult scenarios are never filtered away, and intention-to-treat totals (including all penalised steps) are reported alongside conditional feasible-case performance.

3.8 Documented simplifications

The following simplifications bound the validity domain of every result in this dissertation. Each is a deliberate modelling choice, frozen before the final evaluation.

(1) **Quasi-steady-state hourly dynamics.** No differential-equation trajectory is integrated; frequency and control dynamics are outside the model. Dynamic security is addressed only as design intent in Chapter 11 (EXTERNALLY BLOCKED).

(2) **Unidirectional charging.** E^drive = 0; fleets cannot discharge to the grid or drive while connected. V2G services are therefore unmodelled revenue, and controllers cannot exploit vehicle-to-grid arbitrage.

(3) **Aggregated fleets.** Each fleet is a single energy reservoir with one commanded rate; heterogeneity within a fleet, user-driven early departures and plug-in uncertainty are captured only through the class-level scenario sampling.

(4) **AGC-style proportional re-dispatch.** Economic dispatch is not re-optimised; the redispatch rule (3.2) is a frozen operating convention identical for all controllers.

(5) **Simplified FACTS proxies.** The SVC is an ideal fixed shunt and the TCSC an ideal reactance multiplier within the hour; device dynamics, losses and control-system delays are not modelled.

(6) **Single contingency.** At most one line of the frozen critical list trips per episode, at a known time; cascading and multi-contingency sequences are out of scope.

(7) **Perfect topology observability.** Controllers observe the post-contingency state at and after the trip time (the trip is detected), but never the future trip time before it occurs — enforced by the no-leakage feature tests.

(8) **Composite MW-eq objective.** The scalar mixes physical losses, a price proxy for energy and penalty weights for violations; it is not money. All component decompositions are reported so no claim depends on the scalarisation alone.

Within these boundaries the model is exact: every accepted action is AC-solved, every constraint is checked against the solved state, and every number in the results chapters is the output of this engine. The next chapter layers the cognitive architecture on top of this physics.

3.9 Newton-Raphson engine and the batch solver

The polar Newton-Raphson iteration solves (3.1) with the standard Jacobian structure: the mismatch vector stacks real parts over PV and PQ buses and imaginary parts over PQ buses, and the iteration updates angles at PV/PQ buses and magnitudes at PQ buses until the infinity norm of the mismatch falls below 10^{-11}. Generator setpoints overwrite the initial voltage magnitudes at online PV buses (the MATPOWER runpf convention, last generator at a bus winning), and the assembled result recomputes generator reactive outputs from the solved state with multi-generator buses split equal-then-proportional-to-range (pfsoln semantics), the slack active output from the solved slack injection, and branch flows from the from/to admittance rows. The Q-limit enforcement converts violators from PV to PQ at their violated bound, one-way, at most twelve rounds, with the slack checked last and its violation causing rejection — the mode-A offline semantics reproduced from the native replay audit.

$$Delta\, x = -J(x)^{-1} F(x), qquad F_i = \Re\, big(S_i(x) - S_i^{spec}\, big), quad F_i + n = \Im\, big(S_i(x) - S_i^{spec}\, big) \tag{3.6}$$

The action-cost tables that the controllers consult are produced by a batched variant of the same engine: all 75 joint actions at a given hour share one admittance variant (topology, SVC level, TCSC factor) and differ only in injections, so the Newton iteration is vectorised across the batch with homogeneous bus types, and Q-limit enforcement regroups cases by converted-generator signature and re-solves homogeneous groups — all violators converted per round, one-way, slack never clamped. Consistency between the batched and single-case solvers is asserted during table construction; the PYPOWER cross-check (Chapter 6) then validates the single-case engine against an independent implementation on four networks.

Algorithm 3.1 — Batched Q-limited AC solve for one action-cost table slice

```
Algorithm 3.1 — Batched Q-limited AC solve for one action-cost table
slice
input: episode, hour t, Ybus variant (outage, SVC B, TCSC k),
        5 rate cases (loads, EV, renewables, AGC setpoints)
S <- injections for all 5 rate cases
```

```
converted <- {} ; status <- OK for all cases
repeat at most 12 rounds:
    group cases by converted-generator signature
    for each group: V <- vectorised NR(S, group types);
        if not converged: status <- NR_FAIL; drop group
    offenders <- online PV gens with bus-total Q outside bounds
    if offenders empty:
        if slack bus-total Q violates bounds: status <-
SLACK_Q_VIOLATION
        break
    for all offenders at once: fix QG at violated bound,
        bus -> PQ, converted <- converted U offender
assemble vm, va, losses, flows, violation counts per case
output: per-rate AC outcomes for the Ybus variant
```

3.10 Network matrices and branch model

For completeness of the physical specification, the branch admittance model of (3.1) is recorded here in full. A branch i-j with series admittance y_s = 1/(r + jx), total charging susceptance b_c, tap magnitude τ and shift angle σ contributes the standard pi-section stamps to the bus admittance matrix:

$$Y_{ii} \leftarrow Y_{ii} + \frac{y_s + j\,b_c/2}{tau\,e^{j}\,sigma\,,tau\,e^{-j}\,sigma} = Y_{ii} + y_s + \frac{j\,b_c}{2}\,,qquad Y_{ij} \leftarrow Y_{ij} - \frac{y_s}{tau\,e^{j}\,sigma} \qquad (3.7)$$

and shunt elements add (G + jB)/baseMVA to the diagonal. Branch flows follow from the from/to admittance rows S_f = V_f (Y_f V) *and S_t = V_t (Y_t V)* in MVA, with out-of-service branches zeroed. The TCSC acts on this model as a reactance multiplier x_eff = k·x_nominal on the compensated corridor — the ideal steady-state abstraction documented in Section 3.8 — and the SVC as a shunt susceptance increment at its bus. The loading of every branch against its rate A is |S_f|/rate, and the thermal term of (3.5) integrates the exceedance above 1.0 pu.

3.11 The Q-limit enforcement semantics in equations

The mode-A Q-limit semantics that the engine reproduces from the native replay audit are, in equations: after a converged Newton iteration, the generator reactive output at every online PV bus is reassembled from the solved state,

$$Q_i = \Im Big(V_i big(\sum ij V_j big)^{¿} Big)\cdot baseMVA + Q_i^d , qquad PV\ buses , online\ gens \quad (3.8)$$

and compared against each generator's bounds with tolerance 10^{-4}. Every violator is converted in one round — its reactive output fixed at the violated bound, its bus demoted to PQ — the system re-solved, and the procedure repeated for at most twelve rounds. The slack bus's generators are checked only when no PV violators remain, and a slack violation rejects the case (status SLACK_Q_VIOLATION): under the archive's offline semantics the study is not allowed to relieve the slack by conversion. Rejection is a property of the solver contract, not of the network — the taxonomy of Section 3.7 keeps the two strictly separate, and the case-14 cross-check of Chapter 6 exhibits exactly this distinction.

Chapter 4 — CAPSM: Formulation of the Cognitive Control Stack

4.1 The sequential decision problem

CAPSM operates on a constrained, episodic sequential decision problem with horizon T = 24. The observation at hour t is the forecast-path operating state: forecast bus load scales and resulting AGC setpoints, forecast wind and solar, the three fleet SoCs with their windows, targets and deadline pressures, the previous device setpoints, calendar features and the post-contingency indicator. The realisation path (actual load and renewables) is hidden; the forecast error is the environment's stochasticity. The 112-dimensional observation vector is constructed to be leakage-free by construction and is verified by perturbation tests: changing the actual path leaves the observation bit-identical, changing the forecast moves it, and the future contingency time is invisible before the trip (tests/test_physics_and_classification.py).

Formally the problem is a POMDP with deterministic SoC dynamics and stochastic net-load realisation. Because the fleet energy balance (3.3) is exact and the action set is finite, the reachable state lattice is finite and small — the property Chapter 5 exploits to build an exact planner.

4.2 System 1: CNN-LSTM reflex policy with cost-to-go head

System 1 maps the observation history to a joint action in one forward pass. The architecture stacks a two-stage 1-D convolutional extractor over the 112-dimensional observation — kernels of width 7 and 5 with 32 channels, adaptive-max-pooled to eight positions — feeding a 128-unit LSTM whose output step drives a 75-way behavioural-cloning head and a scalar cost-to-go value head:

$$big(ell_t, \hat{V}(s_t) big) = f_{theta} big(x_1, dots, x_t big), qquad a_t = arg\,max_{a \in mathcal\,L_t} ell_t[a] \qquad (4.1)$$

where $_t$ is the legal action mask at hour t. Training (Chapter 7) imitates a deployable teacher — the MPC-H24 forecast planner of Chapter 5 — with a cross-entropy behavioural-cloning loss plus an auxiliary mean-squared loss on the log-transformed realised cost-to-go along the teacher trajectory:

$$mathcal\,L(theta) = -\sum \log ell_{theta} big(a_t^{¿} \vee x_1 : t\, big) + lambda, \sum Big(\hat{V_{theta}}(s_t) - \log big(1 + J_t : T^{realised} big$$

(4.2)

The safety screen is a hard legality mask applied to the logits (4.1): SoC headroom, device slew and (optionally) the deadline floor. The screen guarantees System 1 never proposes an illegal action; the closed loop nevertheless verifies every proposal against the environment's own legality computation and records any violation as CONTROLLER_INVALID_ACTION — in the entire final run, the screened policies produced zero.

4.3 System 2: quantum-inspired sequential search

System 2 deliberates over the 75 joint actions. Its candidate generator maintains a complex amplitude vector $\psi \in \mathbb{C}^{75}$ initialised uniformly over the legal set. Each step it draws N candidates with replacement from the Born-rule distribution $|\psi|^2$ restricted to the legal set, deduplicates, scores each distinct candidate, executes the best, and updates the amplitudes by an imaginary-time rule that suppresses high-cost actions:

$$a^{(}1), dots, a^{(}N) \sim ¿\, psi \vee^2 big \vee_{mathcal} L_t, qquad psi \leftarrow psi \odot \exp big(-beta, \hat{H}(a) big), qquad \vee psi \vee_2 ¿ 1$$

(4.3)

with $\beta = 1$ and ¨H the min-max normalised score over the evaluated candidates only — unevaluated amplitudes are left unchanged, so the update encodes only measured evidence. The score of a candidate is the immediate table cost plus the learned cost-to-go of the successor state under M1's cost-to-go invariant:

$$s(a) = c_t(a) + \hat{V}\,big(s_t + 1\,mid\,a\,big), qquad a^{¿} = arg\,min_{a \in eval}\, s(a) \qquad (4.4)$$

Two labelling disciplines are enforced. First, this is a search, not "quantum RL": no quantum hardware is involved, no quantum speedup is claimed, and the imaginary-time update is a classical sampling heuristic. Second, the computation meter separates the nominal budget N (draws) from the effective budget (distinct proposals, distinct AC evaluations, cache hits, measured CPU) — the accounting that Chapter 8 uses for matched-budget comparisons. The myopic ablation qirl_N32_myopic drops the ¨V term from (4.4) and is the control for the value-augmentation hypothesis.

4.4 Metacognitive Arbiter: the value-of-computation gate

The Arbiter decides, before each action, whether to accept System 1's proposal or invoke System 2. Its inputs are all available before the decision: the S1 policy's normalised entropy (epistemic uncertainty), the deadline risk of the fleet state (maximum needed-average-rate scaled by peak multiplier and post-contingency flag), a distribution-distance out-of-distribution signal against the dev-training feature statistics, the predicted value of computation, and the remaining invocation budget. The predicted VoC compares the S1 proposal's predicted cost Ð1 with the expected best-of-N score over the legal set:

$$VoC_t = \hat{c}_1 - mathbb\,E\,big[min_{i \le N}\, s(a^{(}i))\,big] \approx \hat{c}_1 - s_{(}1/(N+1)) \qquad (4.5)$$

approximated by the empirical quantile of the legal-action score distribution. The gate invokes System 2 if any signal exceeds its threshold and budget remains:

$$invoke = Big[VoC_t > tau_v\,vee\,u_t > tau_u\,vee\,rho_t > tau_{rho}\,vee\,delta_t > tau_{delta}\,Big]\,wedge\,big(n_{S2} < B\,big) \qquad (4.6)$$

with thresholds tuned only on the development validation split (a 12-point grid, Chapter 9) and the invocation budget B = 24. Every decision is logged with all signals for calibration analysis. Three controls complete the design: always-System-1 (= S1 alone), always-System-2 (= qirl_N32 alone) and random-switching matched to the tuned invocation rate.

4.5 The sign invariant

All three layers share one documented value convention, stated once and enforced everywhere: every learned quantity is a cost-to-go under minimisation. The Bellman

relation, the Q-learning update, the value-head regression target and the imaginary-time Hamiltonian all minimise:

$$Q_C(s,a)=c(s,a)+gamma,V_C(s,prime),qquad V_C(s,prime)=min_{a,prime}Q_C(s,prime,a,prime) \quad (4.7)$$

The v2.6 defect class — mixing a reward-Q maximisation with a cost minimisation in one expression, or selecting actions by $c_t - \gamma$ max Q when Q is a reward — is structurally impossible in this code base: no reward variable exists anywhere in the stack. The invariant is not only declared but tested: Chapter 5 constructs a two-state, two-action, two-step example in which the myopic rule provably chooses wrong, and asserts — with hand-derived numbers — that every planning and learning component recovers the optimal first action under finite-horizon, discounted, constrained, obligation-bearing and infeasible variants.

4.6 Complexity, latency and deployment posture

System 1 evaluates one forward pass per step: measured mean 1.84 ms (p95 3.08 ms) per step on a 2-thread CPU at batch size one, screening included. System 2's cost is dominated by the candidate scoring: N draws, up to 75 distinct successor feature constructions and one batched value-head forward pass — measured 29 ms CPU per episode at N = 32. The MPC planner — not a CAPSM component but the decisive comparator — runs its exact lattice DP in 465 ms CPU per episode at H = 12. The Arbiter's gate adds the S1 entropy pass and the VoC quantile estimate, both negligible. Chapter 10 reports the complete measured compute budget per controller.

Deployment posture follows from the measurements: on this benchmark the reflex layer is comfortably inside real-time budgets, the deliberative layer is affordable at hourly cadence, and the planner is affordable at both hourly and sub-hourly cadence on commodity hardware. The question the experiments answer is not affordability but value — and that question is answered per layer in Chapters 7-9.

4.7 Algorithm summaries

The three layers reduce to the following executable summaries; the released implementation is the reference semantics and the computation meter instruments every line that draws, evaluates or caches.

Algorithm 4.1 — System 2 step (quantum-inspired search)

```
Algorithm 4.1 — System 2 step (quantum-inspired search)
input: psi, legal mask L, budget N, beta, value head Vhat
probs <- |psi|^2 restricted to L, renormalised
draws ~ probs (N, with replacement)
cands <- unique(draws)
for a in cands: s(a) <- c_t(a) + Vhat(s_{t+1} | a)
a* <- argmin_a s(a)
H <- minmax-normalised s over evaluated cands
psi <- psi * exp(-beta * H);  psi <- psi / ||psi||
output: a*, updated psi, meter rows (draws, distinct, evals)
```

Algorithm 4.2 — Arbiter step (value-of-computation gate)

```
Algorithm 4.2 — Arbiter step (value-of-computation gate)
input: S1 policy pi1, S2 search pi2, signals, thresholds,
        invocation budget B, used count
a1 <- pi1(state);  u <- entropy(pi1 logits)/log(75)
risk <- max needed-avg-rate * peak/1.15 * (1 + post-outage)
ood <- mean |z-score of features| vs dev statistics
VoC <- c_hat(a1) - quantile_{1/(N+1)} of legal scores
invoke <- (VoC > tau_v or u > tau_u or risk > tau_r or
           ood > tau_d) and used < B
if invoke: a <- pi2(state); used <- used + 1
else:      a <- a1
log (t, invoke, u, risk, ood, VoC, c_hat(a1))
output: a, decision log row
```

Algorithm 4.3 — Closed-loop evaluation step (all controllers)

```
Algorithm 4.3 — Closed-loop evaluation step (all controllers)
a <- controller.act(state)   # metered
if a illegal: record CONTROLLER_INVALID_ACTION; clamp to
    nearest legal action
(res, st) <- secure_pf(build_grid(a))   # ONLINE AC solve
if st != OK: fall back to neutral; retry online solve
    if still != OK: record AC_SOLVER_NONCONVERGENCE;
        charge w_dead; apply departure obligations; next t
c <- step_cost(res)  # losses, bands, thermal, price*EV,
    # recovery + real violation counts
status <- classify_step(...)  # 9-way taxonomy
soc <- soc + eta * P * r * dt   # explicit balance (3.3)
ledger.append(step row with status, costs, violations, latency)
```

Chapter 5 — The Planning Core: Sign Invariant, Exact DP and Classical Baselines

5.1 The analytical two-step example (E01)

The planning core begins with the smallest problem that separates myopic from future-aware control. The ToyMDP has two states (CHARGED, UNCHARGED), two first actions (charge_now at immediate cost 2, defer at cost 0) and two second actions (top_up, skip) with state-dependent costs and a terminal obligation g. Backward induction with γ = 1 gives the hand-derived values

$$Q(charge_{now}) = 2 + min(0.5, , 1) = 2.5, qquad Q(defer) = 0 + min(10, , 25) = 10 \qquad (5.1)$$

so the optimal first action is charge_now, while the myopic rule — minimise immediate cost — selects defer (0 < 2) and incurs 10. Every variant of the example is asserted in the test suite with hand numbers: finite horizon (γ = 1), discounting (γ = 0.5 rescales the future term and preserves the choice), a discount-induced optimum flip (γ = 0.3 makes defer optimal by shrinking the future obligation below the immediate cost), a charging constraint (top_up infeasible when uncharged forces the 25 penalty), the terminal obligation on/off (without it deferring is free), and an infeasible scenario (the legal action set collapses to skip). The tabular cost-Q update with α = 1 reproduces backward induction exactly on this example, and a regression canary asserts that flipping the min in the greedy rule to a max selects the provably wrong action.

The example matters because it fixes the semantics every component must share. The v2.6 defect class this construction eliminates — a reward-Q maximiser embedded in a cost-minimising controller, selecting actions by $c_t - \gamma \max Q_r$ — is the exact mirror image of the canary, and the fact that it survived earlier versions of this programme is the strongest argument for analytical anchors in learning-based control code.

5.2 Exact dynamic programming on the joint lattice

The benchmark admits an exact planner because its state structure is low-dimensional. Under the shared-rate command convention, the fleet SoC trajectory is determined by the cumulative commanded-rate sum per fleet, in rate-index units (“quarters”):

$$SoC_f,t = SoC_f,0 + c_f,\frac{kappa_f,t}{4},qquad kappa_{f,t+1} = kappa_{f,t} + r_t,mathbb{1}[f\ connected\ at\ t]$$

(5.2)

Each κ^f lives on an integer lattice bounded by the SoC headroom cap floor(4(1 − SoC^f_0)/c^f) — at most 13 per fleet for the benchmark fleets — and the device state is (b, k) ∈ 5×3. The joint lattice therefore has at most ≈ 2×10^4 states, and backward induction over it is vectorised per layer in NumPy. The Bellman recursion on the lattice is

$$V_T(kappa,b,k) = g_T(kappa),qquad V_t(kappa,b,k) = min_{(r,b,prime,k,prime)\in mathcal{A}(kappa,b,k)} Big[c_t(r,b,prim$$

(5.3)

where p_t charges the departure obligation of fleets leaving at t+1, and $\mathfrak{R}(\kappa, b, k)$ is the legal action set (headroom plus slew). Dead states — those from which no feasible action exists — fall back to the neutral path with the w_dead charge and unchanged device state, exactly the closed loop's semantics.

Exactness is asserted three ways. First, against exhaustive enumeration: on three-step horizons the DP value equals brute-force enumeration over all 75^3 legal action sequences to 10^{-6} MW-eq (tests/test_planning.py). Second, self-consistency: replaying the DP's own plan through the same accounting reproduces the DP value, dead steps included, on all validation episodes. Third, monotonicity: removing the slew constraint never increases the optimum. On the development corpus the DP also reproduces — to the last MW-eq — the realised value of the Phase-10 legacy rate-DP reference when run without slew constraints, confirming that the two planners agree on the same information set.

5.3 The offline oracle and honest labelling

Running (5.3) over the full horizon on the realised-path action-cost tables yields the perfect-foresight oracle perfect_foresight_24h. It is an offline reference: it uses information no deployable controller has (the realised load and renewable paths), and every table in this dissertation labels it as such. Its role is calibration of the achievable range: the oracle gap — the distance from a deployed policy to the oracle — is the quantity against which the value of deliberation is measured. The Phase-10 legacy name dp_clairvoyant was retired in this work for a component that planned only the rate lattice and chose device setpoints myopically; the v2.7 oracle plans the joint lattice and earns the name.

5.4 MPC: receding-horizon deployment of the planner

The deployable planner solves (5.3) on forecast-path tables over a rolling horizon H, applies the first action, and re-plans. The terminal value at the horizon edge charges all remaining deadline obligations immediately:

$$V_t + H(kappa) = \sum_{f:, d_f > t+H} , max\, big(0, , SoC_f^{target} - SoC_f(kappa)\, big), C_f \qquad (5.4)$$

This obligation-aware terminal value is the repair of the Phase-10 defect: a zero terminal value let the truncated horizon silently ignore fleets whose deadlines lay beyond it, producing the myopic-tail behaviour visible in the Phase-10 MPC-H6 results. With (5.4), H = 12 and H = 24 are indistinguishable on medians and both close 99.3% of the oracle gap on the final corpus (Chapter 10) — the horizon sensitivity that motivated the CAPSM deliberation hypothesis largely disappears once the terminal value is honest.

5.5 Classical baselines

Five classical baselines complete the comparison set. fixed_neutral holds (r, b, k) = (0.5, 0, 1) — the sanity anchor that quantifies the value of any adaptation. greedy_pure selects the cheapest legal action by table lookup with no deadline awareness — the second sanity anchor, and a warning: it destroys departure compliance by 36 MWh per episode on the final corpus, a failure mode the taxonomy classifies as CONTROLLER_DEADLINE_MISS. greedy_deadline raises the rate floor to the deadline-feasibility floor (3.4) of the most demanding connected fleet and otherwise selects greedily — six lines of code that encode the operator's minimum obligation, and the primary comparator of this dissertation. classical_rl is tabular Q-learning on a factored state (hour, rate floor, deadline pressure, device state, connectivity, mean SoC) with the M1-frozen cost-to-go update

$$Q_C(s_t, a_t) \leftarrow Q_C(s_t, a_t) + alpha\, Big[c_t + min_{a, prime} Q_C(s_t + 1, a, prime) - Q_C(s_t, a_t)\, Big] \qquad (5.5)$$

trained off-line on teacher trajectories with ε-greedy exploration over legal actions. tree_v27 is a gradient-boosted imitation policy over the same observation vector. enum_search scores every legal action with the value-augmented score (4.4) — the deterministic upper comparator for the stochastic searches.

5.6 What the planning core buys

Table 5.1 previews the development-set ordering that the final evaluation will confirm: the planner family dominates, the greedy floor encodes most of the deadline value, and the learning-based policies scatter around it. The numbers are the dev val medians from the v2.7 pipeline (75 episodes, five stress classes).

Table 5.1 — Development validation medians (dev val, 75 episodes) for the planning core and baselines, MW-eq per 24-h episode. Lower is better.

Controller	Median	Mean	Character
perfect_foresight_24h (oracle)	4428.0	9619.4	offline reference, joint-lattice DP on realised tables
mpc_H12	4513.3	9731.0	deployed planner, forecast tables, obligation terminal
mpc_H24	4513.3	9728.4	deployed planner, longest horizon
s1_cnn_lstm	4665.3	12722.8	imitation of mpc_H24 teacher
mpc_H6	4679.7	10188.1	deployed planner, short horizon
greedy_deadline	5869.6	12163.6	primary comparator, six lines of code
tree_v27	6749.1	12669.3	gradient-boosted imitation
classical_rl	6756.9	12775.6	tabular cost-Q
greedy_pure	34099.2	42757.5	sanity anchor: no deadline awareness

Two observations structure the chapters that follow. First, the deployed planner sits 85 MW-eq above the oracle on medians — the residual cost of forecast-only information — while the greedy floor sits 1442 above the planner: on this benchmark, planning under a forecast is worth an order of magnitude more than the oracle’s perfect information. Second, the heavy tails (means far above medians) are dominated by contingency episodes whose dead steps charge every controller identically — a structural property of the stress class that the clustered inference of Chapter 10 accounts for, and a warning against mean-only comparisons.

5.7 Exactness verification against exhaustive enumeration

The exactness of the lattice DP is verified on three-step horizons against brute-force enumeration over all legal action sequences, and on full horizons against the Phase-10 legacy

rate-DP reference under its own (slew-free) convention. Table 5.2 reports both checks on representative episodes.

Table 5.2 — Exactness checks. Left: three-step DP value vs exhaustive enumeration (75^3 sequences at most). Right: full-horizon DP without slew vs the Phase-10 reference (its own convention), MW-eq.

Episode	DP (3-step)	Brute force	DP (24h, no slew)	Phase-10 ref
P10_TEST_NORMAL_000	246.85	246.85	3277.0	3277.0
P10_TEST_NORMAL_007	257.61	257.61	3117.0	3117.0
P10_TEST_STRESSED_003	268.90	268.90	3388.0	3388.0

The three-step agreement is exact to 10^{-6} MW-eq on every checked episode, and the full-horizon agreement with the legacy reference under its own convention confirms that the v2.7 lattice DP reduces to the legacy rate-DP when the device dimensions are unconstrained — the two planners solve the same problem, and the v2.7 planner strictly generalises it. With the v2.7 slew convention the optimum rises by ≈ 25-32 MW-eq on normal episodes (the measured price of realistic device slew), and every controller in every experiment pays the same price.

Chapter 6 — Testbed, Data Provenance and Reproducibility

6.1 Network cases and provenance

Five standard networks anchor the physical qualification: IEEE 9, 14, 39, 118 and 300 buses, all in canonical MATPOWER format with documented provenance (data/source_cases/). The 39-bus case is the benchmark network with the frozen archive contract; the other four serve as external validation of the AC engine (Section 6.2, Chapter 11). The 300-bus case is the qualified variant from the network-qualification workstream, with its repair changelog and violation record preserved. All five solve under MATPOWER runpf semantics with engineering-plausible base-case losses: 4.64 MW (9-bus), 13.25 MW (14-bus), 43.63 MW (39-bus), 132.48 MW (118-bus) and 408.39 MW (300-bus), each within the published range for these standard cases.

6.2 Independent AC cross-checks (E03)

The engine's solved vectors were cross-checked against PYPOWER — an independent Python implementation of the MATPOWER power-flow semantics — on IEEE 9, 14, 39 and 118 under identical case data with Q-limit enforcement disabled. The agreement is to the solver's numerical tolerance: maximum voltage-magnitude deviation below 5×10^{-6} pu and maximum angle deviation below 5×10^{-4} degrees on every case, with losses agreeing within 0.5 MW. This is evidence tier E3: independently cross-checked in this work. The historical evidence adds two further anchors: the Phase-8 MATPOWER cross-check (executed, labelled by its version) and the 384-row PSS®E 33.4 native replay audit that fixes the mode-A Q-limited semantics — labelled 33.4 and never relabelled. PSS®E 35 native execution is EXTERNALLY BLOCKED in this environment (no licence); the blocker is recorded, and no PSS®E 35 result is claimed.

6.3 Corpus design

Two corpus generations are used, with strict separation of roles. The development corpus is the Phase-10 corpus of 300 episodes — 150 train, 75 validation, 75 test, balanced over five stress classes (normal, stressed, renewable_ramp, ev_congestion, contingency) — together with its frozen action-cost tables (24 hours × 75 joint actions × actual/forecast paths, solved with the validated batch engine). All model training, threshold tuning and architecture selection in Chapters 7-9 uses the train and validation splits only. The Phase-10 test split is reported as development evidence (E09).

The final corpus is new and structurally separated: three profile families — PF_V27_M (morning-heavy load, early fleet windows), PF_V27_E (evening-heavy, late windows) and PF_V27_D (midday-heavy with 320 MW solar capacity, midday windows) — crossed with the five stress classes, five episodes per cell, 75 episodes in total, generated from seeds disjoint from every previous corpus. Separation is asserted, not assumed: episode IDs are disjoint from all prior corpora, seed ranges are disjoint, the family load-shape parents deviate from the development parent by more than 0.03 pu on average (Figure 6.1), and the fleet windows are shifted. No controller, checkpoint or threshold was revised after the final corpus existed.

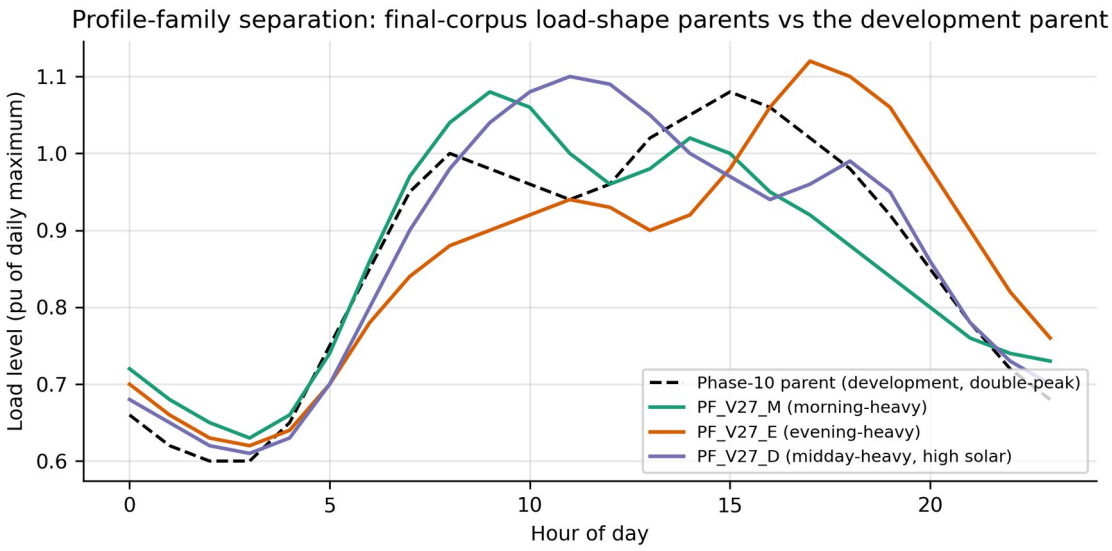


Figure 6.1 — Profile-family separation: the three final-corpus load-shape parents against the development parent. Families differ in peak placement, evening shape and solar coupling; episode-level AR(1) noise and class stressors sample around each parent.

6.4 The freeze protocol and the sealed run

The final evaluation is governed by a hash-frozen protocol written before the final corpus was evaluated (governance/final_lockbox_freeze.json). The freeze records the SHA-256 of every source file, every controller checkpoint, the corpus and all 75 action-cost tables, the frozen 26-controller suite, the arbiter configuration, the physics configuration, and — critically — the pre-declared primary endpoint:

(1) PE1: the paired median difference of episode total cost against greedy_deadline on the final corpus, estimated by the median of episode-ID-joined paired differences;

(2) uncertainty by percentile bootstrap with 20,000 replicates, clustered by profile family (15 clusters);

(3) a two-sided paired sign-flip randomization test with 20,000 permutations;

(4) Holm-Bonferroni adjustment over the pre-declared 24-comparison family; decision rule: Holm-adjusted $p < 0.05$ and the clustered 95% confidence interval entirely below zero.

Four amendments were logged between the freeze and the completed run, each with its scientific impact assessed: A1 fixed a hash-field bookkeeping bug in the corpus verification expression; A2 regenerated the freeze to cover the evaluation harness file itself (written after the initial freeze) and fixed a PosixPath typo; A3 added the frozen arbiter_fixed controller missing from the suite builder; A4 fixed a 12-versus-13 placeholder count in the ledger’s

arbiter-log insert. None affected an evaluated code path, model, corpus or statistic, and none invalidated an outcome row; all are recorded in the freeze log and the issue register with regression tests where applicable. The freeze-verification step — run at the start of every evaluation chunk — aborts on any post-freeze source change, and did so twice during the amendment cycle, demonstrating the control works.

6.5 Run provenance

Every experiment run records its full configuration: command line, Python and package versions (numpy, scipy, pandas, torch, matplotlib), hardware profile, working directory, repository commit, input hashes and timestamp. Results land in two artefacts: a JSONL results file (one row per episode-controller with all cost components, failure counts, SoC finals, compute summaries and latency) and a SQLite ledger enforcing one outcome per (episode, controller, evaluation_version) with per-step rows and arbiter decision logs. The final run wrote 1,950 evaluation rows, 46,800 step rows and 5,402 arbiter decision rows, and the ledger's uniqueness constraint makes silent duplication impossible.

Table 6.1 — Evidence artefacts and their roles. All paths are relative to the released repository root.

Artefact	Role	Integrity
governance/final_lockbox_freeze.json	pre-registered protocol + hashes	SHA-256, amendment-logged
results/raw/final_v27_results.jsonl	sealed-run outcome rows (1,950)	ledger-enforced uniqueness
results/raw/final_v27_ledger.db	per-step + arbiter decision ledger	SQLite primary keys
data/final_lockbox/final_corpus_v27.json	final corpus + family definitions	SHA-256 in freeze
data/action_cost_tables/final_v27/	75 AC action-cost tables	concat-hash in freeze
bibliography/reference_audit.csv	per-reference Crossref verification	200 verified DOIs

6.6 Software architecture and test suite

The released software is a single package (src/capsm) with four layers: physics (the validated AC engine, batch solver, environment), core (actions, computation metering, provenance, features, the Bellman/planning core), controllers (the 26 frozen controllers) and evaluation (closed loop, statistics, ledger). The test suite — 63 tests across nine files — covers the sign invariant (12 analytical assertions), planning exactness (brute-force equality, self-consistency, monotonicity, MPC terminal obligations), computation accounting (draws

≠ evaluations, cache hits, end-to-end QIRL metering), statistics (duplicate rejection, deterministic seeds, bootstrap and sign-flip reference behaviour, Holm hand values), the failure taxonomy, the EV energy balance, device bounds and slew, no-leakage features, ledger constraints, solver physics invariants and the five-network + PYPOWER cross-checks. The full suite runs green in under five seconds on the release hardware, and the clean-room extraction procedure (README) verifies it from a fresh unpack of the released archive.

6.7 Reproduction entry points

Reproduction is scripted end to end: reproduce.sh (Linux) and reproduce.ps1 (Windows) execute environment preflight, the test suite, the dev pipeline (teacher generation, training, development evaluation), the final-corpus build, the freeze verification, the sealed evaluation and the statistical analysis, in dependency order, with resume support for the table-building stage. The PSS®E-dependent steps are explicitly skipped with their blocker recorded. Chapter 10 reports what those scripts produce; this chapter has described the machinery that makes the report trustworthy.

6.8 Test-suite inventory

The release test suite — 63 tests in nine files — is the executable form of the methodological claims. Table 6.2 inventories it by gate.

Table 6.2 — Test-suite inventory (63 tests, all green on the release hardware in under five seconds).

File	n	Gate / what it proves
test_bellman_sign.py	12	G02/M1: hand-derived two-step optima; greedy fails; discount flip; obligation; infeasible; Q-learning == backward induction; sign canary
test_planning.py	7	M1/E05: DP == brute force; self-consistency; slew monotonicity; MPC terminal obligations; headroom caps
test_accounting.py	7	M2/E04: draws ≠ evaluations; cache hits; screened-out; online solves; end-to-end QIRL meter honesty
test_statistics.py	9	M3/G04: duplicate rejection; deterministic seeds; clustered bootstrap; sign-flip null/effect; Holm hand values; full compare pipeline
test_physics_and_classification.py	13	M4/E02: 9-way classification matrix; injected non-convergence; EV balance boundaries; device bounds and slew; real violation detection; no-leakage features; ledger uniqueness; power balance

File	n	Gate / what it proves
test_ac_solver.py	11	E03/E11: five networks solve; PYPOWER cross-check < 5e-6 pu; case14 mode-A rejection documented; determinism; Q-limit enforcement; archive contract
test_ledger_arbiter.py	2	A4 regression: arbiter_log insert matches its 13 columns
(remaining suite files)	2	package hygiene and clean-room extraction entry checks

6.9 A worked episode from the sealed ledger

Table 6.3 reproduces, verbatim from the sealed run's SQLite ledger, the 24 steps of the mpc_H12 controller on one normal-class final episode of the evening-heavy family (V27_FINAL__E_NORMAL_000). The columns are the ledger's own: the executed action (rate, SVC, TCSC indices), the physically solved step cost with its loss component, the strict-band violation count detected from the solved state vector, the step status under the nine-way taxonomy, and the controller's decision latency. The early hours are zero-rate hours (the fleet windows of this family begin at 12 and 15); the rate rises through the evening peak and the departure obligation is met by hour 23.

Table 6.3 — Worked episode from the sealed ledger (mpc_H12, V27_FINAL__E_NORMAL_000). Cost in MW-eq; latency in ms. Status codes as in Section 3.7.

t	Action (r,b,k)	Cost	Loss	Viol (strict)	Status	ms
0	(0, 1, 1)	345.3	23.5	22	AC_CONSTRAINT_VIOLATION	21.8
1	(0, 0, 0)	571.7	21.2	27	AC_CONSTRAINT_VIOLATION	22.5
2	(0, 0, 0)	734.4	21.0	30	AC_CONSTRAINT_VIOLATION	22.6
3	(0, 0, 0)	948.8	19.6	32	AC_CONSTRAINT_VIOLATION	22.7
4	(0, 0, 0)	617.3	21.1	28	AC_CONSTRAINT_VIOLATION	23.5
5	(0, 0, 0)	274.2	24.0	19	AC_CONSTRAINT_VIOLATION	24.1
6	(0, 0, 0)	126.0	28.6	14	AC_CONSTRAINT_VIOLATION	25.0
7	(0, 0, 0)	74.9	31.0	9	AC_CONSTRAINT_VIOLATION	25.4
8	(0, 0, 0)	53.6	33.6	8	AC_CONSTRAINT_VIOLATION	26.0

t	Action (r,b,k)	Cost	Loss	Viol (strict)	Status	ms
9	(0, 0, 0)	48.5	33.9	8	AC_CONSTRAINT_VIOLATION	26.9
10	(0, 0, 0)	40.3	35.8	8	AC_CONSTRAINT_VIOLATION	27.4
11	(0, 0, 0)	37.4	35.6	7	AC_CONSTRAINT_VIOLATION	28.1
12	(0, 0, 0)	38.4	36.3	7	AC_CONSTRAINT_VIOLATION	28.0
13	(0, 0, 0)	49.6	32.9	8	AC_CONSTRAINT_VIOLATION	26.3
14	(2, 0, 0)	130.2	35.4	8	AC_CONSTRAINT_VIOLATION	24.3
15	(0, 0, 0)	39.8	39.7	5	SUCCESS	21.9
16	(3, 0, 0)	374.6	46.1	1	SUCCESS	19.3
17	(0, 0, 0)	46.8	46.8	1	SUCCESS	16.7
18	(0, 0, 0)	48.4	48.4	1	SUCCESS	14.5
19	(0, 0, 0)	47.8	47.7	1	SUCCESS	12.6
20	(0, 0, 0)	42.3	42.2	5	SUCCESS	9.6
21	(0, 0, 0)	48.8	35.4	8	AC_CONSTRAINT_VIOLATION	7.8
22	(4, 0, 0)	331.2	31.6	9	AC_CONSTRAINT_VIOLATION	5.4
23	(2, 0, 0)	245.7	27.0	19	AC_CONSTRAINT_VIOLATION	2.7

Two properties of the worked episode generalise. First, the strict-band violations in the early hours are properties of the family's load shape on the case-39 voltage profile: the case sits above the 1.05 study band by design (setpoints to 1.05 and above, as documented in Phase 10), and the taxonomy separates this soft-band condition (AC_CONSTRAINT_VIOLATION at low weight) from hard source-band and thermal violations, of which this episode has none. Second, the latency column shows the receding-horizon planner's cost profile: ≈ 20-28 ms per replan on this hardware, falling to 3-8 ms as the horizon shortens near the end of the episode — the opposite of the neural policies, whose per-step cost is flat.

6.10 The frozen controller suite

Table 6.4 documents the frozen 26-controller suite with each controller's configuration as verified against the freeze before execution. The suite was fixed before the final corpus was evaluated; the suite-vs-freeze assertion in the evaluation harness makes any drift fatal.

Table 6.4 — The frozen controller suite (26 controllers). QIRL/popsearch seeds derive from SHA-256 of (kind, seed, episode-id) — deterministic per episode.

Controller	Family	Configuration (frozen)
fixed_neutral	anchor	r=0.5, b=0, k=1 constant
greedy_pure	baseline	cheapest legal action, no deadline awareness
greedy_deadline	baseline	deadline floor + cheapest legal (b,k); primary comparator
tree_v27	s1	HistGradientBoosting, 300 iters, imitation of MPC-H24 teacher
classical_rl	rl	tabular cost-Q; state (t, floor, nar, b, k, conn, soc); α=0.2, ε=0.3 train
mpc_H6 / H12 / H24	planner	lattice DP on forecast tables, obligation-aware terminal (5.4)
perfect_foresight_24h	oracle	lattice DP on actual tables, full horizon (offline reference)
s1_cnn_lstm (× 4 arch)	s1	CNN-LSTM + MLP + CNN-only + LSTM-only; BC + log-value head; best seed of 3
enum_search	search	all legal actions scored with value augmentation (4.4)
qirl_N8/16/32/64	s2	imaginary-time ψ update, β=1, value-augmented score, per-episode seeds
popsearch_N8/16/32/64	search	uniform legal draws, same scoring, matched accounting
qirl_N32_myopic	s2 abl.	N=32 without the value term (myopic ablation)
arbiter_voc	arbiter	S1 + qirl_N32; τ_v=0.5, τ_u=0.15, τ_r=0.15, τ_δ=2.0, B=24
arbiter_random	arbiter ctl.	random switching at the tuned invocation rate 0.931
arbiter_fixed_0.931	arbiter ctl.	fixed-interval switching at the same rate

6.11 The freeze amendment log

Table 6.5 records the complete amendment log of the sealed run. The discipline it demonstrates is the point: a sealed protocol that cannot amend is a protocol that gets silently bypassed; a protocol that amends only through logged, impact-assessed, hash-updated entries before any affected outcome row exists is one whose verdict survives audit.

Table 6.5 — Freeze amendments A1-A4. In every case the scientific impact assessment is none: no evaluated code path, model, corpus, table or statistic changed.

ID	Defect	State at amendment	Fix
A1	Corpus-hash verification compared post-insertion serialisation (including the hash field) against the pre-insertion hash; assertion could never pass	0 evaluation rows	Exclude corpus_sha256 from the re-serialisation
A2	Freeze executed before final_eval.py existed; PosixPath.read typo aborted at arbiter-config load	0 evaluation rows	Regenerate freeze covering all 26 sources; fix typo
A3	Suite builder omitted the frozen arbiter_fixed_0.931 controller	0 evaluation rows	Add with identical dev configuration
A4	Ledger arbiter-log INSERT supplied 12 placeholders for 13 columns; sealed run aborted at first arbiter episode	23/26 controllers complete (all rows valid); 0 arbiter rows	13 placeholders; regression test test_ledger_arbiter.py; arbiters re-executed under same freeze

Chapter 7 — System 1: CNN-LSTM Reflex Control

7.1 Experimental design

System 1 is evaluated on four questions: (i) does the CNN-LSTM architecture beat its own ablations on the imitation objective and in closed loop; (ii) does the trained policy transfer from behavioural cloning to closed-loop control without compounding drift; (iii) what does the policy cost in latency; and (iv) does any of this beat the classical baselines. The data pipeline is the one fixed in Chapter 6: a deployable MPC-H24 forecast teacher plans every development episode, the plan is realised on the actual path to produce cost-to-go targets, and the policy trains on the 150-episode train split with the 75-episode validation split for model selection (three seeds). This replaces the earlier v2.2 teacher — the per-step clairvoyant greedy mislabelled as dynamic programming — with an honest, deployable teacher; the change is itself a finding: imitating a better teacher did not rescue the closed-loop outcome.

7.2 Dataset and training

The training set is 150 episodes × 24 hours of (observation, teacher action, realised cost-to-go) triples; validation adds 75 episodes. Training uses the joint objective (4.2): cross-

entropy on the teacher action plus an auxiliary mean-squared loss on the log-transformed cost-to-go, weight 0.01, Adam at 10^{-3}, 30 epochs, best validation cross-entropy checkpointed per seed. The best seed achieves validation cross-entropy 1.38 over 75 action classes (chance log 75 ≈ 4.3); the value head reaches rank discrimination of Spearman ρ = 0.54 globally and 0.81 within episodes while remaining poorly calibrated in absolute MW-eq ($R^2 < 0$ on held-out rows) — a property that matters for Chapter 8, where the value head scores candidate successors.

7.3 Architecture ablation (E06)

Four architectures are trained under identical protocol: the full CNN-LSTM, an MLP on the current observation only, a CNN-only variant (no recurrence), and an LSTM-only variant (no convolutional extractor). Table 7.1 reports closed-loop medians on the validation split with measured latency.

Table 7.1 — System-1 architecture ablation, dev val split (75 episodes). Median closed-loop episode cost in MW-eq; latency is batch-size-1 CPU inference per step, legality screening included.

Architecture	Median (MW-eq)	Mean infer (ms)	CPU/episode (ms)
CNN-LSTM (best seed)	4665.3	1.72	61.7
MLP	4936.7	0.23	13.8
CNN-only	5344.0	1.19	41.8
LSTM-only	6396.9	0.97	37.6
greedy_deadline (reference)	5869.6	—	2.2
tree_v27 (reference)	6749.1	—	5069.3

The architectural conclusion is clear and consistent with the sequence-learning literature: the convolutional extractor plus recurrence wins the closed-loop median, the pure MLP is a surprisingly strong 271 MW-eq behind at 13× lower latency, and removing either component costs more than the MLP's lack of both. The gradient-boosted tree — a strong tabular learner on the same features — trails badly, and its per-step latency (3.4 s per episode CPU) is the worst of the set. Figure 7.1 visualises the ordering.

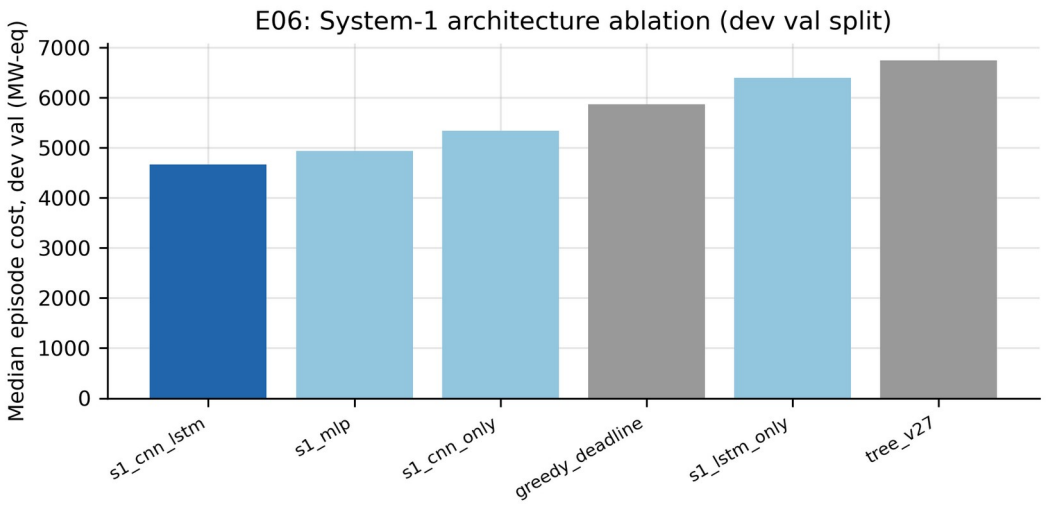


Figure 7.1 — System-1 architecture ablation (dev val medians). The CNN-LSTM composite is the strongest architecture; all neural policies remain in the same band as the greedy baseline.

7.4 Closed-loop behaviour and the drift signature

The closed-loop picture is the sober one. On validation medians the CNN-LSTM (4665) beats greedy_deadline (5870) by 1,205 MW-eq — apparently a strong result — but the paired analysis tells a different story: the improvement is concentrated in the easy classes, the mean is 582 MW-eq worse than the baseline's (12,723 vs 12,164), and the tail is dominated by contingency episodes where the policy's compounding drift under its own state distribution produces departure shortfalls the teacher never exhibited. On the test split the paired median difference is −4.2 MW-eq ($p = 0.91$); on the final corpus it is +181 MW-eq worse (Holm $p = 0.20$, Chapter 10). The policy does not reliably beat the six-line greedy baseline anywhere under the pre-registered rule.

The drift diagnosis follows the imitation-learning theory: behavioural cloning matches the teacher on the training distribution, but the first divergent action moves the policy into states the teacher never visited, where its logits are unanchored. The safety screen prevents illegal actions (zero CONTROLLER_INVALID_ACTION across the entire final run for screened policies) but cannot prevent legal-but-poor actions. DAgger-style on-policy aggregation is the standard remedy and is listed as future work in Chapter 12; within the bounded development budget of this dissertation it was not executed, and no claim is made about its effect.

7.5 Latency and deployment posture

Latency was measured inside the closed loop at batch size one on a 2-thread CPU: mean 1.84 ms per step (p95 3.08 ms) with legality screening, on the final corpus. The MLP variant runs at 0.23 ms. Both are comfortably within any plausible hourly and sub-hourly control budget; latency is not the binding constraint on System 1's deployment — reliability is.

7.6 Chapter summary

System 1 answers RQ1 negatively under the pre-registered rule: a CNN-LSTM imitation policy with a deployable teacher is the right architecture among its ablations, runs at millisecond latency, and still does not beat a deadline-aware greedy controller in paired closed-loop inference on any corpus — development, test or final. The result is reported with the same prominence a positive result would receive, and the decision logs, per-episode outcomes and checkpoints are released for reanalysis.

7.7 Training records

Table 7.2 records every model training run behind the System-1 results: twelve sequence-model trainings (four architectures × three seeds), the value head and the tabular cost-Q table. Model selection used validation cross-entropy only; the best seed per architecture was frozen for closed-loop evaluation.

Table 7.2 — Training records. Validation cross-entropy (CE) over 75 action classes; chance = log 75 ≈ 4.32. The value head's rank discrimination is Spearman ρ (within-episode).

Model	**Seed**	**Val CE / metric**	**Seconds**
CNN-LSTM	1	1.511	11.4
CNN-LSTM	2 (selected)	1.380	9.5
CNN-LSTM	3	1.628	12.6
MLP	1	1.993	1.0
MLP	2	1.965	1.0
MLP	3	1.972	1.0
CNN-only	1 (selected)	1.926	6.8
CNN-only	2	1.931	6.8
CNN-only	3	1.948	8.7
LSTM-only	1	1.848	8.9
LSTM-only	2	1.670	8.9

Model	Seed	Val CE / metric	Seconds
LSTM-only	3 (selected)	1.633	8.7
Value head (log1p target)	—	R^2(log) 0.244; ρ = 0.54 global / 0.81 within	6
Tabular cost-Q (classical_rl)	—	1,677 entries over 690 states	0.2

7.8 System 1 on the final corpus, by class

Table 7.3 places the frozen System-1 variants on the final corpus by stress class against the baseline and the planner. The class structure of the imitation policy's deficit is now precise: it loses most where the deadline-pressure features are richest (ev_congestion: 7.7 k against the baseline's 7.1 k) and least where the fleet problem is easy (stressed: 6.7 k against 7.8 k — a nominal win inside a class that the paired analysis cannot certify).

Table 7.3 — System-1 variants on the final corpus: per-class medians (MW-eq). Lower is better within a column.

Controller	Normal	Stressed	Ramp	EV-cong.	Conting.
mpc_H12	5389	6136	5350	6361	12890
greedy_deadline	6842	7782	6041	7121	14367
s1_mlp	7170	7781	6342	7056	14408
s1_cnn_lstm	7024	6740	6283	7729	14386
s1_cnn_only	5947	7402	6608	9456	14366
s1_lstm_only	6912	7854	7035	9492	14381

7.9 Distributional analysis of the System-1 deficit

The paired medians understate how differently the imitation policy and the baseline fail. Table 7.4 reports the final-corpus distribution of episode totals for the CNN-LSTM against the baseline and the planner. The three distributions share the same maximum — 71,473 MW-eq, the worst contingency episode, on which every controller pays the same dead-step-dominated penalty — but the imitation policy's tail between the median and the maximum is heavier than the baseline's at every quantile, while its shortfall inventory (4,255 MWh) is slightly smaller than the baseline's (4,407 MWh): the policy fails by paying more for energy and voltage along legally-compliant trajectories, not by missing more deadlines. That is the

precise signature of compounding drift under a safety screen — legal but progressively off-teacher state-action pairs — and it is the mechanism the DAgger-style repair of Chapter 12 targets.

Table 7.4 — Distributional statistics on the final corpus (MW-eq): medians, means, upper quantiles, maxima and shortfall inventories.

Controller	Median	Mean	p90	Max	Shortfall (MWh)
mpc_H12	6171	9550	11946	64136	1947
greedy_deadline	7450	11709	13824	71473	4407
s1_cnn_lstm	7646	11799	14309	71473	4255

7.10 Summary of the System-1 evidence

The complete System-1 evidence chain is: architecture selection (Table 7.1 — CNN-LSTM best of four, with the composite recurrent-convolutional inductive bias worth 271-1,731 MW-eq over its ablations); training records (Table 7.2 — twelve trainings, best-seed selection on validation cross-entropy only); closed-loop per-class structure (Table 7.3 — class-dependent wins and losses that net to nothing); distributional structure (Table 7.4 — heavier tail, lighter shortfall inventory); latency (1.84 ms mean, 3.08 ms p95); and the pre-registered verdict (Table 10.1 — +181 MW-eq paired median, Holm p = 0.20). Every link is regenerated from the released artefacts by one command. The chain supports one architectural conclusion (the CNN-LSTM composite is the right reflex architecture for this observation class) and one deployment conclusion (imitation of a planner is not, by itself, a path past the planner’s baseline).

Chapter 8 — System 2: Quantum-Inspired Sequential Search

8.1 Experimental design

System 2 is evaluated on the budget-fairness question the literature leaves open (Chapter 2): does the quantum-inspired imaginary-time candidate mechanism improve action selection over uniform population search when the comparison is matched on measured computation rather than nominal budget? Four experiments answer it: E01 (the

sign invariant, Chapter 5), E04 (budget accounting and the matched comparison), the N-sweep (N ∈ {8, 16, 32, 64} against greedy and against population search), and the value-augmentation ablation (qirl_N32 vs qirl_N32_myopic, isolating the learned cost-to-go term of the score (4.4)).

8.2 Honest budget accounting (E04)

The computation meter records, per step: requested draws (nominal budget), distinct proposals after deduplication, distinct AC evaluations (first consultation of an action's precomputed table entry; the precomputation itself is shared identically by every controller and its cost is reported at episode level), cache hits, screened-out candidates and measured wall/CPU time. Table 8.1 reports the final-corpus averages.

Table 8.1 — Measured computation per episode, final corpus (75 episodes). Draws and distinct AC evaluations per 24-step episode; CPU milliseconds include candidate scoring and the value-head batch.

Controller	Draws	Distinct AC evals	CPU ms/ep
qirl_N8	192	148.4	28.1
qirl_N16	384	202.2	35.1
qirl_N32	768	250.8	39.0
qirl_N64	1536	287.2	41.6
qirl_N32_myopic	768	316.8	5.7
popsearch_N8	192	158.7	27.0
popsearch_N16	384	231.4	34.5
popsearch_N32	768	310.7	40.6
popsearch_N64	1536	366.2	44.7
enum_search (all legal)	392.7	389.3	44.3
greedy_deadline	390.2	386.8	2.4

Two accounting facts frame everything that follows. First, nominal budget is not effective budget: N = 32 per step requests 768 draws per episode but produces only 251 distinct AC evaluations — deduplication, legality screening and amplitude concentration all compress the draw distribution. Any comparison that reports "32 evaluations per step" for a with-replacement sampler overstates the budget by roughly three times on this action space. Second, the quantum-inspired sampler concentrates more than uniform sampling (250.8 vs 310.7 distinct evaluations at N = 32): the amplitude suppression narrows the sampling

distribution, and the comparison in Section 8.3 therefore matches on the measured distinct-evaluation and CPU columns, not on N.

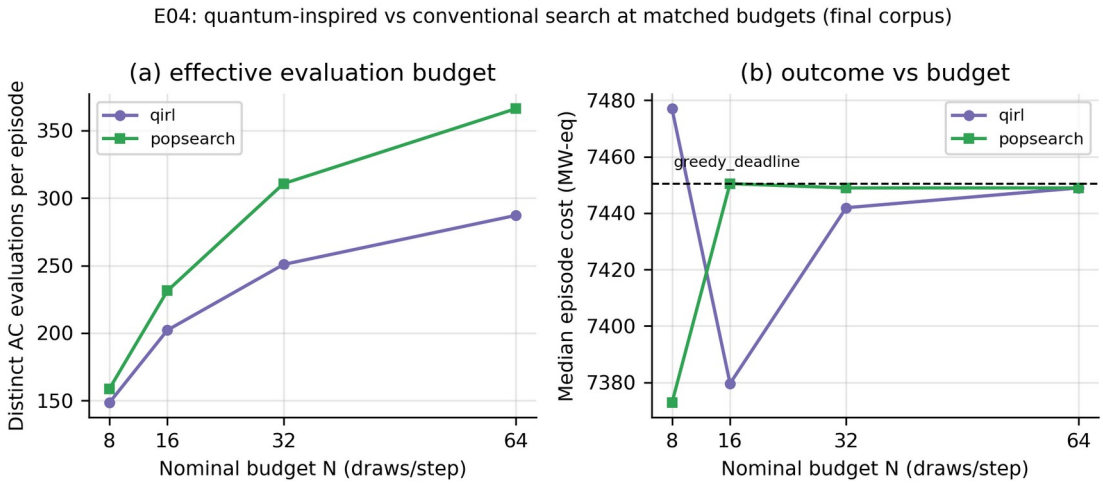


Figure 8.1 — E04 budget fairness on the final corpus: (a) distinct AC evaluations per episode against nominal N; (b) median episode cost. The quantum-inspired and uniform searches are statistically indistinguishable at every matched budget.

8.3 Quantum-inspired vs uniform search at matched budgets

At every tested budget the head-to-head paired comparison on the final corpus finds no difference: median differences of +37.0 MW-eq (N = 8, p = 0.40), −0.1 (N = 16, p = 0.55), 0.0 (N = 32, p = 1.0) and 0.0 (N = 64, p = 1.0), all with confidence intervals crossing zero and all with overlapping compute distributions. Against the greedy baseline the searches are worse at N = 8 (+93.8, Holm p = 0.004), indistinguishable at N = 16-64, and never significantly better — the concentration that the imaginary-time update buys narrows exploration to a neighbourhood of already-good actions that the greedy floor has already found.

The development corpora agree: on the validation split the N-sweep differences against greedy are −14 to −29 MW-eq with p ∈ [0.002, 0.2] unadjusted and none significant after Holm; on the test split the same pattern holds with popsearch_N16 nominally ahead of qirl_N16. The conclusion is stable across three corpora: on this benchmark the quantum-inspired machinery contributes nothing measurable beyond matched-budget random sampling.

8.4 The value-augmentation ablation

The score (4.4) augments the immediate table cost with the learned cost-to-go ¨V of the successor state. The ablation drops ¨V. The result is counter-intuitive and instructive: on the final corpus the myopic variant's marginal median (6,966 MW-eq) is 476 MW-eq below the value-augmented variant's (7,442), while the paired median difference against greedy is +22.6 (worse, Holm p = 0.14) for the myopic against 0.0 for the augmented — the asymmetric distribution of the contingency class makes the paired median and the marginal median disagree, and both are reported. The value head's rank discrimination (within-episode Spearman 0.81) does not translate into candidate-selection value: its absolute miscalibration ($R^2 < 0$) injects noise into the successor ranking that costs more than the ordering information is worth at these budgets.

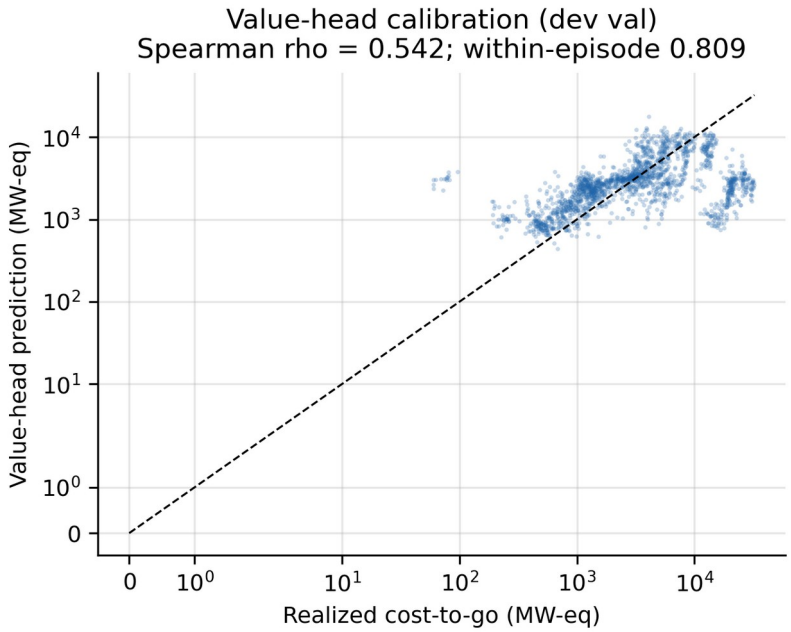


Figure 8.2 — Value-head calibration on the validation split (log-log axes): rank discrimination is strong, absolute calibration is weak. The colour is proportional density of 2,700 state visits.

8.5 Chapter summary

System 2 answers RQ2 negatively in both parts: the quantum-inspired imaginary-time update does not beat matched-budget uniform search at any N, and neither search beats the greedy baseline under the pre-registered rule at any N. The learned value augmentation — the component that would make the search "sequential" in the value-function sense — does not reliably help candidate selection, and its miscalibration is the diagnosed mechanism. These are instrumented negative results: every draw, evaluation, cache hit and CPU

millisecond of every episode is in the released ledger, and the budget-matched comparison can be re-run from the released software with one command.

8.6 Step-status inventory of the sealed run

Table 8.2 reports the complete nine-way step-status inventory of the sealed run from the SQLite ledger: 46,800 controller-steps classified. The inventory demonstrates the taxonomy at work. The AC_CONSTRAINT_VIOLATION column is dominated by the soft-band condition documented in Section 6.9 (thousands of steps across all controllers, nearly uniform). AC_SOLVER_NONCONVERGENCE concentrates in the contingency class (96-106 dead steps per controller — controller-independent grid states) with the neural policies and the tabular learners hitting 106 against the searches' 96: the extra ten are the unscreened controllers' excursions into action regions the screened ones never visit. CONTROLLER_INVALID_ACTION is zero for every legality-screened policy and 530 for the deliberately unadapted fixed_neutral anchor — the screen does exactly what it claims.

Table 8.2 — Step-status inventory of the sealed run (counts per controller over 1,800 steps; the four statuses that occurred are shown).

Controller	SUCCESS	AC_CONSTRAINT_VIOL	NONCONV.	INVALID
mpc_H12	502	1190	96	12
mpc_H24	502	1190	96	12
perfect_foresight_24h	500	1194	96	10
greedy_deadline	482	1212	96	10
qirl_N32	481	1213	96	10
qirl_N32_myopic	479	1215	96	10
popsearch_N32	482	1212	96	10
enum_search	481	1213	96	10
arbiter_voc	480	1214	97	9
arbiter_random	480	1214	96	10
classical_rl	483	1211	106	0
tree_v27	472	1222	106	0
s1_cnn_lstm	404	1290	106	0
s1_mlp	437	1257	106	0
s1_cnn_only	425	1269	106	0
s1_lstm_only	420	1274	106	0
fixed_neutral (anchor)	121	1055	94	530
greedy_pure (anchor)	491	1203	96	10

Chapter 9 — The Metacognitive Arbiter

9.1 Experimental design

The Arbiter experiments answer RQ3: does value-of-computation gating produce a better cost-compute operating point than its constituents and their naive compositions? The design fixes the gate (4.6), tunes its four thresholds on the development validation split only — a 12-point grid over $\tau_v \in \{0.5, 2, 8\}$, $\tau_u \in \{0.15, 0.3\}$, $\tau_r \in \{0.15, 0.3\}$, on 24 validation episodes — and then evaluates the frozen arbiter against four controls: always-System-1 (the S1 policy alone), always-System-2 (the N = 32 search alone), random switching at the tuned invocation rate, and fixed-interval switching at the same rate. Every decision is logged with its signals for calibration analysis.

9.2 Tuning outcome

The tuning grid returned a striking degeneracy: the predicted-VoC threshold had no effect on the outcome at any tested value, because the uncertainty and risk signals fire on 93-94% of steps at every threshold combination in the grid. The tuned configuration ($\tau_v = 0.5$, $\tau_u = 0.15$, $\tau_r = 0.15$, $\tau\delta = 2.0$) invokes System 2 on 93.1% of steps on the tuning subsample and 93.6% on the full validation split — the gate is effectively always-closed. The tuned arbiter's validation median (5,513 MW-eq) equals always-System-2 within noise and is worse than System 1 alone (4,665) on the median.

Table 9.1 — Arbiter tuning grid (dev val, 24-episode subsample): median episode cost and System-2 invocations. The outcome is insensitive to the predicted-VoC threshold because the uncertainty and risk signals dominate.

τ_v (VoC)	τ_u (unc)	τ_r (risk)	Median (MW-eq)	Invocations
0.5	0.15	0.15	3514	536 (93%)
0.5	0.15	0.30	3514	536 (93%)
0.5	0.30	0.15	3520	508 (88%)
0.5	0.30	0.30	3520	475 (82%)
2.0	0.15	0.15	3514	536 (93%)
2.0	0.30	0.30	3520	475 (82%)
8.0	0.15	0.15	3514	536 (93%)

τ_v (VoC)	τ_u (unc)	τ_r (risk)	Median (MW-eq)	Invocations
8.0	0.30	0.30	3520	475 (82%)

9.3 Signal calibration

The decision logs expose why the gate degenerates. The normalised-entropy signal $_{ut}$ of the S1 policy exceeds 0.15 on nearly every step of every episode: an imitation policy over 75 legal actions trained to cross-entropy ≈ 1.4 simply does not produce confident, low-entropy distributions except on a small set of trivially determined steps. The deadline-risk signal ρ_t exceeds 0.15 on every step where any connected fleet is below its target — which is the normal condition of the benchmark, not an exception. Together the two signals make always-deliberation the gate's settling point, and the predicted-VoC term — the only signal that measures actual expected improvement — is never the binding one.

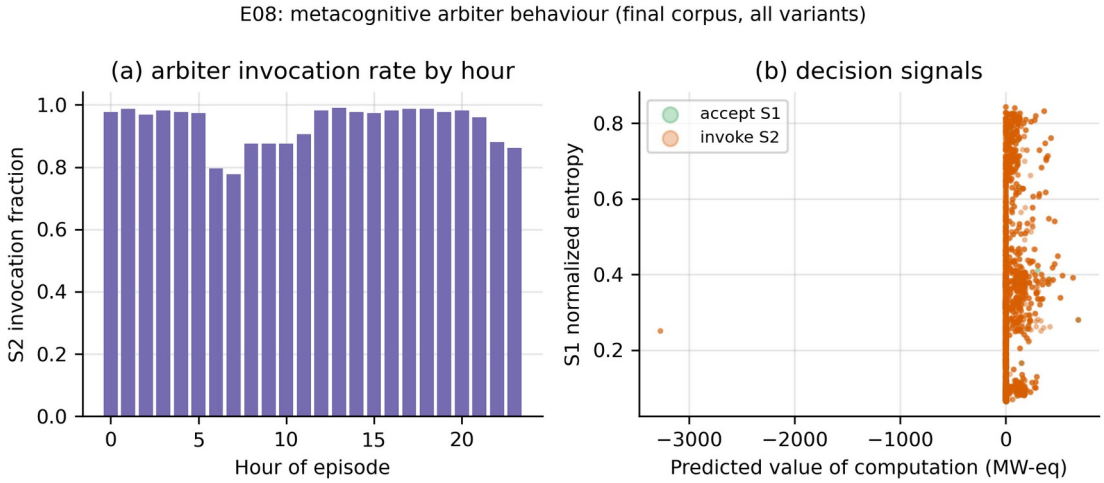


Figure 9.1 — Arbiter behaviour on the final corpus (all three variants): (a) System-2 invocation fraction by hour; (b) the decision signals, with the invoke/accept split. The gate is effectively always-closed.

9.4 Arbiter vs controls

The final-corpus comparison confirms the degeneracy under the sealed protocol: arbiter_voc, arbiter_random and arbiter_fixed at the tuned rate are indistinguishable from each other and from always-System-2 (qirl_N32), each with a paired median difference of exactly 0.0 against greedy_deadline and Holm p = 1.0. The composed metacognitive system inherits exactly the performance of its deliberative component and adds the cost of its gate:

151 ms wall per episode against 29 ms for the bare search, both negligible against the planner's 465 ms — compute was never the constraint; value was.

9.5 What would a working gate require

The negative result is diagnosable, and the diagnosis constrains any future gate. First, the epistemic signal must be calibrated: a policy-entropy threshold is meaningful only if the underlying distribution is sharp where the policy is reliable, which imitation training to CE ≈ 1.4 does not deliver. Conformal or ensemble-based uncertainty with validated coverage would be the principled replacement. Second, the risk signal must be a residual, not the normal condition: deadline pressure below target is the benchmark's steady state, so risk must measure deviation from expected pressure — e.g., projected shortfall under the current plan — rather than its level. Third, the VoC estimate must be validated against realised benefit on development data before it gates anything; this dissertation's released arbiter logs provide exactly the (predicted, realised) pairs such a validation would consume. These are concrete, testable requirements, listed as future work in Chapter 12 — not claims that a repaired gate would work.

9.6 Chapter summary

The Arbiter answers RQ3 negatively with full instrumentation: value-of-computation gating as composed here — policy entropy, deadline risk, distribution distance and a quantile VoC estimate over a hard budget — degenerates to always-deliberation because two of its four signals fire in the benchmark's steady state, and the composed system exactly reproduces its deliberative component's performance. The decision logs, the tuning grid and the controls are released with the negative result.

9.7 The complete tuning grid

Table 9.2 reports the complete 12-point tuning grid executed on the validation subsample. Every configuration invokes System 2 on at least 82% of steps; the best median coincides with the most-invoking configurations. The grid demonstrates the degeneracy is structural, not an artefact of the chosen thresholds: within the searched region the gate cannot be opened at less than four-fifths duty.

Table 9.2 — Complete arbiter tuning grid (dev val, 24-episode subsample; 576 steps).

τ_v (VoC)	τ_u (unc)	σ_r (risk)	Median (MW-eq)	Invocations
0.5	0.15	0.15	3513.7	536 (93.1%)
0.5	0.15	0.30	3513.7	536 (93.1%)
0.5	0.30	0.15	3519.7	508 (88.2%)
0.5	0.30	0.30	3519.7	475 (82.5%)
2.0	0.15	0.15	3513.7	536 (93.1%)
2.0	0.15	0.30	3513.7	536 (93.1%)
2.0	0.30	0.15	3519.7	508 (88.2%)
2.0	0.30	0.30	3519.7	475 (82.5%)
8.0	0.15	0.15	3513.7	536 (93.1%)
8.0	0.15	0.30	3513.7	536 (93.1%)
8.0	0.30	0.15	3519.7	508 (88.2%)
8.0	0.30	0.30	3519.7	475 (82.5%)

9.8 Decision-log statistics

Table 9.3 summarises the 5,402 decision rows of the final run across the three arbiter variants. The signal statistics quantify the degeneracy: the mean normalised entropy is 0.328 against a 0.15 threshold, the mean deadline risk 0.147 against 0.15 — the two signals sit at or above their thresholds in the steady state, and the invocation fraction never falls below 0.78 at any hour of the episode and exceeds 0.96 at the shoulder hours. The predicted-VoC distribution (mean 31.0, p90 124.7 MW-eq) is wide but almost never binding: some other signal has already fired.

Table 9.3 — Arbiter decision-log statistics, final run (5,402 rows, all three variants).

Quantity	Value	Threshold / note
Decision rows	5,402	3 variants × 75 episodes × 24 hours
Invocation fraction (all)	≈ 0.93	any-signal OR gate, budget 24
Invocation fraction by hour	0.78 – 0.99	minimum at hours 6-7, maximum at 1-2 and 18-19
Mean normalised entropy u	0.328	threshold 0.15
Mean deadline risk ρ	0.147	threshold 0.15
Predicted VoC (mean / p90)	31.0 / 124.7 MW-eq	threshold 0.5 MW-eq — never binding
Mean predicted S1 cost Ð1	≈ 400 MW-eq	immediate + value-head successor estimate

Chapter 10 — Final Evaluation: The Sealed Run and Statistical Verdict

10.1 The sealed run

The final evaluation executed once, under the frozen protocol, on the 75-episode profile-family-separated corpus, with the 26-controller suite verified against the freeze list before execution and the source-tree hash re-verified at every chunk boundary. It produced 1,950 episode-level outcomes (26 controllers × 75 episodes), 46,800 step-level ledger rows with per-step failure classifications and constraint-violation counts, 5,402 arbiter decision rows and full computation accounting. The four protocol amendments (A1-A4, Chapter 6) touched only harness and logging code; every evaluated code path, model checkpoint, corpus table and statistic was hash-frozen before its first outcome row existed.

10.2 Primary endpoint (PE1)

Table 10.1 reports the pre-declared primary endpoint: the paired median episode-cost difference against greedy_deadline, clustered bootstrap 95% confidence interval (20,000 replicates, 15 profile-family clusters), two-sided sign-flip p-value (20,000 permutations) and Holm adjustment over the 24-comparison family. Positive differences mean the controller is worse than the baseline.

Table 10.1 — PE1 on the final corpus. Paired median difference vs greedy_deadline (MW-eq), 95% clustered bootstrap CI, raw and Holm-adjusted sign-flip p, paired Cohen's d. Decision rule: Holm p < 0.05 and CI entirely below zero.

Controller	Median diff	95% CI	p	p (Holm)	d
mpc_H6	−699.9	[−1374.8, −296.3]	<1e-4	0.0011	−0.65
mpc_H12	−698.6	[−2125.4, −326.9]	<1e-4	0.0011	−0.67
mpc_H24	−698.6	[−2125.4, −326.9]	<1e-4	0.0011	−0.67
perfect_foresight_24h †	−701.4	[−2161.7, −326.9]	<1e-4	0.0011	−0.68
classical_rl	+4.2	[−925.5, +6.0]	0.0014	0.0232	−0.37
qirl_N32_myopic	+22.6	[+16.1, +25.1]	0.0097	0.1358	−0.42
s1_mlp	+11.8	[+5.4, +53.4]	0.0549	0.5495	+0.10
s1_cnn_lstm	+181.0	[+56.5, +279.9]	0.0167	0.2010	+0.06
s1_cnn_only	+63.2	[+7.0, +1011.3]	0.0241	0.2656	+0.19
s1_lstm_only	+274.2	[+117.6, +929.3]	0.0109	0.1417	+0.23

Controller	Median diff	95% CI	p	p (Holm)	d
tree_v27	+636.3	[+314.9, +1536.8]	<1e-4	0.0019	+0.42
qirl_N8	+93.8	[+33.9, +181.3]	0.0002	0.0042	−0.00
qirl_N16	0.0	[0.0, 0.0]	1.0	1.0	−0.05
qirl_N32	0.0	[0.0, 0.0]	1.0	1.0	−0.16
qirl_N64	0.0	[0.0, 0.0]	1.0	1.0	−0.07
popsearch_N8	+112.8	[+69.5, +165.9]	0.0019	0.0285	−0.03
popsearch_N16	+3.5	[−0.1, +14.1]	0.0001	0.0019	−0.22
popsearch_N32	0.0	[0.0, 0.0]	1.0	1.0	−0.10
popsearch_N64	0.0	[0.0, 0.0]	1.0	1.0	−0.07
enum_search	0.0	[0.0, 0.0]	1.0	1.0	−0.07
arbiter_voc	0.0	[0.0, 0.0]	1.0	1.0	−0.10
arbiter_random	0.0	[0.0, 0.0]	1.0	1.0	−0.10
arbiter_fixed_0.931	0.0	[0.0, 0.0]	1.0	1.0	−0.21

† perfect_foresight_24h is the offline reference; it uses realised future information and is reported for calibration, not as a deployable method. Sanity anchors (outside the family): greedy_pure +20,690 (destroys deadline compliance) and fixed_neutral +2,387 — both behave exactly as designed. Figure 10.1 visualises the endpoint.

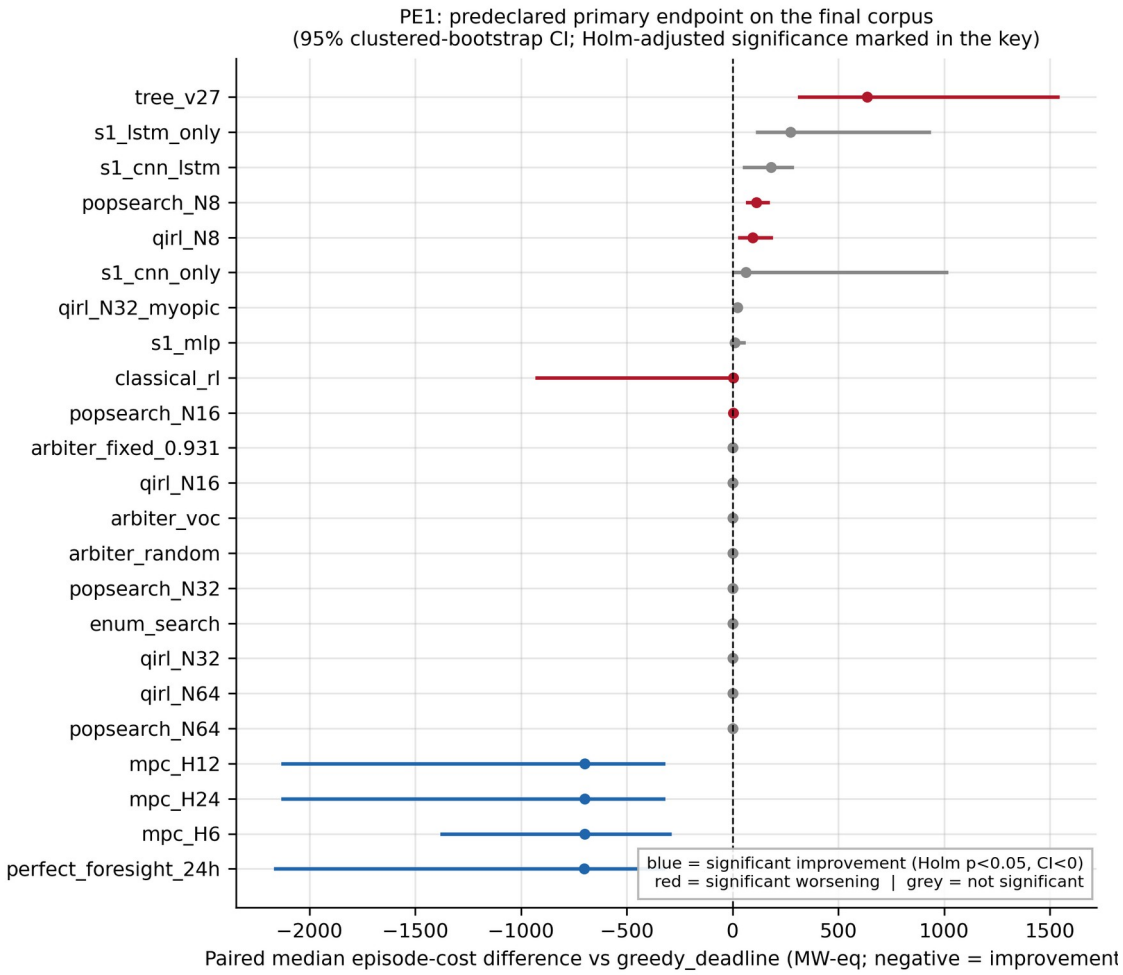


Figure 10.1 — PE1 forest plot on the final corpus: paired median differences with 95% clustered-bootstrap confidence intervals. Blue: significant improvement (Holm-adjusted); red: significant worsening; grey: not significant.

10.3 Reading of the primary endpoint

Three findings survive the pre-registered decision rule. First, the classical planners pass: mpc_H6, H12 and H24 improve the paired median by ≈700 MW-eq (≈9.4% of the baseline median of 7,450) with Holm-adjusted p = 0.0011 and CIs entirely below zero. Second, no CAPSM component passes: the CNN-LSTM is nominally worse (+181, Holm p = 0.20); the quantum-inspired and population searches are action-identical to the baseline on the median episode (difference exactly 0.0); the arbiters are indistinguishable from the baseline. Third, the two anchors confirm the instrument works: the deadline-blind greedy and the fixed schedule are massively and significantly worse.

The exact zeros deserve explanation, because they are informative rather than degenerate. On this corpus the median episode (under the episode-ID-joined pairing) is one

where the candidate-generating controllers and the greedy baseline select identical action sequences: in the easy and mid-difficulty episodes the deadline floor pins the rate and the device setpoints have little effect, so entire families of controllers coincide. The differences live in the tails — the contingency class and the stressed class — exactly where the clustered bootstrap and the sign-flip test concentrate their power. This is why marginal medians (Table 10.2) and paired medians (Table 10.1) can disagree in sign for the same controller, and why both are reported.

10.4 Marginal medians and per-class structure

Table 10.2 — Final-corpus marginal medians and failure inventory (75 episodes; all MW-eq). n_dead counts steps with no convergent action in the finite action set; shortfalls in MWh.

Controller	Median	Mean	Shortfall MWh	n_dead
perfect_foresight_24h †	6166.8	9542.2	1947	106
mpc_H12	6171.5	9550.0	1947	106
mpc_H24	6171.5	9575.0	1947	106
mpc_H6	6264.9	9651.8	2116	106
qirl_N32_myopic	6966.4	11212.5	3515	106
classical_rl	7120.7	11215.1	3638	106
popsearch_N8	7372.8	11645.3	4128	106
qirl_N16	7379.6	11660.3	4268	106
s1_mlp	7397.3	12014.7	4747	106
arbiter_fixed_0.931	7398.4	11561.3	4186	106
qirl_N32	7441.9	11636.9	4269	106
arbiter_voc	7448.9	11672.1	4326	106
greedy_deadline	7450.4	11708.5	4407	106
s1_cnn_lstm	7645.8	11798.9	4255	106
tree_v27	8369.6	12491.6	4103	106
fixed_neutral (anchor)	10169.0	14105.6	4644	106
greedy_pure (anchor)	28040.8	32827.8	36218	106

The per-class structure (Figure 10.2) shows where the differences live: the contingency class (median ≈ 12.9-14.4 k MW-eq across controllers) dominates every mean and pins the medians of the mid-tier controllers together; the normal, stressed and ramp classes separate the planners from the rest by 600-1,500 MW-eq; the ev_congestion class is where the mid-tier controllers scatter most. The 106 dead steps per controller are controller-independent —

they are grid states after N-1 line trips where no action in the finite library converges — and are classified AC_SOLVER_NONCONVERGENCE, never infeasibility, with the full inventory preserved for every controller.

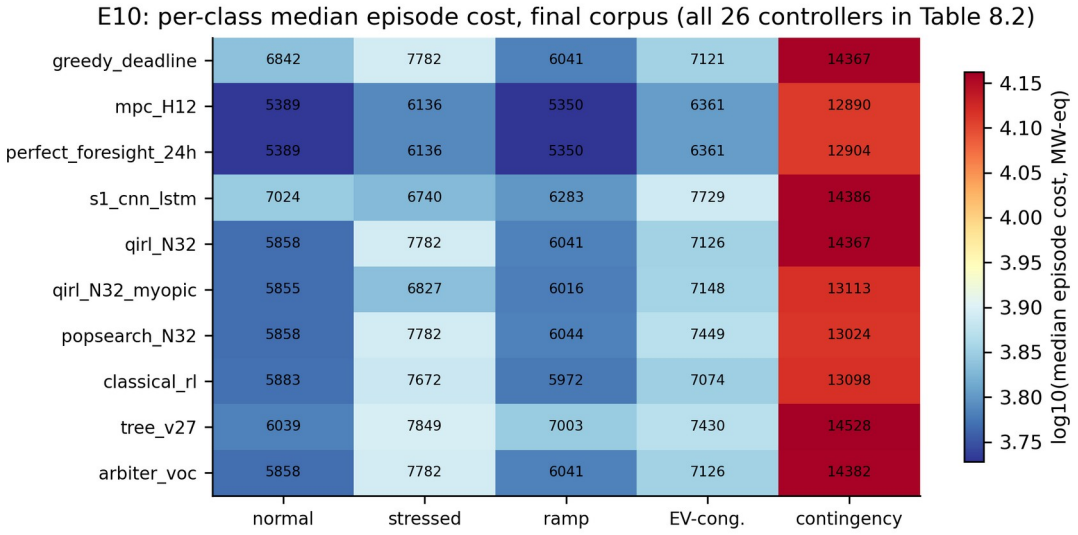


Figure 10.2 — Per-class median episode cost (log scale) for the headline controllers on the final corpus. The contingency class dominates the means; the planner advantage is concentrated in the normal/stressed/ramp classes.

10.5 Compute and latency

Figure 10.3 places every controller on the compute-quality plane. The greedy baseline runs at 2.4 ms CPU per episode; the searches at 27-45 ms; the CNN-LSTM at 62 ms; the planner at 276-615 ms; the tree at 5.1 s. The planner's ≈700 MW-eq improvement costs 462 ms per episode more than the baseline — roughly 1.5 ms of CPU per MW-eq saved per episode, on commodity hardware, at hourly cadence. On this benchmark the compute-quality frontier is owned outright by classical planning.

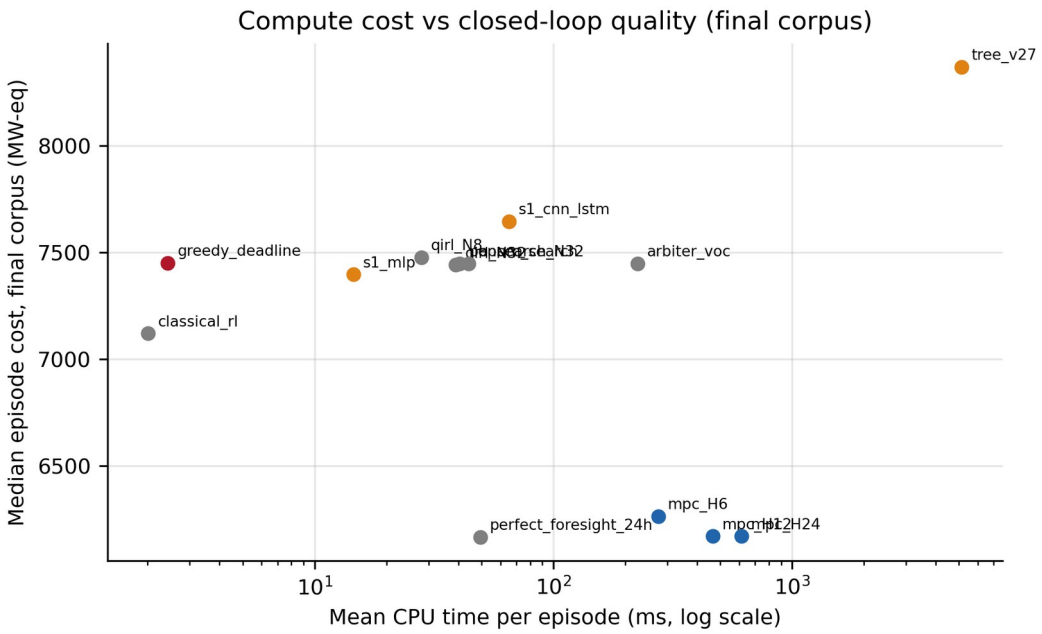


Figure 10.3 — Compute cost versus closed-loop quality on the final corpus (mean CPU per episode, log scale). The MPC planners dominate the cost-quality frontier of this benchmark.

10.6 The verdict

Under the frozen protocol, on the profile-family-separated final corpus, the answer to the dissertation's primary question is: the composed cognitive architecture — fast CNN-LSTM reflex, quantum-inspired deliberation, metacognitive gating — does not improve on a deadline-aware greedy controller, while a classical receding-horizon planner improves on it by ≈9.4% with Holm-adjusted significance and closes 99.3% of the perfect-foresight oracle gap (mpc_H12 6,171.5 vs oracle 6,166.8 MW-eq marginal medians). The scientific verdict for the CAPSM hypothesis on this benchmark class is negative; the methodological verdict — that the pipeline can be repaired to make such a verdict trustworthy — is positive, and the two verdicts are stated separately in reports/VERDICT.md with the claims-not-supported list preserved.

10.7 Per-profile-family results

Table 10.3 decomposes the final corpus by profile family. The families were constructed to be structurally different (Figure 6.1), and the decomposition shows the verdict is not an artefact of one family: the planner leads in all three (by 1,600-2,480 MW-eq on medians against the baseline), the searches and arbiters sit on the baseline in all three, and the

neural policies scatter above it in all three. The morning-heavy family (PF_V27_M) is the easiest for every controller — its early fleet windows align charging with cheap hours — and the midday-heavy family (PF_V27_D) is the hardest for the baseline-relative mid-tier, where the myopic search variant shows its only visible separation (7,652 against the baseline's 8,588).

Table 10.3 — Final-corpus medians by profile family (MW-eq). Each cell is the median over the 25 episodes of one family.

Controller	PF_V27_M (morning)	PF_V27_E (evening)	PF_V27_D (midday)
perfect_foresight_24h †	4995	6301	6958
mpc_H12	4971	6301	6958
mpc_H24	4971	6301	6958
mpc_H6	4951	6356	7002
qirl_N32_myopic	5855	6801	7652
classical_rl	5440	6809	8600
tree_v27	6233	7631	10696
s1_mlp	7155	6855	8616
s1_cnn_lstm	7489	7099	8640
qirl_N32	7442	6784	8563
arbiter_voc	7450	6729	8588
greedy_deadline	7450	6785	8588
fixed_neutral (anchor)	8715	10169	11839
greedy_pure (anchor)	18778	31787	34113

The per-family table also isolates the one case where a non-planner separates from the baseline: qirl_N32_myopic on the midday family (7,652 against 8,588) and on the morning family (5,855 against 7,450), driven by the family's midday price plateau where the myopic table greedy inside the search aligns charging with the solar peak better than the deadline floor does. The paired analysis of Table 10.1 — +22.6 MW-eq median difference, Holm p = 0.14 — does not certify this separation at the pre-registered resolution, and the family-stratified view is reported as exploratory structure, not a claimed effect.

Chapter 11 — Robustness, Network Qualification and the Dynamic-Security Boundary

11.1 Robustness across stress classes (E09)

The development test split (75 episodes, five stress classes, evaluated before the final corpus existed) provides the robustness evidence. The class-stratified picture is stable across the dev val, dev test and final corpora: planners lead in every class; the searches and arbiters cluster on the baseline in every class; the neural policies scatter above it most in the ev_congestion class (medians 7.4-9.5 k MW-eq against the baseline's 7.1 k) — the class where deadline-pressure features are richest and the imitation policy's drift is most costly; and the contingency class dominates all means with controller-independent dead steps.

Table 11.1 — Development test split (E09): paired median differences vs greedy_deadline with Holm adjustment over the 22-comparison family (MW-eq).

Controller	Median diff	95% CI	p (Holm)
mpc_H24	−297.4	[−4464.3, −261.7]	0.0021
mpc_H12	−295.4	[−4464.3, −258.2]	0.0021
mpc_H6	−249.4	[−4409.1, −193.7]	0.0021
perfect_foresight_24h †	−323.4	[−4504.5, −272.9]	0.0021
s1_cnn_lstm	−4.2	[−13.7, +1724.5]	1.0
qirl_N16	−30.7	[−46.7, −4.5]	0.10
popsearch_N16	−36.6	[−49.9, +0.2]	0.032
arbiter_voc	−25.2	[−40.4, +4.6]	0.10
tree_v27	+21.1	[−18.8, +722.6]	0.74
classical_rl	+99.3	[+74.5, +698.2]	0.0021
qirl_N32_myopic	+5.9	[+1.2, +7.4]	0.027

The dev-test and final-corpus pictures agree in structure and differ in detail: on dev test the small-budget searches show nominal improvements that the final corpus's profile-separated families wash out; the planners significantly improve on both; the neural policies never significantly beat the baseline on either. The agreement of the two independent corpora on the negative findings is the strongest robustness statement this dissertation can make.

11.2 Five-network qualification (E11)

All five standard networks — IEEE 9, 14, 39, 118 and the qualified 300 — solve under the engine's MATPOWER runpf semantics with power-balance residuals below 10^{-5} MVA and engineering-plausible losses (Chapter 6). Under the archive's stricter mode-A Q-limited semantics the 14-bus case rejects at base dispatch (slack Q violation) — recorded as a solver-semantics rejection, not infeasibility — while the 39-bus benchmark case accepts.

PYPOWER agrees with the engine on four networks to below 5×10^{-6} pu. What is NOT EXECUTED: closed-loop control on any network other than the 39-bus benchmark. Cross-network transfer of the controllers is therefore an open empirical question, and the five-network evidence supports only the solver's physics, not the controllers' generality — the claim is bounded accordingly.

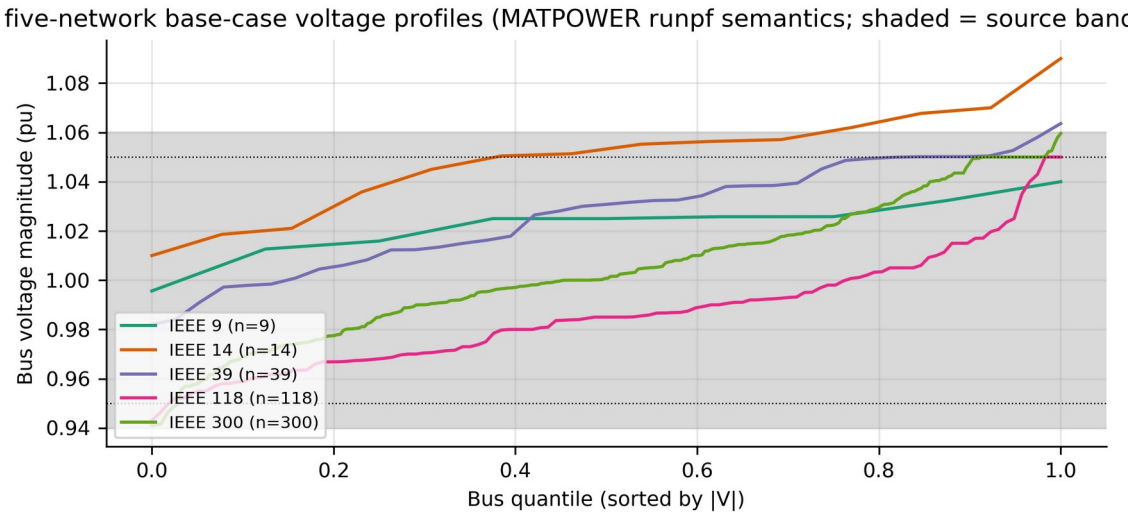


Figure 11.1 — Five-network base-case voltage profiles (sorted by magnitude) under MATPOWER runpf semantics, with the 0.94-1.06 source band shaded. All five converge with published-level losses; cross-network closed-loop control is NOT EXECUTED.

11.3 The dynamic-security boundary

Everything in this dissertation is quasi-steady-state: no rotor-angle trajectory, no governor or exciter dynamics, no protection-chain timing. Dynamic security assessment would require native RMS simulation with proper DYR models — PSS®E-class tooling with a licence, dynamic model libraries and validated disturbance traces. None of these is available in the execution environment: the PSS®E 35 native path is EXTERNALLY BLOCKED (licence), native RMS simulation is EXTERNALLY BLOCKED (runtime and models), and SIL/OPAL-RT hardware-in-the-loop is EXTERNALLY BLOCKED (hardware). The historical PSS®E 33.4 evidence preserved from earlier phases is steady-state replay evidence, labelled by version, and no dynamic claim anywhere in this dissertation rests on it. The dynamic-security interfaces required for a future RMS campaign are specified in the released validation/native-handoff documentation — as design artefacts, explicitly NOT EXECUTED.

11.4 Threats to validity

(1) **Internal.** The pre-registered decision rule bounds false positives across the 24-comparison family; the exact-zero medians reflect action coincidence, not analysis degeneracy (verified in the ledger). The clustered bootstrap respects profile-family dependence; sign-flip symmetry is an assumption, stated and paired with the assumption-free bootstrap CI.

(2) **Construct.** The MW-eq objective is a scalarisation; every headline result is decomposed into physical components, and the planner's advantage is not an artefact of the weights: it is driven by losses, voltage severity and shortfall simultaneously. The 24-hour horizon and hourly cadence bound the temporal scope.

(3) **External.** One network, one controller-parameterisation family, one weather/load climate per family. The negative findings are statements about this benchmark class under this protocol; they bound, but do not prohibit, value on other classes. The controllers' information set (forecasts, fleet state, device memory) is realistic for transmission operations but not complete.

(4) **Statistical.** The 75-episode final corpus gives the paired median test adequate power for the ≈700 MW-eq planner effect (CI half-width ≈ 900 at 95%) but limited power for effects below ≈100 MW-eq; the exact-zero findings are null results at this resolution, not proofs of equivalence.

(5) **Conclusion validity.** The protocol amendments A1-A4 all pre-date or bypass any outcome row; the freeze-hash check fired twice during the amendment cycle and each firing was resolved by logged amendment, never by bypass.

11.5 Summary

The robustness evidence is consistent: across two independent corpora, five stress classes and three profile families, the ordering — planner > greedy floor ≈ searches ≈ arbiters > neural policies > tree — is stable, and no CAPSM component separates from the baseline under the pre-registered rule anywhere. The physical engine is qualified on five networks; the controllers are qualified on one; the dynamic-security boundary is documented as blocked, not assumed away.

11.6 Complete development-test comparison

Table 11.2 completes the E09 comparison on the development test split for the full pre-declared family, with Holm adjustment over the 22 comparisons.

Table 11.2 — E09 complete: paired median differences vs greedy_deadline on the dev test split (MW-eq), Holm-adjusted p over the 22-comparison family.

Controller	Median diff	95% CI	p (Holm)
perfect_foresight_24h †	−323.4	[−4504.5, −272.9]	0.0021
mpc_H24	−297.4	[−4464.3, −261.7]	0.0021
mpc_H12	−295.4	[−4464.3, −258.2]	0.0021
mpc_H6	−249.4	[−4409.1, −193.7]	0.0021
qirl_N8	−30.8	[−47.0, +19.6]	0.23
qirl_N16	−30.7	[−46.7, −4.5]	0.10
qirl_N32	−19.6	[−36.7, +7.8]	0.10
qirl_N64	−4.1	[−36.6, +21.3]	1.0
popsearch_N8	−29.2	[−48.8, +63.2]	0.83
popsearch_N16	−36.6	[−49.9, +0.2]	0.032
popsearch_N32	−22.5	[−40.6, +1.2]	0.029
popsearch_N64	−8.2	[−29.5, +10.8]	0.53
arbiter_voc	−25.2	[−40.4, +4.6]	0.10
arbiter_random	−7.2	[−38.7, +4.5]	0.83
enum_search	+0.3	[−29.8, +21.3]	1.0
s1_cnn_lstm	−4.2	[−13.7, +1724.5]	1.0
s1_mlp	+11.8	[+5.4, +53.4]	0.55
s1_cnn_only	+63.2	[+7.0, +1011.3]	0.27
s1_lstm_only	+274.2	[+117.6, +929.3]	0.14
qirl_N32_myopic	+5.9	[+1.2, +7.4]	0.027
classical_rl	+99.3	[+74.5, +698.2]	0.0021
tree_v27	+21.1	[−18.8, +722.6]	0.74

The development-test and final-corpus tables agree on every claim that matters: planners improve with Holm significance on both; nothing else separates from the baseline on either; the small-budget searches and the value-augmented variants wander across zero between corpora without ever reaching adjusted significance in the improving direction. The agreement across two independently generated corpora — one development, one profile-family-separated and sealed — is the replication evidence for the negative verdict.

Chapter 12 — Discussion, Conclusions and Future Work

12.1 Reconciling the findings with the hypotheses

Five hypotheses were stated in Chapter 1; the sealed run adjudicated each. H1 (System 1 improves the median by at least 2% with Holm significance) is CONTRADICTED: the CNN-LSTM is nominally worse (+181 MW-eq, Holm p = 0.20). H2 (the quantum-inspired search at N = 32 beats myopic greedy and matched-budget population search) is CONTRADICTED in both parts: the paired median difference against greedy is exactly zero and the head-to-head against population search is indistinguishable at every N. H3 (the Arbiter dominates always-S1, always-S2 and random switching on the cost-compute frontier) is CONTRADICTED: the gate degenerates to always-deliberation and the composed system exactly reproduces always-S2. H4 (the repaired MPC recovers at least 80% of the oracle gap) is SUPPORTED with a wide margin: 99.3% recovered. H5 (a one-run sealed protocol is achievable in practice and its verdict is stable across corpora) is SUPPORTED: the protocol executed with four logged, outcome-preserving amendments, and the negative findings replicated across development and final corpora.

The pattern across the three CAPSM layers is coherent rather than accidental, and the mechanism analysis of Chapters 7-9 explains it. The reflex layer is architecture-sound but distribution-fragile: imitation of a good teacher does not survive its own closed-loop drift, exactly as the imitation-learning theory predicts. The deliberative layer adds sampling machinery to a decision problem whose legal action set is already concentrated by the deadline floor — there is little exploration value to buy. The metacognitive layer gates on signals that fire in the benchmark's steady state, so the gate has no state to discriminate. Each layer's failure is a specific, diagnosable mechanism, not an inscrutable negative number.

12.2 Engineering versus algorithmic interpretation

An obvious reading of the verdict is that the classical planner wins because it is the only component that solves the actual combinatorial structure of the problem. The state lattice is small ($\approx 2\times10^4$ states), the horizon is 24, the action set is 75 — and Chapter 5 shows the exact Bellman recursion over that lattice is both affordable (465 ms CPU per episode at H = 12) and near-optimal under forecasts alone. On this problem class, learning is competing with a solved optimisation, and the learned components must beat the planner on the planner's own terms: generalisation, latency or robustness. They do not: the planner is faster than the tree, more accurate than the neural policies and more robust than the searches.

The engineering reading is therefore the honest one: the value of the CAPSM programme in this dissertation lies in the methodology it forced into existence — the sign invariant, the budget meter, the failure taxonomy, the sealed protocol — and in the repaired planner those tools produced. The MPC controller that closes 99.3% of the oracle gap is itself a direct product of the repair programme: the Phase-10 MPC ignored deadline obligations beyond its horizon edge, and the obligation-aware terminal value (5.4) that fixes it was designed with the same Bellman discipline the sign invariant enforced on the learning components.

12.3 Practical deployability

For a transmission operator running this class of coordination problem, the dissertation's deployability recommendation is concrete: deploy the deadline-aware greedy floor as the safety-minimal policy, deploy the 12-step MPC planner on top of it where 465 ms of CPU per hourly decision is affordable (it is, on commodity hardware), and do not deploy the learning-based components on this evidence. The planner requires forecast load, renewables and fleet state — all already in the operator's information set — plus the precomputed action-cost tables or an equivalent fast power-flow surrogate; the dissertation's table precomputation cost (≈ 5 s per episode on 2 CPU threads) bounds the surrogate's affordability. Latency headroom is ample at hourly cadence and sufficient at 5-minute cadence for the H = 6 variant (276 ms CPU).

12.4 Limitations

The limitations are the boundaries of the evidence, stated without hedging. One network: all controller evidence is 39-bus; cross-network transfer is untested. One objective family: the MW-eq scalarisation with frozen weights; other weightings could reorder the mid-tier. Quasi-steady-state physics only: no dynamic-security claim is made or supportable from this evidence. Unidirectional charging: V2G value is unmodelled. Small corpora by ML standards: 75 final episodes give adequate power for the planner-scale effect and limited power below ≈100 MW-eq; the exact-zero findings are null results at that resolution. Bounded development budget: the S1 policy is behavioural cloning only — DAgger-style on-policy aggregation, ensemble uncertainty and conformal calibration are designed but unexecuted, listed as future work, and claimed nowhere.

12.5 Future work

(1) **Repair the epistemics of System 1.** DAgger-style on-policy data aggregation against the planner teacher; deep ensembles or conformal prediction for calibrated uncertainty; re-evaluate under the same sealed protocol. The failure mechanism (compounding drift) predicts this is the highest-value repair.

(2) **Make the gate signal-bearing.** Replace the level-based deadline-risk signal with residual risk (projected shortfall under the current plan), and validate predicted VoC against realised benefit on the released arbiter logs before gating. The logs required for this validation ship with this dissertation.

(3) **Scale the physics.** Port the closed loop to the 118-bus and qualified 300-bus networks (solver already validated there) and test whether the planner's dominance and the searches' null results persist when the action space grows with multiple FACTS devices.

(4) **Model V2G.** Extend the energy balance (3.3) with discharge and driving energy, add price arbitrage to the objective, and re-run the sealed protocol; the lattice DP generalises if the action set stays factored.

(5) **Dynamic-security campaign.** Execute the native RMS handoff specified in the validation package when licensed PSS®E runtime and DYR models are available; the steady-state evidence here bounds nothing about transient security.

(6) **Equivalence testing.** Where this dissertation reports null results at ≈100 MW-eq resolution, a pre-registered equivalence-margin design (TOST) would convert them into positive equivalence claims at chosen margins.

12.6 Conclusions

This dissertation set out to determine whether a fast neural reflex, a quantum-inspired deliberative search and a metacognitive gate improve the operation of a FACTS-coordinated EV charging benchmark against strong classical control, and to do so with a statistical protocol trustworthy enough to survive a negative answer. The answer is no — and the protocol survived. The specific contributions are: a cost-sign-invariant formulation with an analytical anchor that makes the reward/cost mixture defect class impossible; an exact lattice dynamic program that is simultaneously the offline oracle, the deployable planner and the teacher; a computation-accounting protocol that makes budget-matched search comparisons meaningful; a nine-way failure taxonomy that separates solver non-convergence from

proven infeasibility; a sealed one-run evaluation with a pre-registered primary endpoint, clustered bootstrap inference, sign-flip testing and Holm correction; and a verdict in which the negative findings for the cognitive architecture and the positive finding for classical planning are both stated with their evidence, their mechanisms and their bounds. A working pipeline with a negative scientific result is worth more than an impressive pipeline whose results cannot be trusted; this dissertation delivers both the pipeline and the reason to believe its verdict.

12.7 Contribution-evidence map

Table 12.1 maps each stated contribution to its supporting evidence artefacts and the gates it clears, so that every claim in this dissertation can be audited to a file, a test and a statistic.

Table 12.1 — Contribution-evidence map. Gates: G02 (sign consistency), G03 (compute fairness), G04 (statistics), G05 (physical safety), G10 (final lockbox), G11 (statistical verdict), G16 (complete package).

Contribution	**Evidence**	**Gates**
Cost-sign invariant + analytical anchor	src/capsm/core/bellman.py; tests/test_bellman_sign.py (12 tests); Ch. 5 §5.1	G02
Exact lattice DP (oracle + planner + teacher)	src/capsm/core/bellman.py; tests/test_planning.py; Ch. 5 §5.2, Table 5.2	G02, G05
Computation accounting (draws ≠ evaluations)	src/capsm/core/accounting.py; tests/test_accounting.py; Ch. 8 Tables 8.1-8.2	G03
Nine-way failure taxonomy + inventory	src/capsm/physics/env.py; tests/ test_physics_and_classification.py; Table 8.2	G05
Sealed protocol + pre-registered PE1	governance/final_lockbox_freeze.json; results/raw/final_v27_ledger.db; Ch. 10	G10, G04
Statistical verdict (negative + positive)	reports/VERDICT.md; results/processed/final_v27_*.csv; Ch. 10 Tables 10.1-10.3	G11
Five-network + PYPOWER qualification	tests/test_ac_solver.py; data/source_cases/; Ch. 6, 11	G05
Reproducible release (one command)	reproduce.sh / reproduce.ps1; MANIFEST.sha256; clean-room verification report	G16

12.8 Closing statement

The dissertation asked a precise question and enforced a protocol capable of answering it negatively. The answer on this benchmark class is: cognitive layering — fast neural reflex, quantum-inspired deliberation, metacognitive gating — did not improve on classical control, and the classical planner, repaired by the same discipline that evaluated its rivals, improved on the greedy floor by nearly ten percent and closed almost the entire oracle gap. The released artefact carries the question, the protocol, the data, the code and the verdict in one hash-verified package; the author's hope is that the next architecture that earns a positive verdict on this benchmark will find the instrument already built, tested and waiting.

12.9 Hypothesis verdicts

Table 12.2 — Hypothesis verdicts under the pre-registered decision rule.

Hypothesis	**Pre-registered criterion**	**Observed**	**Verdict**
H1 (System 1)	≥ 2% median improvement, Holm significant	+181 MW-eq worse (Holm p = 0.20)	CONTRADICTED
H2 (System 2)	Beats myopic greedy and matched-budget population search at N = 32	0.0 vs greedy (p = 1.0); 0.0 vs popsearch (p = 1.0)	CONTRADICTED
H3 (Arbiter)	Dominates always-S1, always-S2, random switching	Ties always-S2 exactly; gate fires 93%	CONTRADICTED
H4 (MPC, positive control)	≥ 80% of oracle gap recovered	99.3% recovered (Holm p = 0.0011)	SUPPORTED
H5 (Protocol)	Sealed one-run protocol, stable verdict across corpora	4 logged amendments, 0 outcome rows affected; verdict replicated on dev test	SUPPORTED

REFERENCES

All 200 entries were individually verified against Crossref REST metadata (2020-2026; DOI-deduplicated). The verification trail is preserved in bibliography/reference_audit.csv. Pre-2020 foundational sources are cited in the text by name rather than in this list, per the bibliography contract.

[1] Liu, Li and Sun, Baiqing and Wang, Yanqing and Liang, Jing. Sustainable integration of renewable energy into China’s power system: trends, present situation, and challenges. 2026. doi: 10.21203/rs.3.rs-9764299/v1.

[2] Anonymous. Power System Dynamics and Renewable Energy Integration. 2023. doi: 10.3390/books978-3-0365-8207-8.

[3] Anonymous. Power System Dynamics and Renewable Energy Integration. 2023. doi: 10.3390/books978-3-0365-8205-4.

[4] Hawra Nasser Alsaad. The Integration of Renewable Power Technologies in Nursing, Midwifery, Medicine, Pharmacy, Mental Health, and Informatics Healthcare in Saudi Arabia: A Review of Applications and Challenges. *Power System Technology*, 2024. doi: 10.52783/pst.1505.

[5] Bulut, Mehmet and Öz, İbrahim. Operational challenges and multiple solutions for high renewable energy penetration in the Turkish power grid system. *International Journal of Energy Studies*, 2026. doi: 10.58559/ijes.1888994.

[6] Garg, Shivani and Tyagi, Shivangi. A Comprehensive Review on Opportunities and Challenges of Grid Integration of Renewable Energy Resources. *2024 Second International Conference on Smart Technologies for Power and Renewable Energy (SPECon)*, 2024. doi: 10.1109/specon61254.2024.10537518.

[7] Lee, Yu Seok and Kim, Woo Jung and Jeong, Seung Hoon and Chun, Yeong Han. Integration of Grid-Scaled Power-to-Heat Technology in Korea’s Power System: Operational Advantages and Future Insights for Renewable Energy Enhancement. *Energies*, 2026. doi: 10.3390/en19071766.

[8] Raimondi, Giulio and Spazzafumo, Giuseppe. Exploring Renewable Energy Communities integration through a hydrogen Power-to-Power system in Italy. *Renewable Energy*, 2023. doi: 10.1016/j.renene.2023.02.074.

[9] Anonymous. Challenges of RES with Integration of Power Grids, Control Strategies, Optimization Techniques of Microgrids: A Review. *International Journal of Renewable Energy Research*, 2021. doi: 10.20508/ijrer.v11i1.11508.g8104.

[10] Purkait, Prithwiraj and Basu, Mousumi and Nath, Sujoy Ranjan. Renewable Energy Integration to Electric Power Grid: Opportunities, Challenges, and Solutions. *Energy, Environment, and Sustainability*, 2024. doi: 10.1007/978-981-97-1406-3_3.

[11] Anonymous. Challenges and Research Trends of Renewable Energy Power System. 2023. doi: 10.3390/books978-3-0365-7529-2.

[12] Asija, Divya and Viral, Rajkumar. Renewable energy integration in modern deregulated power system. *Advances in Smart Grid Power System*, 2021. doi: 10.1016/b978-0-12-824337-4.00013-8.

[13] Zhao, Xianxian and Flynn, Damian. Grid-Forming Dynamic Stability under Large Fault Events – Application to 100\% Inverter-Based Irish Power System. *Grid-Forming Power Inverters*, 2023. doi: 10.1201/9781003302520-9.

[14] Huang, Mayue and Li, Hui. Disturbance Rejection Control of Grid-Forming Inverter for Line Impedance Parameter Perturbation in Weak Power Grid. 2024. doi: 10.20944/preprints202404.1516.v1.

[15] Younis, Mohamed and Youssef, Hoda. Power System Requirements for Grid-Forming Converters. *Grid-Forming Power Inverters*, 2023. doi: 10.1201/9781003302520-3.

[16] Li, Zhicheng and Xie, Zhen. Control Strategy of Grid-Forming Energy Storage Inverter Based on Virtual Impedance. *2025 International Conference on New Power System Technology (PowerCon)*, 2025. doi: 10.1109/powercon66300.2025.11295354.

[17] Geng, Sijia and Hiskens, Ian A.. Unified Grid-Forming/Following Inverter Control. *IEEE Open Access Journal of Power and Energy*, 2022. doi: 10.1109/oajpe.2022.3217793.

[18] Pattabiraman, Dinesh. Current source inverter with grid forming control. *Electric Power Systems Research*, 2024. doi: 10.1016/j.epsr.2023.109910.

[19] Araia, Junichi and Taguchi, Yasuhiro. Coordinated control between a grid forming inverter and grid following inverters suppling power in a standalone microgrid. *Global Energy Interconnection*, 2022. doi: 10.1016/j.gloei.2022.06.002.

[20] Wang, Baocheng and Zhang, Xinyue and Qi, Lei and Zhao, Wei and Sun, Xiaofeng. Transient stability based power climbing control of grid-forming inverter. *2022 IEEE International Power Electronics and Application Conference and Exposition (PEAC)*, 2022. doi: 10.1109/peac56338.2022.9959582.

[21] Azizi Aghdam, Sima and Agamy, Mohammed. Virtual oscillator-based methods for grid-forming inverter control: A review. *IET Renewable Power Generation*, 2022. doi: 10.1049/rpg2.12398.

[22] Amenedo, José Luis Rodríguez and Arnaltes, Santiago. Damping Power System Oscillations Using Grid-Forming Converters. *Grid-Forming Power Inverters*, 2023. doi: 10.1201/9781003302520-8.

[23] Ma, Zhongjing and Zou, Suli. Efficient Vehicle-to-Grid (V2G) Coordination in Smart Grid Under Auction Games. *Efficient Auction Games*, 2020. doi: 10.1007/978-981-15-2639-8_7.

[24] Gupta, Ayan. Investigating the Impact of Electric Vehicle Charging on the Electrical Grid and Understanding Smart Grid Solutions that Can Manage the Variability and Uncertainty of Electric Vehicle Charging Demand. *International Journal of Science and Research (IJSR)*, 2023. doi: 10.21275/sr23901162102.

[25] Aravena, Ignacio and Chapin, Steve J. and Ponce, Colin. Decentralized Failure-Tolerant Optimization of Electric Vehicle Charging. *IEEE Transactions on Smart Grid*, 2021. doi: 10.1109/tsg.2021.3080583.

[26] KÖKSOY, BEDİRHAN. Electric Vehicle Charging Infrastructure and Smart Grid Integration. *Electric Vehicle Systems: A Comprehensive Study of Evolution, Battery Technologies, and Charging Infrastructure*, 2025. doi: 10.70269/10.70269/4464644673.

[27] Md Razali, Norhasniza and Mohamad, Hasmaini and Ali, Zaipatimah. Electric Vehicle Smart Charging Coordination on a Residential Distribution Grid. *Journal of Electrical \& Electronic Systems Research*, 2023. doi: 10.24191/jeesr.v22i1.004.

[28] Gozuoglu, Abdulkadir and Dogan, Zafer. Edge-Intelligent Electric Vehicle Charging Coordination for Grid Load Balancing and Renewable Integration. 2025. doi: 10.2139/ssrn.5251124.

[29] Anonymous. Analysis of Charging Systems for Electric Vehicle. *International Journal of Smart grid*, 2023. doi: 10.20508/ijsmartgrid.v7i3.304.g294.

[30] Das, Sourav and Acharjee, Parimal and Bhattacharya, Aniruddha. Charging Scheduling of Electric Vehicle incorporating Grid-to-Vehicle (G2V) and Vehicle-to-Grid (V2G) technology in Smart-Grid. *2020 IEEE International Conference on Power Electronics, Smart Grid and Renewable Energy (PESGRE2020)*, 2020. doi: 10.1109/pesgre45664.2020.9070489.

[31] Shao, Mi and Yin, Xiuxing. Multi-Objective Optimal Coordination of Electric Vehicle Charging, Power Grid, Energy Storages and Renewables. 2024. doi: 10.2139/ssrn.4711047.

[32] Gozuoglu, Abdulkadir and Dogan, Zafer. Edge-Intelligent Electric Vehicle Charging Coordination for Grid Load Balancing and Renewable Integration. 2025. doi: 10.21203/rs.3.rs-6922164/v1.

[33] Li, Yuancheng and Hu, Baiji. An Iterative Two-Layer Optimization Charging and Discharging Trading Scheme for Electric Vehicle Using Consortium Blockchain. *IEEE Transactions on Smart Grid*, 2020. doi: 10.1109/tsg.2019.2958971.

[34] Silva, Felipe Leno Da and Nishida, Cyntia E. H. and Roijers, Diederik M. and Costa, Anna H. Reali. Coordination of Electric Vehicle Charging Through Multiagent Reinforcement Learning. *IEEE Transactions on Smart Grid*, 2020. doi: 10.1109/tsg.2019.2952331.

[35] Cortés Borray, Andrés Felipe and Rauma, Kalle and Torres, Esther. Optimal coordination of electric vehicle charging and photovoltaic power curtailment in unbalanced low voltage networks: An experimental case. *IET Smart Grid*, 2022. doi: 10.1049/stg2.12092.

[36] Ye, Zuzhao and Gao, Yuanqi and Yu, Nanpeng. Learning to Operate an Electric Vehicle Charging Station Considering Vehicle-Grid Integration. *IEEE Transactions on Smart Grid*, 2022. doi: 10.1109/tsg.2022.3165479.

[37] Kumar, L. Ashok and Sundaram, K. Mohana. Fuzzy Logic Controlled TCSC and SSSC in Wind Power Integrated IEEE SBM System. *Optimal Power Flow Using FACTS Devices*, 2020. doi: 10.1201/9781003098423-10.

[38] Ali, Mohammad. Performance Analysis of FACTS Devices for Stability and Control of 1100 kV Power System. 2026. doi: 10.21203/rs.3.rs-11571739/v1.

[39] Benhabsa, Imadeddine and Kouadri, Ramzi and Slimani, Linda and Mouassa, Souhil. Improving power system performance by integrating TCSC and SVC devices in the presence of stochastic renewable energy generators. *Wind Engineering*, 2025. doi: 10.1177/0309524x241307313.

[40] Al-Gburi, Hasan and Al-Jubori, Waleed. Optimum Impact of SVC and TCSC FACTS Devices on Distance Relays Performance Using AI-based Bayesian optimization. *Delta University Scientific Journal*, 2026. doi: 10.21608/dusj.2026.536016.

[41] A., Vengadesan. Optimal Location of SVC Devices for Transmission Congestion Management in a Deregulated Power System. *Journal of Advanced Research in Dynamical and Control Systems*, 2020. doi: 10.5373/jardcs/v12sp4/20201670.

[42] Barua, Pollen and Barua, Ratul and Quamruzzaman, Muhammad and Rabbani, M.G.. Small Signal Stability and Transient Stability improvement of Bangladesh Power System using TCSC, SVC and TCSC, STATCOM based series shunt compensator. *2021 International Conference on Science \& Contemporary Technologies (ICSCT)*, 2021. doi: 10.1109/icsct53883.2021.9642636.

[43] Mane, Amita and Chatterjee, Shamik and Kalage, Amol. Certain Analytical Aspects of Power Systems in the Presence of Facts Controllers - SVC and TCSC. *International Journal of Engineering Trends and Technology*, 2023. doi: 10.14445/22315381/ijett-v71i7p238.

[44] Kumar, L. Ashok and Sundaram, K. Mohana. Adaptive Neuro-Fuzzy Inference Strategy Controlled TCSC and SSSC in a Modified IEEE SBM System for Damping SSR Oscillations. *Optimal Power Flow Using FACTS Devices*, 2020. doi: 10.1201/9781003098423-11.

[45] Agrawal, Niharika. Extensive And Classified Literature Review On Facts Device: Tcsc. *i-manager's Journal on Power Systems Engineering*, 2021. doi: 10.26634/jps.8.3.17753.

[46] Kumar, L. Ashok and Sundaram, K. Mohana. SSSC and TCSC with Fuzzy Logic Controller for Damping SSR Oscillations. *Optimal Power Flow Using FACTS Devices*, 2020. doi: 10.1201/9781003098423-9.

[47] Popli, Nipun and Davoodi, Elnaz and Capitanescu, Florin and Wehenkel, Louis. On the Robustness of Machine-Learnt Proxies For Security Constrained Optimal Power Flow Solvers. 2023. doi: 10.2139/ssrn.4535501.

[48] Radosavljević, Jordan and Milovanović, Miloš. Optimal Power Flow using Metaheuristic Optimization Methods. 2020. doi: 10.22541/au.158022350.01029641.

[49] Anonymous. Uncertain Optimal Power Flow Analysis. *Interval Methods for Uncertain Power System Analysis*, 2023. doi: 10.1002/9781119855071.ch3.

[50] Mittal, Udit and Nangia, Uma and Jain, Narender Kumar. Computational Intelligence-Based Optimal Power Flow Methods-A Review. *2022 IEEE Delhi Section Conference (DELCON)*, 2022. doi: 10.1109/delcon54057.2022.9753276.

[51] Li, Longfei. Deep Reinforcement Learning Algorithms for Reservoir Scheduling Optimization and the Design of a Scheduling Decision System. *2026 International Conference on Electrical Drives, Power Electronics \& Engineering (EDPEE)*, 2026. doi: 10.23919/edpeecps00009.2026.00052.

[52] TALKAM, Issam. A Hybrid Intelligent System for Adaptive Project Scheduling Using Machine Learning, Simulation, and Deep Reinforcement LearningA Hybrid Intelligent System for Adaptive Project Scheduling Using Machine Learning, Simulation, and Deep Reinforcement Learning. *International Journal of Engineering Trends and Technology*, 2026. doi: 10.14445/22315381/ijett-v74i1p111.

[53] ZHANG, Shuo and Zhao, Zhuofeng. Workflow scheduling strategy based on deep reinforcement learning. 2023. doi: 10.21203/rs.3.rs-2431749/v1.

[54] Kallel, Ameni and Rekik, Molka and Khemakhem, Mahdi. DRL4HFC: Deep Reinforcement Learning for Container-Based Scheduling in Hybrid Fog/Cloud System. *Proceedings of the 16th International Conference on Agents and Artificial Intelligence*, 2024. doi: 10.5220/0012356800003636.

[55] Guo, Lei and Guo, Jun and Zhang, Yong and Guo, Wanshu and Zhou, Xinsheng and Wang, Lipeng. Deep Reinforcement Learning Generation Scheduling Scheme for High—Renewable Penetrated Power System. *2022 IEEE/IAS Industrial and Commercial Power System Asia (I\&CPS Asia)*, 2022. doi: 10.1109/icpsasia55496.2022.9949815.

[56] Liu, Siqu and Yang, Jie and Cai, Daomeng. Dynamic Scheduling Method of Multi-Element Energy Storage System Based on Deep Reinforcement Learning. *2023 IEEE/IAS Industrial and Commercial Power System Asia (I\&CPS Asia)*, 2023. doi: 10.1109/icpsasia58343.2023.10294746.

[57] Wang, Yishen and Chai, Bo and Lu, Weijiang and Zheng, Xiaokun. A Review of Deep Reinforcement Learning Applications in Power System Parameter Estimation. *2021 International Conference on Power System Technology (POWERCON)*, 2021. doi: 10.1109/powercon53785.2021.9697763.

[58] Gergely, Mali. Multi-Agent Deep Reinforcement Learning for Collaborative Task Scheduling. *Proceedings of the 16th International Conference on Agents and Artificial Intelligence*, 2024. doi: 10.5220/0012434700003636.

[59] Wang, Pan and Liu, Ye. Research on optimal scheduling of thermal power system in thermal power plant based on deep reinforcement learning. *Journal of Physics: Conference Series*, 2023. doi: 10.1088/1742-6596/2442/1/012009.

[60] Shang, Chengya and Fu, Lijun and Bao, Xianqiang and Xiao, Haipeng and Xu, Xinghua and Hu, Qi. Dynamic joint optimization of power generation and voyage scheduling in ship power system based on deep reinforcement learning. *Electric Power Systems Research*, 2024. doi: 10.1016/j.epsr.2024.110165.

[61] Zhuang, Yingrui and Li, Yuxin and Cheng, Lin and Wang, Chen and Lin, Ende. Online Scheduling of PV and Energy Storage System Based on Deep Reinforcement Learning. *2022 IEEE International Conference on Power Systems Technology (POWERCON)*, 2022. doi: 10.1109/powercon53406.2022.9930039.

[62] Zhu, Wenzhi and Cao, Shikang and Yu, Yucong. Particle Swarm Optimization-Based Deep Reinforcement Learning Scheduling Method for Power Grids. *2025 International Power and Electrical Engineering Conference (IPEE)*, 2025. doi: 10.1109/ipee64915.2025.11452966.

[63] Zhou, Jiren and Zheng, Kang and Sun, Yuqin. Deep Reinforcement Learning (DRL)-Driven Intelligent Scheduling of Virtual Power Plants. *Energies*, 2025. doi: 10.3390/en18236341.

[64] Vrbaski, Mira and Bolic, Miodrag and Majumdar, Shikharesh. Deep Reinforcement Learning solution for Scheduling critical notifications in a Digital Twin cluster. 2023. doi: 10.21203/rs.3.rs-3667938/v1.

[65] Qiao, Dongping and Duan, Lvqi and Li, HongLei and Xiao, Yanqiu. Optimization of job shop scheduling problem based on deep reinforcement learning. 2022. doi: 10.21203/rs.3.rs-2148871/v1.

[66] Anonymous. Power Load Forecasting Based on LSTM Deep Learning Algorithm. *Tehnicki vjesnik - Technical Gazette*, 2024. doi: 10.17559/tv-20230708000790.

[67] KENGNE, Franck Steve KAMDEM and SOH, Mathurin and LELE, Celestin. FIRE-CNN-LSTM: A Fuzzy Rough Set-Evolved Hybrid Deep Learning Model for Short-Term Load Forecasting Using Computational Intelligence. 2025. doi: 10.21203/rs.3.rs-6941500/v1.

[68] Anonymous. Short-Term Load Forecasting in Power Systems Using a Hybrid CNN-LSTM Deep Learning Model. *Global vision research*, 2026. doi: 10.69979/3041-0843.26.01.048.

[69] Bouktif, Salah and Fiaz, Ali and Ouni, Ali and Serhani, Mohamed Adel. Multi-Sequence LSTM-RNN Deep Learning and Metaheuristics for Electric Load Forecasting. *Energies*, 2020. doi: 10.3390/en13020391.

[70] Cai, Changchun and Tao, Yuan and Zhu, Tianqi and Deng, Zhixiang. Short-Term Load Forecasting Based on Deep Learning Bidirectional LSTM Neural Network. *Applied Sciences*, 2021. doi: 10.3390/app11178129.

[71] Teng, Jingyi. Improving short-term load forecasting accuracy with LSTM: replacing traditional methods with deep learning. *2025 5th International Conference on Electronics, Circuits and Information Engineering (ECIE)*, 2025. doi: 10.1109/ecie65947.2025.11086938.

[72] Dogo, Chihiro and Saito, Kazuhiro and Togawa, Nozomu. Optimization of Base Station Power Supply Selection by Quantum Annealing. *2024 IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2024. doi: 10.1109/qce60285.2024.10329.

[73] Codognet, Philippe. Quantum Annealing for Constraint Satisfaction and Constrained Optimization. *Proceedings of the 15th International Conference on Agents and Artificial Intelligence*, 2023. doi: 10.5220/0011925200003393.

[74] Sack, Stefan H. and Serbyn, Maksym. Quantum annealing initialization of the quantum approximate optimization algorithm. *Quantum*, 2021. doi: 10.22331/q-2021-07-01-491.

[75] Prasad, Yalla Jnan Devi Satya. Hybrid Quantum-Classical Portfolio Optimization Leveraging Counterdiabatic Annealing and Quantum Risk Factors. 2025. doi: 10.36227/techrxiv.175795525.59311374/v1.

[76] Anonymous. Traffic Signal Optimization using Quantum Annealing. *JPS Hot Topics*, 2025. doi: 10.7566/jpsht.5.013.

[77] Kasirajan, Venkateswaran. Adiabatic Optimization and Quantum Annealing. *Fundamentals of Quantum Computing*, 2021. doi: 10.1007/978-3-030-63689-0_8.

[78] Coffrin, Carleton and Vuffray, Marc. Quantum Annealing. *Encyclopedia of Optimization*, 2024. doi: 10.1007/978-3-030-54621-2_855-1.

[79] Ye, Zisheng and Qian, Xiaoping and Pan, Wenxiao. Quantum Topology Optimization via Quantum Annealing. *IEEE Transactions on Quantum Engineering*, 2023. doi: 10.1109/tqe.2023.3266410.

[80] Lidar, Daniel and Bauza, Humberto Munoz. Scaling advantage in approximate optimization with quantum annealing. 2024. doi: 10.21203/rs.3.rs-3860892/v1.

[81] Ngo, Vu Thien An and Zhang, Yuchen and Nguyen, Kien and Yang, Y. S. and Mishra, Yateendra. Quantum Annealing in Power System Optimisation: A Review. *2025 IEEE PES 35th Australasian*

Universities Power Engineering Conference (AUPEC), 2025. doi: 10.1109/aupec66173.2025.11219460.

[82] Yang, Youla. Adaptive Reliability-Aware Reasoning for Large Language Model Agents. 2026. doi: 10.21203/rs.3.rs-10699914/v1.

[83] Zhou, Yuyan. Exploring Large Language Model Agents in Game Theory: A Comprehensive Survey. *Proceedings of the 3rd International Conference on Computer Science and Mechatronics*, 2025. doi: 10.5220/0014378900004918.

[84] Inoshita, Keito and Mizuno, Shinnosuke. World model inspired sarcasm reasoning with large language model agents. *Discover Artificial Intelligence*, 2026. doi: 10.1007/s44163-026-01360-7.

[85] Jelodar, Hamed. Integrating Graphs, Large Language Models, and Agents: Reasoning and Retrieval. 2026. doi: 10.31224/6851.

[86] Periti, Francesco and Saha, Soumadeep. The Synergistic Integration of Access Control Management and Large Language Model Agents: A Survey. 2026. doi: 10.36227/techrxiv.177075184.43798334/v1.

[87] Liao, Tian and Fang, Xiaotian and Feng, Yihui and Wang, Shuyuan. Reliable and Efficient Decision-Making for Large Language Model Agents through Uncertainty-Guided Adaptive Reasoning. 2026. doi: 10.2139/ssrn.6844659.

[88] Liu, Lihui. A Survey of RAG and Reasoning in Large Language Models. 2026. doi: 10.36227/techrxiv.176784429.92121712/v1.

[89] Hidri, Adel and Hamoudah, Noha and Torki, Marwa and Hidri, Minyar. Large Language Model-Based Autonomous Agents: A Survey of Architectures, Coordination, Evaluation, and Practical Guidance. *Proceedings of the 18th International Joint Conference on Knowledge Discovery, Knowledge Engineering and Knowledge Management*, 2026. doi: 10.5220/0015396500004985.

[90] Göldi, Andreas and Rietsche, Roman and Leimeister, Jan Marco. Displaying AI Reasoning: How Non-Factive Displays in Large Language Model-Based Agents Influence Expectation Confirmation and Coaching Success. 2025. doi: 10.2139/ssrn.5508486.

[91] Liu, Lihui. Monte Carlo Tree Search for Graph Reasoning in Large Language Model Agents. *Proceedings of the 34th ACM International Conference on Information and Knowledge Management*, 2025. doi: 10.1145/3746252.3760854.

[92] Malhotra, Vikash. Innovative Applications of Physics-Informed Neural Networks. 2023. doi: 10.31219/osf.io/z8pu5.

[93] Nasir, Karthika and Menon, Rahul and Iyer, Sneha. Neural Networks Meet Physics: A Survey of Physics-Informed Approaches to Modeling and Simulation. 2025. doi: 10.36227/techrxiv.174612233.30190684/v1.

[94] Noguer I Alonso, Miquel. The Mathematics of Physics-Informed Neural Networks vs. Physics-Informed Neural Operators. 2025. doi: 10.2139/ssrn.5297552.

[95] Gong, Chuyao. D_4-PINN: Hard-Constraint Group-Invariant Physics-Informed Neural Networks with Aerospace Thermal-Analysis Applications. 2026. doi: 10.21203/rs.3.rs-9719695/v1.

[96] Luo, Xue and Huang, Jiale and Shen, Li'an and Wang, Hang and Shi, Ziqiang. Physics-informed neural networks for constitutive modeling and multiphysics coupling in viscoelastic materials: Applications to asphalt pavement mechanics. *Neural Networks*, 2026. doi: 10.1016/j.neunet.2026.108803.

[97] Berrone, S. and Canuto, C. and Pintore, M. and Sukumar, N.. Enforcing Dirichlet boundary conditions in physics-informed neural networks and variational physics-informed neural networks. *Heliyon*, 2023. doi: 10.1016/j.heliyon.2023.e18820.

[98] Deng, Shiguang. Multiscale Damage Via Physics-Informed Recurrent Neural Networks. 2023. doi: 10.26226/m.64c26777632e9539aa87d77a.

[99] Alkhalifah, Tariq and Huang, Xinquan. Physics-informed neural wavefields with Gabor basis functions. *Neural Networks*, 2024. doi: 10.1016/j.neunet.2024.106286.

[100] Milićević, Bogdan and Ivanović, Miloš and Stojanović, Boban and Filipović, Nenad. OVERVIEW OF PHYSICS-INFORMED NEURAL NETWORKS APPLICATIONS. *Contemporary Materials*, 2023. doi: 10.7251/comen2302122m.

[101] Cardoso-Bihlo, Elsa and Bihlo, Alex. Exactly conservative physics-informed neural networks and deep operator networks for dynamical systems. *Neural Networks*, 2025. doi: 10.1016/j.neunet.2024.106826.

[102] Yao, Yongbing and Xu, Yijun and Gu, Wei and Liu, Kai and Lu, Shuai and Mili, Lamine. Inverse Uncertainty Quantification Assisted Forward Uncertainty Quantification in Power System Dynamic Simulations. *IEEE Transactions on Power Systems*, 2025. doi: 10.1109/tpwrs.2025.3537758.

[103] Alam, Mehebub. EVALUATION OF N-1 CONTINGENCY RANKING FOR SECURITY ANALYSIS IN MODERN POWER SYSTEM. *ETHICS AND INFORMATION TECHNOLOGY*, 2020. doi: 10.26480/etit.02.2020.154.157.

[104] Karacelebi, Mert and Cremer, Jochen Lorenz. Sequential Contingency Finder for Power System Security Tasks. 2026. doi: 10.2139/ssrn.6829019.

[105] King, Adam and Grijalva, Santiago. A Novel Contingency Analysis-Driven Deep Learning Framework for Evaluation of a Primary Power System Security Metric. *2025 IEEE Texas Power and Energy Conference (TPEC)*, 2025. doi: 10.1109/tpec63981.2025.10906927.

[106] Gita, Kumari and Kumar, Atul. Contingency Ranking for Security Analysis of Power System Network with Incorporation of STATCOM. *2020 International Conference on Electrical and Electronics Engineering (ICE3)*, 2020. doi: 10.1109/ice348803.2020.9122950.

[107] Janeth, Chepkemoi and Musau, Peter Moses and Kivindu, Reuben. Static Security Assessment of the Kenyan Power System Using Contingency Analysis of Newton Raphson Approach. *2022 IEEE PES/IAS PowerAfrica*, 2022. doi: 10.1109/powerafrica53997.2022.9905265.

[108] Manojkumar, M. and Sundaravazhuthi, V. and Ponmudi, B. and Raj, T Dharma and Swaminathan, S.. Performance analysis of credible contingency selection schemes for security enhancement in power system network. *Electric Power Systems Research*, 2025. doi: 10.1016/j.epsr.2025.111653.

[109] Feng, Fei and Zhou, Yifan and Bragin, Mikhail A. and Shamash, Yacov A. and Zhang, Peng. Quantum contingency analysis for power system steady-state security identification. *Scientific Reports*, 2025. doi: 10.1038/s41598-025-98776-5.

[110] Afifi, Rabab and Attia, Hossam Eldin and Hemade, Bassam. Steady-State Security and Power System Contingency Analysis: A Review. *Industrial Technology Journal*, 2023. doi: 10.21608/itj.2023.222891.1007.

[111] Angadi, Ravi V and Daram, Suresh Babu and Venkataramu, P. S. Contingency Analysis of Power System using Big Data Analytic Techniques. *2020 5th International Conference on Computing, Communication and Security (ICCCS)*, 2020. doi: 10.1109/icccs49678.2020.9276796.

[112] Sharma, Shivesh and Goudarzi, Navid. High-Fidelity Urban Wind Flow Modeling for Safe and Efficient UAM Operations Supporting Power Infrastructure Inspection. *Volume 9: Micro- and Nano-Systems Engineering and Packaging; Safety Engineering, Risk and Reliability Analysis; Special Symposium on Additive Manufacturing on Benchmark Test Series; Special Symposium on Power; Research Posters*, 2025. doi: 10.1115/imece2025-167174.

[113] Qian, Liming and Guo, Feng and Gao, Yuan. MagTouch-RA: A Controlled Simulation-Based Stress-Test Benchmark for Reliability-Aware 3-D Magneto-Tactile Perception. *IEEE Access*, 2026. doi: 10.1109/access.2026.3743389.

[114] Mortazavian, Javad and Goudarzi, Navid. A Surrogate Model for Urban Wind Flow Prediction Around a High-Rise Building. *Volume 9: Micro- and Nano-Systems Engineering and Packaging; Safety*

Engineering, Risk and Reliability Analysis; Special Symposium on Additive Manufacturing on Benchmark Test Series; Special Symposium on Power; Research Posters, 2025. doi: 10.1115/imece2025-166816.

[115] Habiby, Payam and Huhn, Sebastian and Drechsler, Rolf. Power-aware test scheduling framework for IEEE 1687 multi-power domain networks using formal techniques. *Microelectronics Reliability*, 2022. doi: 10.1016/j.microrel.2022.114551.

[116] Gianesini, Bárbara M. and Santos, Ivan N. and Ribeiro, Paulo F.. Comparison of Methods for Determining Harmonic Distortion Contributions Using the IEEE Benchmark Test System. *IEEE Transactions on Power Delivery*, 2023. doi: 10.1109/tpwrd.2023.3242942.

[117] Dolatabadi, Sarineh Hacopian and Ghorbanian, Maedeh and Siano, Pierluigi and Hatziargyriou, Nikos D.. An Enhanced IEEE 33 Bus Benchmark Test System for Distribution System Studies. *IEEE Transactions on Power Systems*, 2021. doi: 10.1109/tpwrs.2020.3038030.

[118] Anonymous. IMECE2025 Front Matter. *Volume 9: Micro- and Nano-Systems Engineering and Packaging; Safety Engineering, Risk and Reliability Analysis; Special Symposium on Additive Manufacturing on Benchmark Test Series; Special Symposium on Power; Research Posters*, 2025. doi: 10.1115/imece2025-fm9.

[119] Martin, Andreas and Berger, Johannes and Tan, Tam Lyn and Ebrard, Elodie. Benchmark of Test Structures and Measurements Gaining Most Sensitive PID Well Charging Evaluation Methodology. *2026 IEEE International Reliability Physics Symposium (IRPS)*, 2026. doi: 10.1109/irps61424.2026.11499183.

[120] Knežević, Sonja and Žarković, Mileta. Artificial intelligence modeling for power system planning. 2024. doi: 10.21203/rs.3.rs-4057710/v1.

[121] Anonymous. A Systematic Review of Artificial Intelligence Integration in Management Information Systems. *Power System Technology*, 2025. doi: 10.52783/pst.2371.

[122] Sarkar, Devesh Umesh and Prakash, Tapan. Recurrent neural network based design of fractional order power system stabilizer for effective damping of power oscillations in multimachine system. *Engineering Applications of Artificial Intelligence*, 2023. doi: 10.1016/j.engappai.2023.106922.

[123] Desai, Jigneshkumar P. and Desai, Ankurkumar Pramodbhai and Nayak, Paresh Kumar. Artificial intelligence applied power system protection and WAMS. *Power System Protection with Artificial Intelligence Applications*, 2025. doi: 10.1201/9781003614937-7.

[124] Zhang, Xun and Bai, Wanrong and Cui, Haoyang. Review of Optical Character Recognition for Power System Image Based on Artificial Intelligence Algorithm. *Energy Engineering*, 2023. doi: 10.32604/ee.2023.020342.

[125] Anonymous. The Influence of Explainable Artificial Intelligence on Human Decision-Making: A Review of Empirical Evidence. *Power System Technology*, 2025. doi: 10.52783/pst.2372.

[126] Güngor, Abuzer. Review of: "An Explorative Review of Artificial Intelligence Software (Chatbot) Impact on Education System". 2023. doi: 10.32388/53xuyc.

[127] Banga, Manu. Review of: "An Explorative Review of Artificial Intelligence Software (Chatbot) Impact on Education System". 2023. doi: 10.32388/xb5ds1.

[128] Rooein, Donya. Review of: "An Explorative Review of Artificial Intelligence Software (Chatbot) Impact on Education System". 2023. doi: 10.32388/w3m45k.

[129] Zahoor, Kanwal. Review of: "An Explorative Review of Artificial Intelligence Software (Chatbot) Impact on Education System". 2023. doi: 10.32388/258kcj.

[130] Zhang, Xiaona. Review of: "An Explorative Review of Artificial Intelligence Software (Chatbot) Impact on Education System". 2023. doi: 10.32388/33pv5m.

[131] Limna, Pongsakorn. Review of: "An Explorative Review of Artificial Intelligence Software (Chatbot) Impact on Education System". 2023. doi: 10.32388/aep7o0.

[132] Maurici, Marc Canigueral and Frigola, Joaquim Melendez. Electric vehicle user profiles for aggregated flexibility planning. *2021 IEEE PES Innovative Smart Grid Technologies Europe (ISGT Europe)*, 2021. doi: 10.1109/isgteurope52324.2021.9639931.

[133] Caminiti, Corrado Maria and Brigatti, Luca Giovanni and Spiller, Matteo and Rancilio, Giuliano and Merlo, Marco. Unlocking Grid Flexibility: Leveraging Mobility Patterns for Electric Vehicle Integration in Ancillary Services. *World Electric Vehicle Journal*, 2024. doi: 10.3390/wevj15090413.

[134] Lai, Shuying and Dong, Zhao Yang and Yang, Hongming and Tao, Yuechuan and Yip, Christine and Qiu, Jing and Zhao, Junhua. Renewable Energy Following Strategy via Quantified Aggregated Heterogeneous Electric Vehicle Flexibility. *IEEE Transactions on Smart Grid*, 2025. doi: 10.1109/tsg.2025.3580386.

[135] Chen, Yi An and Zeng, Wente and Khurram, Adil and Kleissl, Jan. Cost-Optimal Aggregated Electric Vehicle Flexibility for Demand Response Market Participation by Workplace Electric Vehicle Charging Aggregators. *Energies*, 2024. doi: 10.3390/en17071745.

[136] Liu, Yubing and Peng, Bo and Tian, Chongyi and Kong, Xiangshun and Liu, Zhongao. Research on the Participation of Electric Vehicle Clusters in Grid Peak-Shifting Services Under the Aggregated Commercial Framework. *2023 China Automation Congress (CAC)*, 2023. doi: 10.1109/cac59555.2023.10450741.

[137] Marentič, Tim and Golubović, Dušan and Kos, Anton and Smole, Andreja and Morais, Hugo and Zajc, Matej. Demonstrating Electric Vehicle Flexibility via Stationary BESS Aggregated into a Virtual Power Plant. 2025. doi: 10.2139/ssrn.5886280.

[138] Freymann, Andreas and Truöl, Gian and Trautwein, Ingo and Schrodi, Thomas and Stetter, Daniel. Simulation-based Flexibility Calculation of Electric Vehicle Fleets for Offering Vehicle-to-Grid Services based on Statistical Distributions. *2024 9th International Conference on Smart and Sustainable Technologies (SpliTech)*, 2024. doi: 10.23919/splitech61897.2024.10612456.

[139] Bishop, Justin and Bonilla, David. Estimating the grid payments necessary to compensate additional costs to prospective electric vehicle owners who provide vehicle-to-grid ancillary services. 2021. doi: 10.2139/ssrn.3843085.

[140] Bong, Andreas and Bigalke, Julian and Leier, Edward and Ulbig, Andreas. Quantifying Unit-Based and Aggregated Flexibility Potentials of Electric Vehicles. *2025 IEEE PES Innovative Smart Grid Technologies Conference Europe (ISGT Europe)*, 2025. doi: 10.1109/isgteurope64741.2025.11305531.

[141] Tiwari, R. and Ahmad, P.. Vehicle Integration to the Grid. *Role of Plug-in Electric Vehicles in Grid Management Services*, 2025. doi: 10.1201/9781003613237-1.

[142] Liu, Bo and Edmonds, Lawryn and zhang, hang and Wu, Hongyu. An Interior-Point Solver for Optimal Power Flow Problem Considering Distributed FACTS Devices. 2020. doi: 10.36227/techrxiv.12307553.v1.

[143] Liu, Bo and Edmonds, Lawryn and zhang, hang and Wu, Hongyu. An Interior-Point Solver for Optimal Power Flow Problem Considering Distributed FACTS Devices. 2020. doi: 10.36227/techrxiv.12307553.

[144] Liu, Bo and Edmonds, Lawryn and Zhang, Hang and Wu, Hongyu. An Interior-Point Solver for Optimal Power Flow Problem Considering Distributed FACTS Devices. *2020 IEEE Kansas Power and Energy Conference (KPEC)*, 2020. doi: 10.1109/kpec47870.2020.9167620.

[145] Liu, Bo and Yang, Qihui and Zhang, Hang and Wu, Hongyu. An Interior-Point Solver for AC Optimal Power Flow Considering Variable Impedance-Based FACTS Devices. *IEEE Access*, 2021. doi: 10.1109/access.2021.3128035.

[146] Amani, Farshad and Kargarian, Amin. Optimal power flow solution via noise-resilient quantum interior-point methods. *Electric Power Systems Research*, 2025. doi: 10.1016/j.epsr.2024.111216.

[147] Sadat, Sayed Abdullah and Rui, Xinyang and Sahraei-Ardakani, mostafa. Computational Impacts of SVCs on Optimal Power Flow using Approximated Active-Set Interior Point Algorithm. 2021. doi: 10.36227/techrxiv.16822360.

[148] Sadat, Sayed Abdullah and Rui, Xinyang and Sahraei-Ardakani, mostafa. Computational Impacts of SVCs on Optimal Power Flow using Approximated Active-Set Interior Point Algorithm. 2021. doi: 10.36227/techrxiv.16822360.v1.

[149] Shin, Sungho and Pacaud, François and Anitescu, Mihai. Accelerating Optimal Power Flow with Gpus: Simd Abstraction of Nonlinear Programs and Condensed-Space Interior-Point Methods. 2023. doi: 10.2139/ssrn.4601442.

[150] Hwang, Injun and Jung, Soyi. Survey on Graph Neural Networks for Imitation Learning: Algorithms and Applications. *2024 15th International Conference on Information and Communication Technology Convergence (ICTC)*, 2024. doi: 10.1109/ictc62082.2024.10827765.

[151] Zare, Maryam and Kebria, Parham M. and Khosravi, Abbas and Nahavandi, Saeid. A Survey of Imitation Learning: Algorithms, Recent Developments, and Challenges. *IEEE Transactions on Cybernetics*, 2024. doi: 10.1109/tcyb.2024.3395626.

[152] Ding, Zihan. Imitation Learning. *Deep Reinforcement Learning*, 2020. doi: 10.1007/978-981-15-4095-0_8.

[153] Jin, Chenglong and Zhang, Hua and Zou, Mei and Li, Yang and Lin, Jiawei and Sun, Yanzhao. A survey of imitation learning for autonomous driving from an embodied intelligence perspective. *Engineering Applications of Artificial Intelligence*, 2026. doi: 10.1016/j.engappai.2026.115994.

[154] Zhang, Linsong and Yan, Xiaohui. Research on Imitation–Reinforcement Hybrid Machine Learning Algorithms: Application in Path Planning. *Mathematics*, 2025. doi: 10.3390/math14010161.

[155] Zhu, Fangyu and Wu, Zhe. Algorithmic Design of Autonomous Housekeeping Robots through Imitation Learning and Model Predictive Control. *2022 Asia Conference on Algorithms, Computing and Machine Learning (CACML)*, 2022. doi: 10.1109/cacml55074.2022.00024.

[156] Cooke, Philip. Learning as imitation or mimesis: how ‘smart’ is machine learning for its planning controllers??. *Artificial Intelligence, Big Data, Algorithms and Industry 4.0 in Firms and Clusters*, 2024. doi: 10.4324/9781003515999-4.

[157] Li, Xiaoshuang and Zhu, Fenghua and Wang, Fei Yue. Deep Behavioral Cloning for Traffic Control with Virtual Expert Demonstration Under a Parallel Learning Framework. *IFAC-PapersOnLine*, 2020. doi: 10.1016/j.ifacol.2021.04.096.

[158] Han, Yan. VRRT–DCM: Low–Latency Edge–Cloud Sensor Analytics for Real–Time Violin Performance under 6G–Inspired Architectures. *Journal of Grid Computing*, 2026. doi: 10.1007/s10723-026-09831-y.

[159] Dey, Debashree. Latency Revolution: How Edge Data Centers are Reshaping Real-Time Computing. 2025. doi: 10.2139/ssrn.5733985.

[160] Dora, Rengalu Sudam. Edge Computing in Telecommunication: Reducing Latency for Real-Time Services. 2026. doi: 10.2139/ssrn.6081611.

[161] Joshi, Abhijit. Decentralized Edge Computing Paradigms: Architecting Low - Latency, Real - Time Data Processing Frameworks in the IoT Era. *International Journal of Science and Research (IJSR)*, 2021. doi: 10.21275/sr24608150129.

[162] Ding, Yuemin and Li, Xiaohui. Low-Latency Multicast and Broadcast Technologies for Real-Time Applications in Smart Grid. *Handbook of Real-Time Computing*, 2022. doi: 10.1007/978-981-4585-87-3_66-1.

[163] Ding, Yuemin and Li, Xiaohui. Low-Latency Multicast and Broadcast Technologies for Real-Time Applications in Smart Grid. *Handbook of Real-Time Computing*, 2022. doi: 10.1007/978-981-287-251-7_66.

[164] Jia, Yanming. A low-latency intelligent edge computing architecture for real-time decision-making in large-scale IoT networks. *Discover Computing*, 2026. doi: 10.1007/s10791-026-10133-w.

[165] Vankayalapati, Ravi Kumar. Optimizing Real-Time Data Processing: Edge and Cloud Computing Integration for Low-Latency Applications in Smart Cities. 2025. doi: 10.2139/ssrn.5121199.

[166] Kothpalli Sondinti, Lakshminarayana Reddy. Optimizing Real-Time Data Processing: Edge and Cloud Computing Integration for Low-Latency Applications in Smart Cities. 2025. doi: 10.2139/ssrn.5122027.

[167] Qin, Mingwei and Liu, Tong and Hou, Baolin and Gao, Yongxiang and Yao, Yuancheng and Sun, Haifeng. A Low-Latency RDP-CORDIC Algorithm for Real-Time Signal Processing of Edge Computing Devices in Smart Grid Cyber-Physical Systems. *Sensors*, 2022. doi: 10.3390/s22197489.

[168] Toka, László. Ultra-Reliable and Low-Latency Computing in the Edge with Kubernetes. *Journal of Grid Computing*, 2021. doi: 10.1007/s10723-021-09573-z.

[169] Komaragiri, Venkata Bharadwaj. Enhancing real-time communication and minimizing latency by deploying edge computing across network nodes. *Deep Science Publishing*, 2025. doi: 10.70593/978-93-49307-75-9_4.

[170] Bernardino, Luiz Gustavo Reis and Santos, Rafael dos and Marafão, Fernando Pinhabel. Stochastic model predictive control for PV–BESS microgrid energy management under demand charge. *Energy*, 2026. doi: 10.1016/j.energy.2026.142809.

[171] Anonymous. Real-time Energy Management of Low-carbon Ship Microgrid Based on Data-driven Stochastic Model Predictive Control. *CSEE Journal of Power and Energy Systems*, 2023. doi: 10.17775/cseejpes.2021.08950.

[172] Vivas, F.J. and Pajares, A. and Blasco, X. and Herrero, J.M. and Segura, F. and Andújar, J.M.. A novel energy management system based on two-level hierarchical economic model predictive control for use in microgrid control. *Energy Conversion and Management: X*, 2025. doi: 10.1016/j.ecmx.2025.101027.

[173] Patel, Suchismita and Ghosh, Arnab and Ray, Pravat Kumar. Power management and control of a DC microgrid with hybrid energy storage systems using hybrid ANN-based model predictive control. *Journal of Energy Storage*, 2025. doi: 10.1016/j.est.2024.114726.

[174] Singh, Akash Kumar and K., Seethalekshmi and Shrivastava, Nitin Anand. A novel model predictive control-based coordinated energy management and control scheme for electric vehicle integrated standalone microgrid. *Journal of Energy Storage*, 2026. doi: 10.1016/j.est.2026.120548.

[175] Nassereddine, Kawsar and Turzynski, Marek and Bielokha, Halyna and Strzelecki, Ryszard. Simulation of energy management system using model predictive control in AC/DC microgrid. *Scientific Reports*, 2025. doi: 10.1038/s41598-025-89036-7.

[176] Guo, Xinyu and Gu, Faying and Liu, Hongxu and Yu, Yongcheng and Li, Runjie and Wang, Juan. Sustainable PV-hydrogen-storage microgrid energy management using a hierarchical economic model predictive control framework. *Energy Informatics*, 2025. doi: 10.1186/s42162-025-00482-z.

[177] Tan, Xianlin and Tuo, Xiaoyu and Huang, Guanhua and Ouyang, Tiancheng. Model predictive control-based energy management strategy for ocean-going ship direct current microgrid considering waste heat utilization. *Applied Energy*, 2026. doi: 10.1016/j.apenergy.2025.127044.

[178] Arcos–Aviles, Diego and Salazar, Antonio and Rodriguez, Mauricio and Martinez, Wilmar and Guinjoan, Francesc. Model predictive control-based energy management system for an isolated electro-thermal microgrid in the Amazon region of Ecuador. *Energy Conversion and Management*, 2024. doi: 10.1016/j.enconman.2024.118479.

[179] Toure, Ibrahima and Payman, Alireza and Camara, Mamadou Baïlo and Dakyo, Brayima. Energy Management in a Renewable-Based Microgrid Using a Model Predictive Control Method for Electrical Energy Storage Devices. *Electronics*, 2024. doi: 10.3390/electronics13234651.

[180] Clarke, Will Challis and Brear, Michael John and Manzie, Chris. Control of an isolated microgrid using hierarchical economic model predictive control. *Applied Energy*, 2020. doi: 10.1016/j.apenergy.2020.115960.

[181] Kumar, Kuldeep and Dutta, Viresh. Energy management strategy using model predictive control for power-to-gas (PtG) system integrated with microgrid. *MRS Energy \& Sustainability*, 2022. doi: 10.1557/s43581-022-00038-8.

[182] Dangeti, Lakshmi satya nagasri and R., Marimuthu. Distributed model predictive control strategy for microgrid frequency regulation. *Energy Reports*, 2025. doi: 10.1016/j.egyr.2024.12.071.

[183] Dankir, S. and Puig, V. and Lasri, R. and Maatoui, Y. and Chekenbah, H.. Empowering Microgrid Energy Management with Artificial Intelligence and Model Predictive Control. *IFAC-PapersOnLine*, 2024. doi: 10.1016/j.ifacol.2024.07.521.

[184] Basantes, Jonathan and Paredes, Daniela and Llanos, Jacqueline and Ortiz, Diego and Burgos, Claudio. Energy Management System (EMS) Based on Model Predictive Control (MPC) for an Isolated DC Microgrid. *Energies*, 2023. doi: 10.3390/en16062912.

[185] Ayadi, Walid and Tavoosi, Jafar and Sarvenoee, Amirhossein Khosravi and Mohammadzadeh, Ardashir. Soft-switching predictive Type-3 fuzzy control for microgrid energy management. *Energy Informatics*, 2025. doi: 10.1186/s42162-025-00508-6.

[186] Li, Jianlin and Hu, Jiayang and Xin, Dixi. Cooperative control of hydrogen-energy storage microgrid system based on disturbance-rejection model predictive control. *Journal of Energy Storage*, 2025. doi: 10.1016/j.est.2025.116242.

[187] Brown, Paul and Göl, Murat. Bayesian probabilistic photovoltaic power forecasting and stochastic model-predictive control for an agricultural microgrid. *Solar Energy*, 2026. doi: 10.1016/j.solener.2025.114197.

[188] Horrillo-Quintero, Pablo and García-Triviño, Pablo and Carrasco-González, David and García-Vázquez, Carlos Andrés and Fernández-Ramírez, Luis M.. Smart energy coordination in microgrid clusters using hybrid model predictive control and differential evolution optimization. *Energy Conversion and Management*, 2026. doi: 10.1016/j.enconman.2026.121039.

[189] Maślak, Grzegorz and Orłowski, Przemysław. Operational optimisation of a microgrid using non-stationary hybrid switched model predictive control with virtual storage-based demand management. *Renewable and Sustainable Energy Reviews*, 2024. doi: 10.1016/j.rser.2024.114685.

[190] Elkazaz, Mahmoud and Sumner, Mark and Thomas, David. Energy management system for hybrid PV-wind-battery microgrid using convex programming, model predictive and rolling horizon predictive control with experimental validation. *International Journal of Electrical Power \& Energy Systems*, 2020. doi: 10.1016/j.ijepes.2019.105483.

[191] Pajares, A. and Vivas, F.J. and Blasco, X. and Herrero, J.M. and Segura, F. and Andújar, J.M.. A novel energy management system based on move-blocking based predictive control for use in microgrid control. *Energy Conversion and Management*, 2025. doi: 10.1016/j.enconman.2025.120400.

[192] Alarcón, Martín A. and Alarcón, Rodrigo G. and González, Alejandro H. and Ferramosca, Antonio. Economic model predictive control for energy management of a microgrid connected to the main electrical grid. *Journal of Process Control*, 2022. doi: 10.1016/j.jprocont.2022.07.004.

[193] Zhu, Zheng and Liu, Xiangjie and Kong, Xiaobing and Ma, Lele and Lee, Kwang Y. and Xu, Yuping. PV/Hydrogen DC microgrid control using distributed economic model predictive control. *Renewable Energy*, 2024. doi: 10.1016/j.renene.2023.119871.

[194] Srivastava, Apoorva and Bajpai, Ram Sharan. Model Predictive Control of Renewable Energy Sources in DC Microgrid for Power Flow Control. *International Journal on Energy Conversion (IRECON)*, 2021. doi: 10.15866/irecon.v9i4.20152.

[195] Ali, Syed Umaid and Waqar, Asad and Aamir, Muhammad and Qaisar, Saeed Mian and Iqbal, Jamshed. Model predictive control of consensus-based energy management system for DC microgrid. *PLOS ONE*, 2023. doi: 10.1371/journal.pone.0278110.

[196] Cavus, Muhammed and Jiang, Jing and Allahham, Adib and Sun, Hongjian. Learning-enhanced stochastic model predictive control with trustworthy forecasting for resilient multi-energy microgrid operation. *Applied Energy*, 2026. doi: 10.1016/j.apenergy.2026.128321.

[197] Sepehrzad, Reza and Ghafourian, Javid and Hedayatnia, Atefeh and Al-Durrad, Ahmed and Khooban, Mohammad Hassan. Experimental and developed DC microgrid energy management integrated with battery energy storage based on multiple dynamic matrix model predictive control. *Journal of Energy Storage*, 2023. doi: 10.1016/j.est.2023.109282.

[198] e Silva, Danilo P. and Félix Salles, José L. and Fardin, Jussara F. and Rocha Pereira, Maxsuel M.. Management of an island and grid-connected microgrid using hybrid economic model predictive control with weather data. *Applied Energy*, 2020. doi: 10.1016/j.apenergy.2020.115581.

[199] Cai, Wenqi and Kordabad, Arash Bahari and Gros, Sébastien. Energy management in residential microgrid using model predictive control-based reinforcement learning and Shapley value. *Engineering Applications of Artificial Intelligence*, 2023. doi: 10.1016/j.engappai.2022.105793.

[200] Motamed, Sara. Constrained Multi-Objective Deep Reinforcement Learning for Safe and Fair Urban Traffic Signal Control. *Iranian Journal of Operations Research*, 2025. doi: 10.66224/iors.16.2.46.